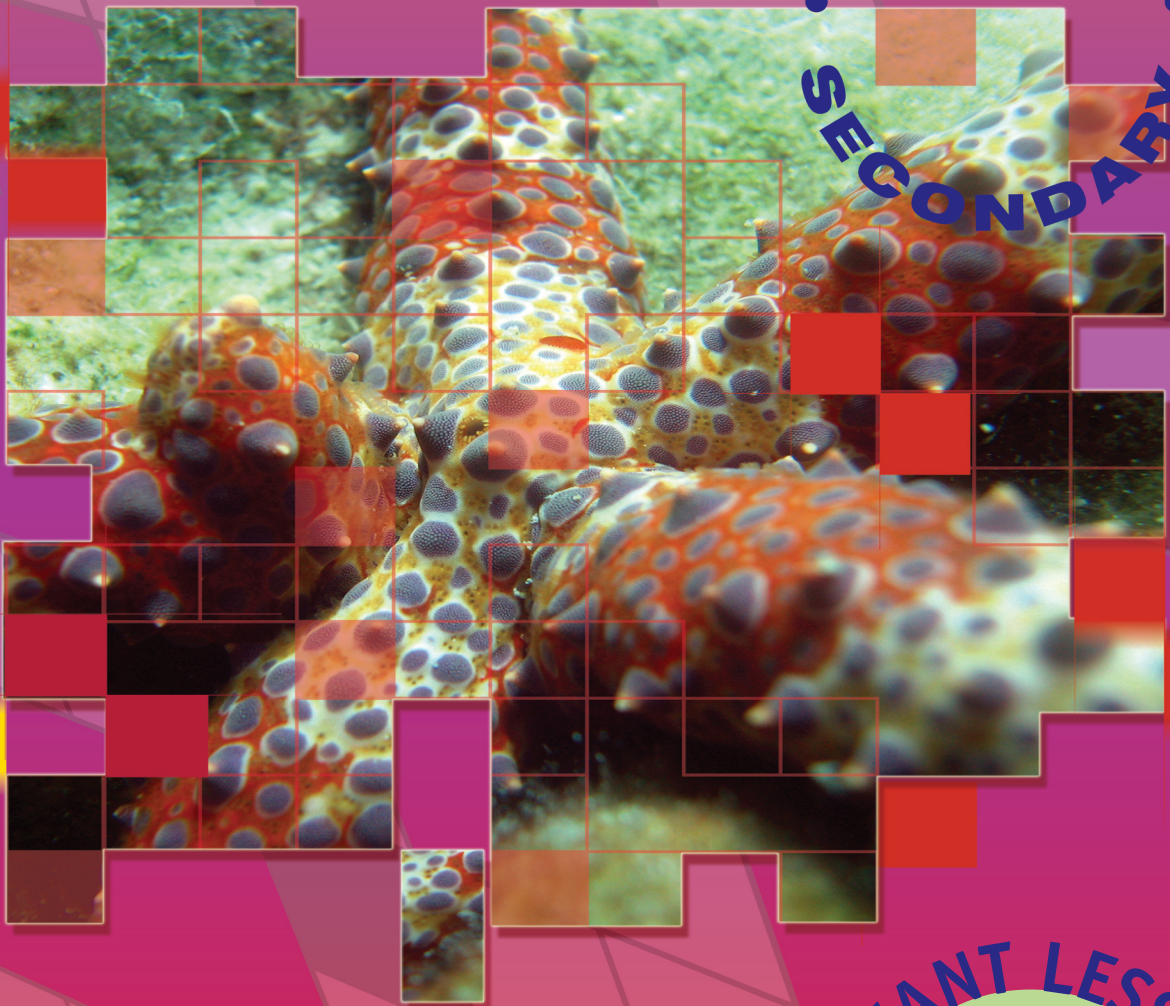


CHALLENGING SCIENCE

CROSSWORDS

MIDDLE

• SECONDARY •



Dr Raimund Pohl

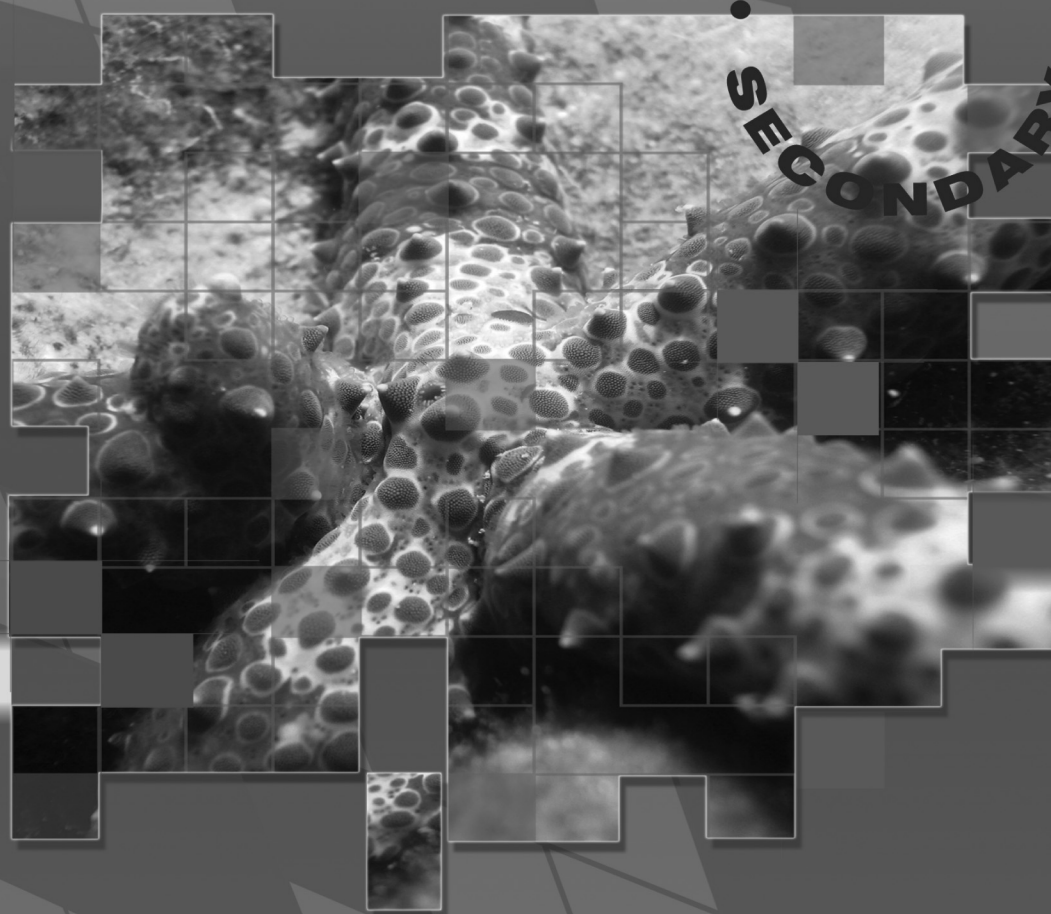


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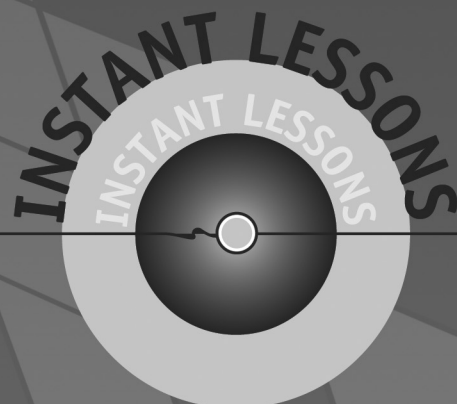
CHALLENGING SCIENCE CROSSWORDS

MIDDLE

SECONDARY



Dr Raimund Pohl



Acknowledgements

I would like to thank the students of Cheltenham Girls' High School for trialling these crosswords, the teachers who also assisted by their positive criticisms and last but not least my wife, who gave up her precious time in proofing these crosswords.

Dr Raimund Pohl

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INTRODUCTION



This series of *Challenging Science Crosswords* is a unique, student-friendly approach to increasing science literacy. They have been thoroughly trialled in classrooms, with positive feedback from students who find the crosswords great for revision and fun to do. Teachers have found the crosswords to be excellent springboards for promoting discussion of science concepts and issues, and for introducing or revising core curriculum. They also love the opportunity to target every student's ability and confidence level – the crosswords contain a wordlist at the bottom of the page. This list can easily be covered when photocopying to create a more challenging activity or left on to provide helpful hints for students.

Challenging Science Crosswords need no further preparation, and are stimulating, educational activities, perfect for last-minute teaching or integration into a science unit.

What the students said: *Interesting and not too difficult ... Great for revision ... Learnt something new ... Challenging – I love science crosswords ... They are fun ... Good material to study with ... Educational! ... The word bank helped a lot ... I tried these and I remember.*

What the teachers said: *A real help for revision and reinforcing of ideas ... Good to see something that makes revision enjoyable ... Why didn't I think of that? ... The word bank makes it great for the junior years.*

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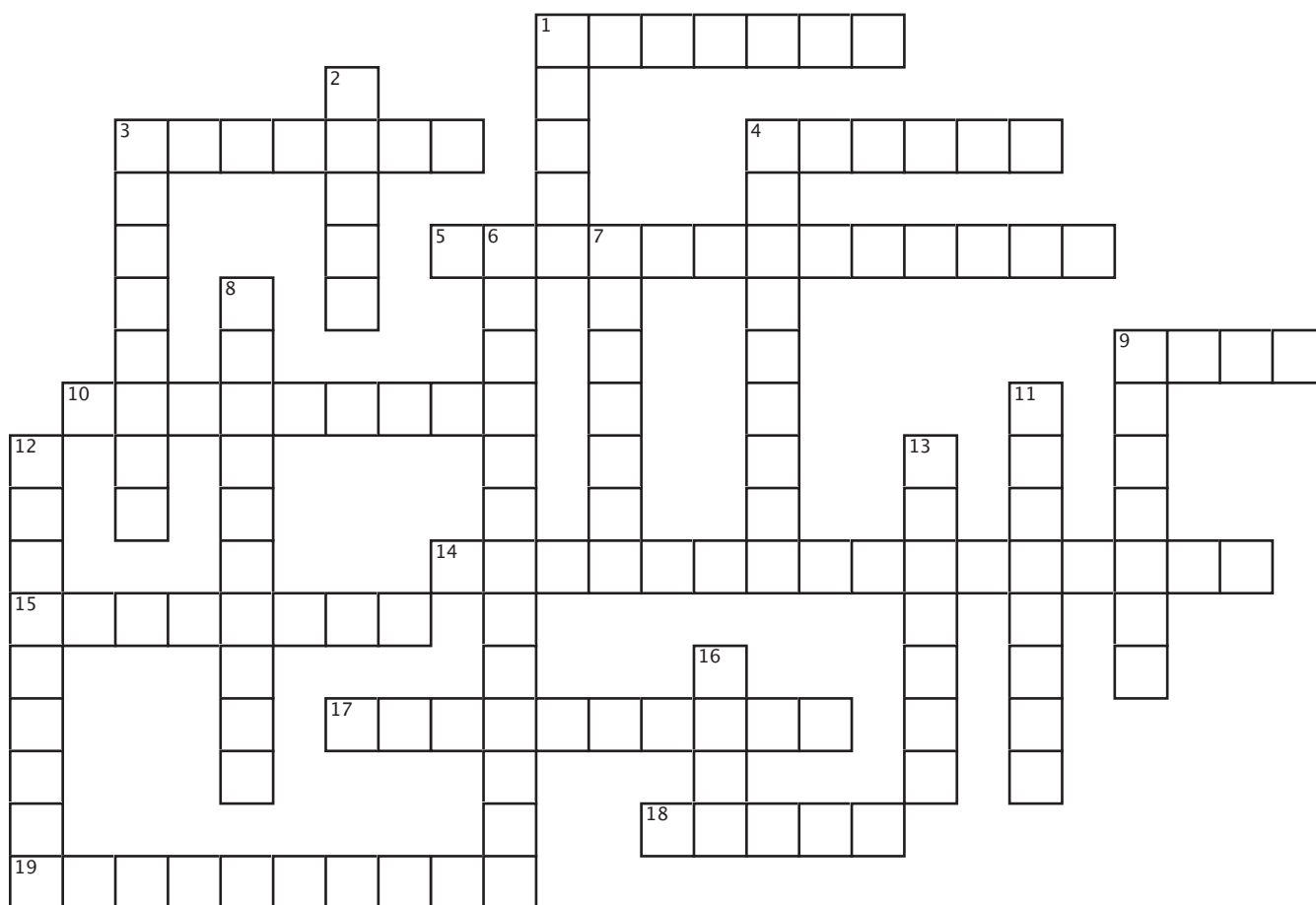
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ACROSS

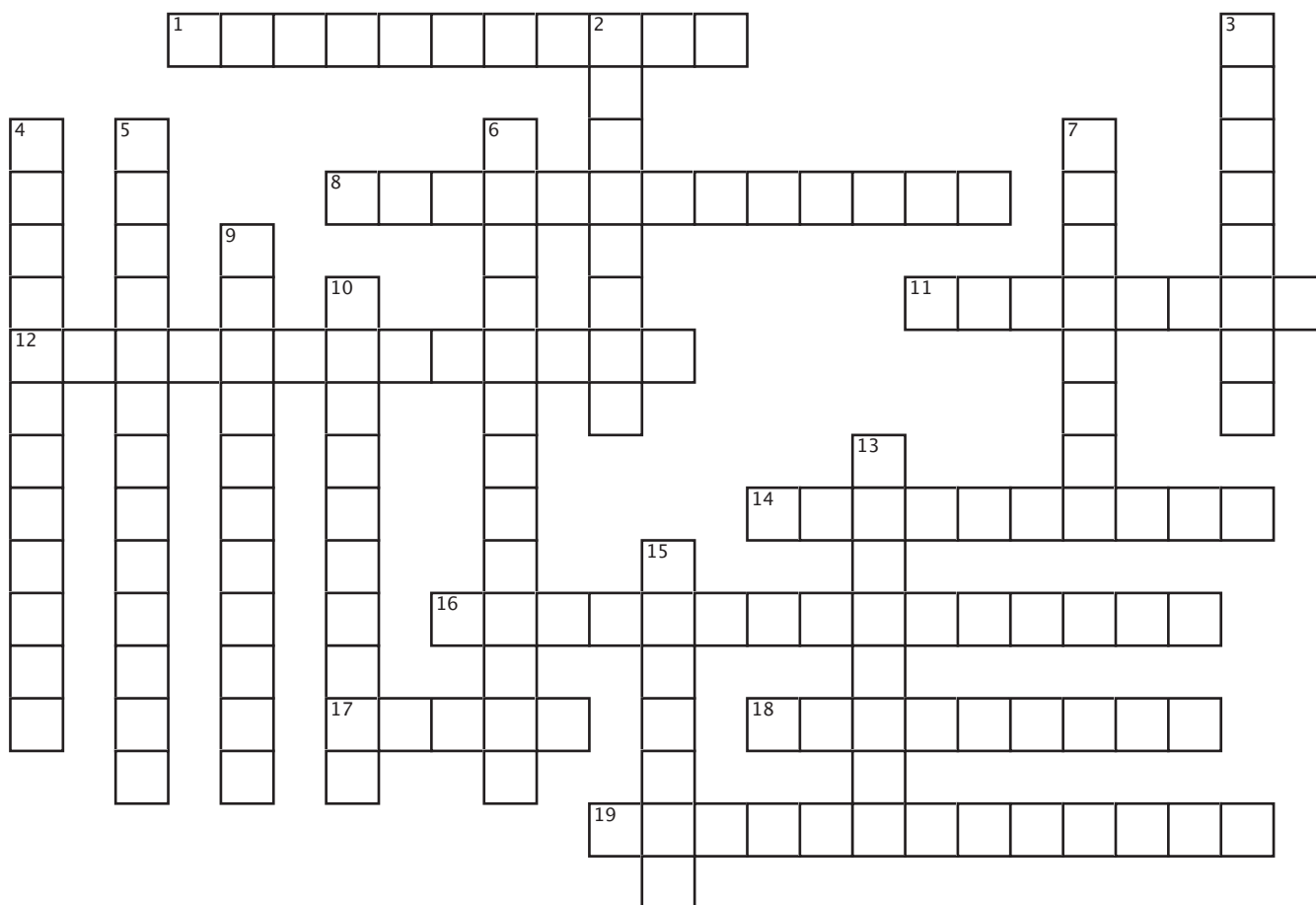
1. A natural event which forms over water, caused by the rapid inward circulation of air masses and heat
3. Correct term for tidal wave
4. Oceanic crust is made from this rock
5. Layer beneath the lithosphere that behaves plastically
9. A bend in rock layers
10. An up-fold or arch of rock layers
14. Movement of plates
15. Term used for determining precise geological dating
17. Sudden release of tensional force along a fault
18. Molten rock inside the Earth
19. The sliding of one plate under another

DOWN

1. Least dense layer of the Earth
2. A fracture in bedrock along which movement has occurred
3. Force that causes uplift, movement and deformation of the crust
4. Sphere that contains all living things
6. The law that states that the youngest layer of rock is always at the top and the oldest at the bottom
7. Hawaiian islands are formed from this
8. Science that has unravelled the structure of the Earth
9. Previous living organisms used to date rock layers
11. A super continent made up of Australia, Antarctica, Africa, South America and India
12. Collision of two continental plates formed these mountains
13. Force that pulls plates under another plate
16. Molten material extruded by a volcano

WORD LIST

☆ seismology	☆ earthquake	☆ gravity	☆ tectonic	☆ hotspot
☆ continental drift	☆ superposition	☆ fossils	☆ crust	☆ Himalayas
☆ tsunami	☆ subduction	☆ basalt	☆ asthenosphere	☆ fault
	☆ magma	☆ absolute	☆ Gondwana	☆ anticline
	☆ lava	☆ cyclone	☆ biosphere	☆ fold



ACROSS

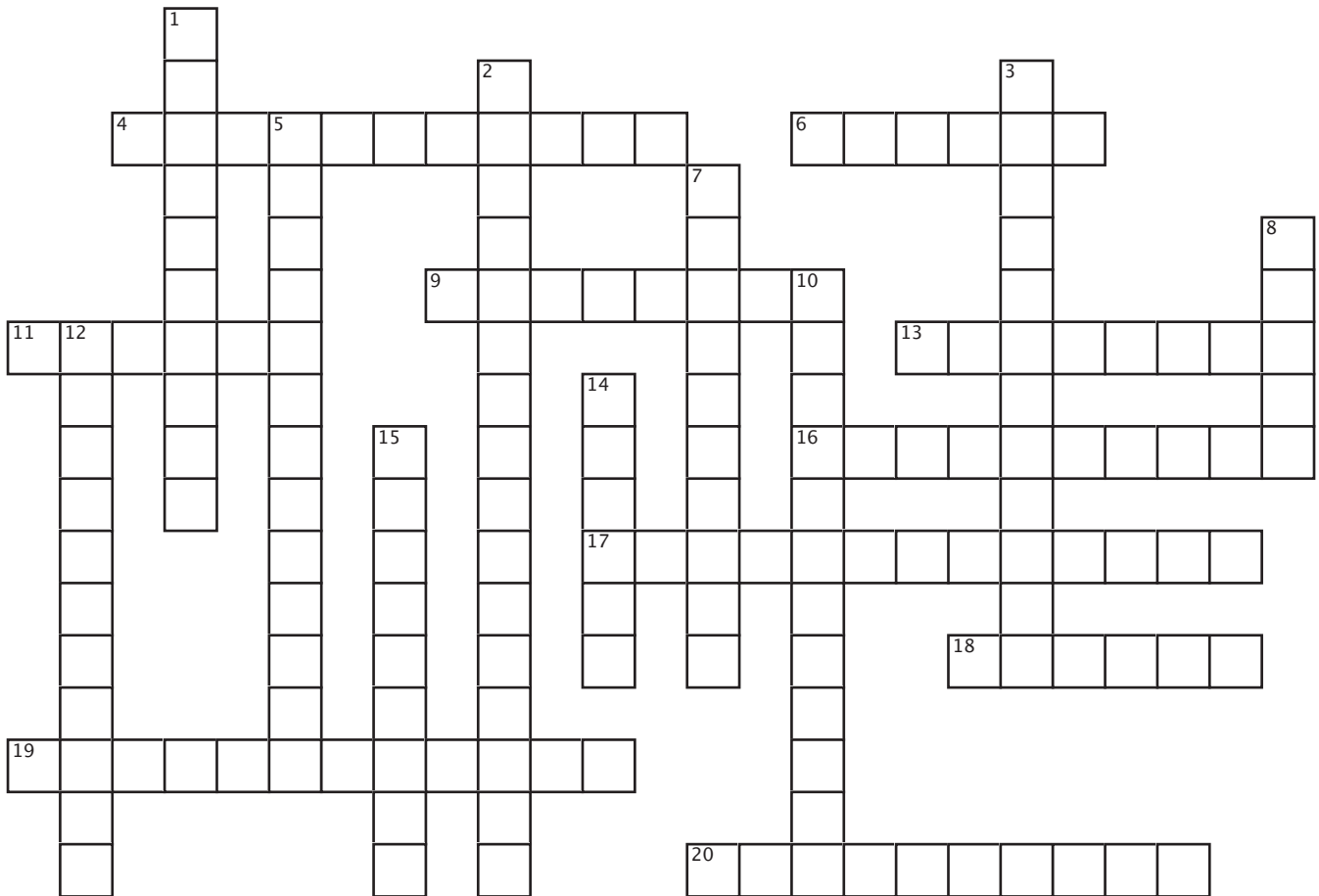
1. An extinct species of humans, regarded as an ancestor of Homo sapiens
8. Mats of cyanobacteria deposited with calcium carbonate
11. Means middle-life period and includes the Triassic, Jurassic and Cretaceous periods
12. Oldest oxygen-producing organisms
14. Another name for chalk period
16. Scientist who studies ancient life
17. Type of fossil that has short geologic time range but wide distribution
18. Era after Mesozoic named after humans
19. Age of amphibians, first winged insects and seed ferns

DOWN

2. Mammals first appeared in this period
3. First land animals and seedless vascular plants appeared in this period
4. A small, active, carnivorous dinosaur related to the ancient birds; featured in the film 'Jurassic Park'
5. Name of ancient bird
6. Jellyfish and similar fossils found in Ediacra, South Australia
7. Belonging to the coiled chambered fossil shells of extinct molluscs
9. First life period
10. Means 'ancient life'
13. Period known as the Age of Fishes
15. Period when almost all life was wiped out

WORD LIST

☆ Mesozoic	☆ homoerectus	☆ archaeopteryx	☆ stromatolites	☆ Permian
☆ ammonite	☆ Proterozoic	☆ Devonian	☆ index	☆ palaeontologist
☆ velociraptor	☆ Cainozoic	☆ Carboniferous	☆ Triassic	☆ Silurian
☆ Palaeozoic	☆ coelenterates	☆ cyanobacteria	☆ Cretaceous	



ACROSS

4. Dinosaur with large plates above its backbone
6. The name of the super-continent where dinosaurs evolved
9. Long-necked and long-tailed, herbivorous dinosaur
11. Queensland town that has sandstone fossilised footprints of stampeding dinosaurs
13. Marine reptile associated with the Loch Ness monster
16. Egg-eating dinosaurs
17. Bird-like dinosaur whose name means 'ancient feather'
18. North western, Australian town where dinosaur footprints have been found
19. A bipedal, plant-eating dinosaur named after an Australian airline
20. Geologic period when dinosaurs became extinct

DOWN

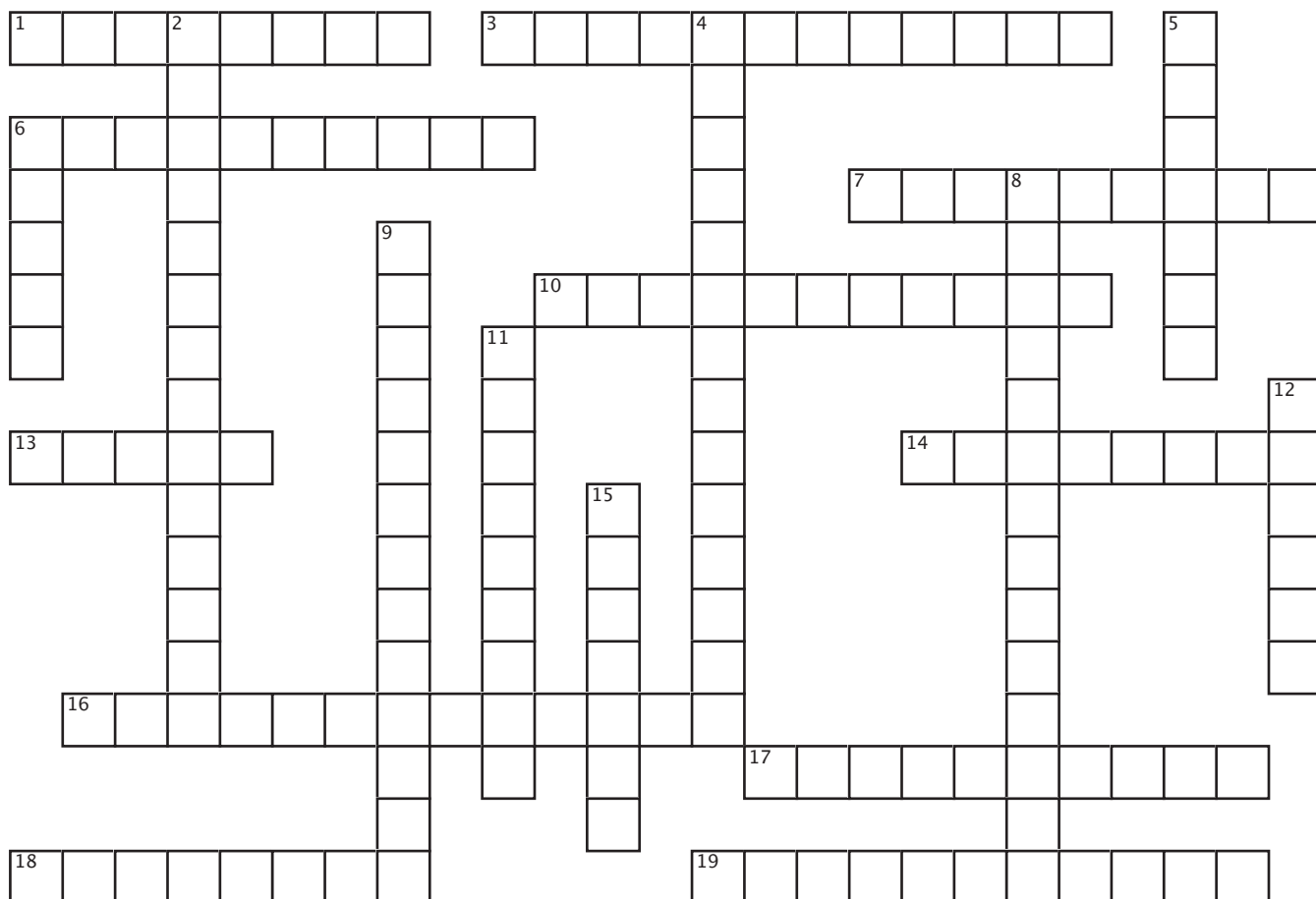
1. Carnosaurus was found on this continent
2. Thumbed, plant-eating dinosaur that lived in cretaceous Australia
3. The name of this dinosaur means 'fast robber'
5. Largest, dagger-toothed dinosaur that roamed Gondwanan forests
7. Longest of all known dinosaurs
8. Modern descendants of the dinosaurs
10. Largest dinosaur site in Australia
12. Fast swimming reptile similar to today's dolphins
14. Fern-like food for plant-eating dinosaurs
15. Flying reptile

WORD LIST

☆ birds	☆ Ichthyosaur	☆ Velociraptor	☆ Diplodocus	☆ Oviraptors
☆ Muttabutt	☆ Pliosaur	☆ Broome	☆ Archaeopteryx	☆ Antarctica
☆ asaurus	☆ Pterosaur	☆ Dinosaur Cove	☆ Winton	☆ Pangea
☆ Gigantosaurus	☆ Sauropod	☆ Stegosaurus	☆ Cretaceous	☆ Cycads
	☆ Qantassaurus			



AUSTRALIAN MEGA FAUNA AND SITES



ACROSS

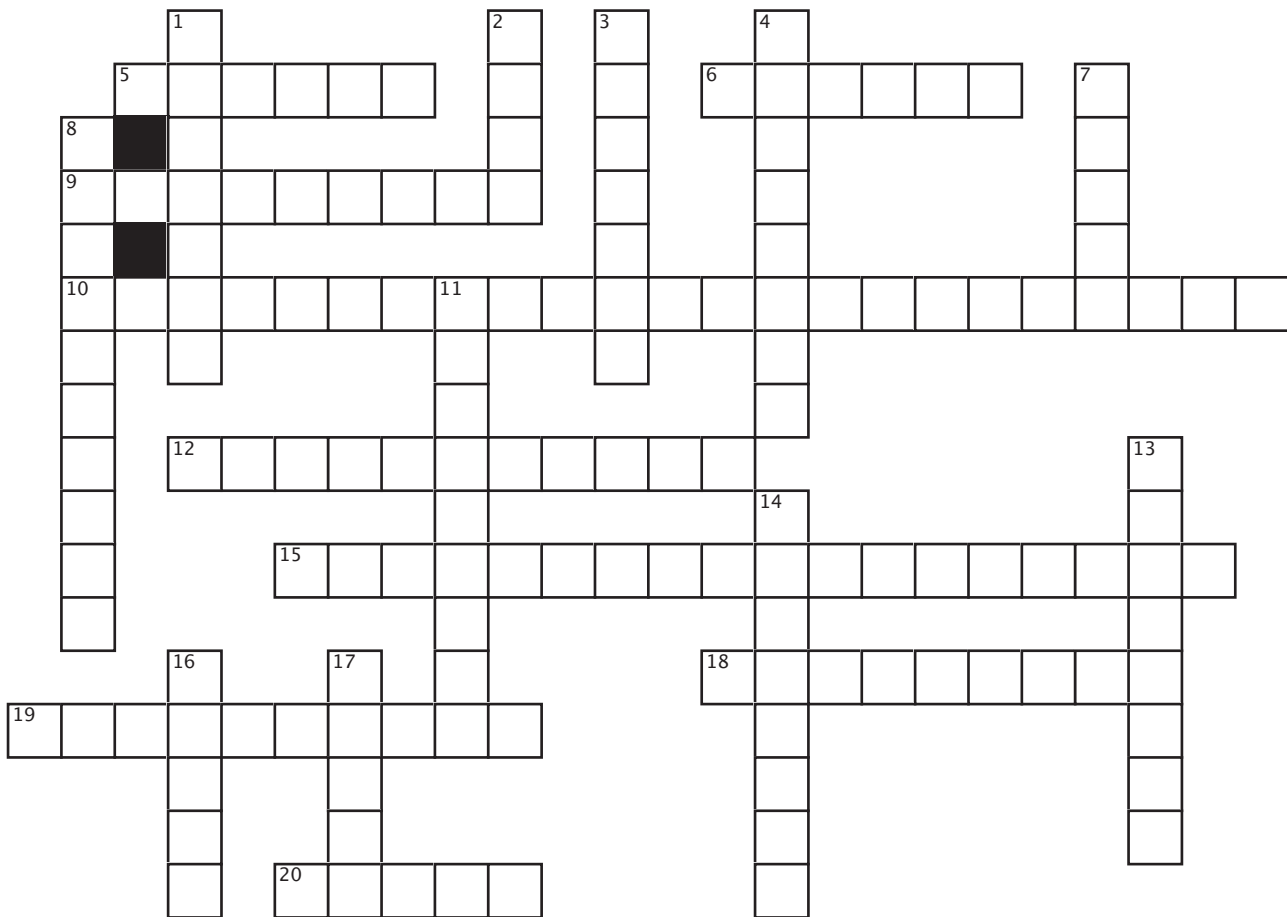
1. A large, wading bird with long legs and a flexible neck, found in tropical regions. It became extinct 50 000 years ago
3. A very large, horned, toothless, jawed reptile with a bony shell
6. Largest marsupial that ever lived
7. Name of the marsupial tiger
10. Fossil site in Queensland
13. Gumleaf-eating marsupial that evolved 25 million years ago
14. A 20-million-year-old, hopping marsupial with fangs
16. Carnivorous marsupial that became extinct during the last ice age
17. Site of the largest known snake, which lived 4 million years ago
18. Monotreme fossil found in Australia, Antarctica and South America
19. Height of the largest kangaroo

DOWN

2. A pony-sized, pouched animal with a trunk and narrow tongue
4. Ancient fish that could walk on mudflats
5. Fossil, sheep-sized monotreme that ate ants
6. The ancestor of this animal is the Asian grey wolf
8. Opalised fossils have been found at this place
9. NSW fossil site of mega fauna and humans
11. One of the largest, carnivorous, freshwater reptiles found, which has dagger-like teeth
12. Terrestrial, carnivorous lizard that grew to a length of 7 metres
15. Nickname for giant, flightless bird

WORD LIST

☆ Diprotodon	☆ marsupial lion	☆ big bird	☆ crocodile	☆ echidna
☆ Eusthenopteron	☆ Riversleigh	☆ dingo	☆ fangaroo	☆ goanna
☆ platypus	☆ thylacine	☆ horned turtle	☆ flamingo	☆ Bluff Downs
☆ three metres	☆ koala	☆ marsupial tapir	☆ Cuddie Springs	☆ Lightning Ridge



ACROSS

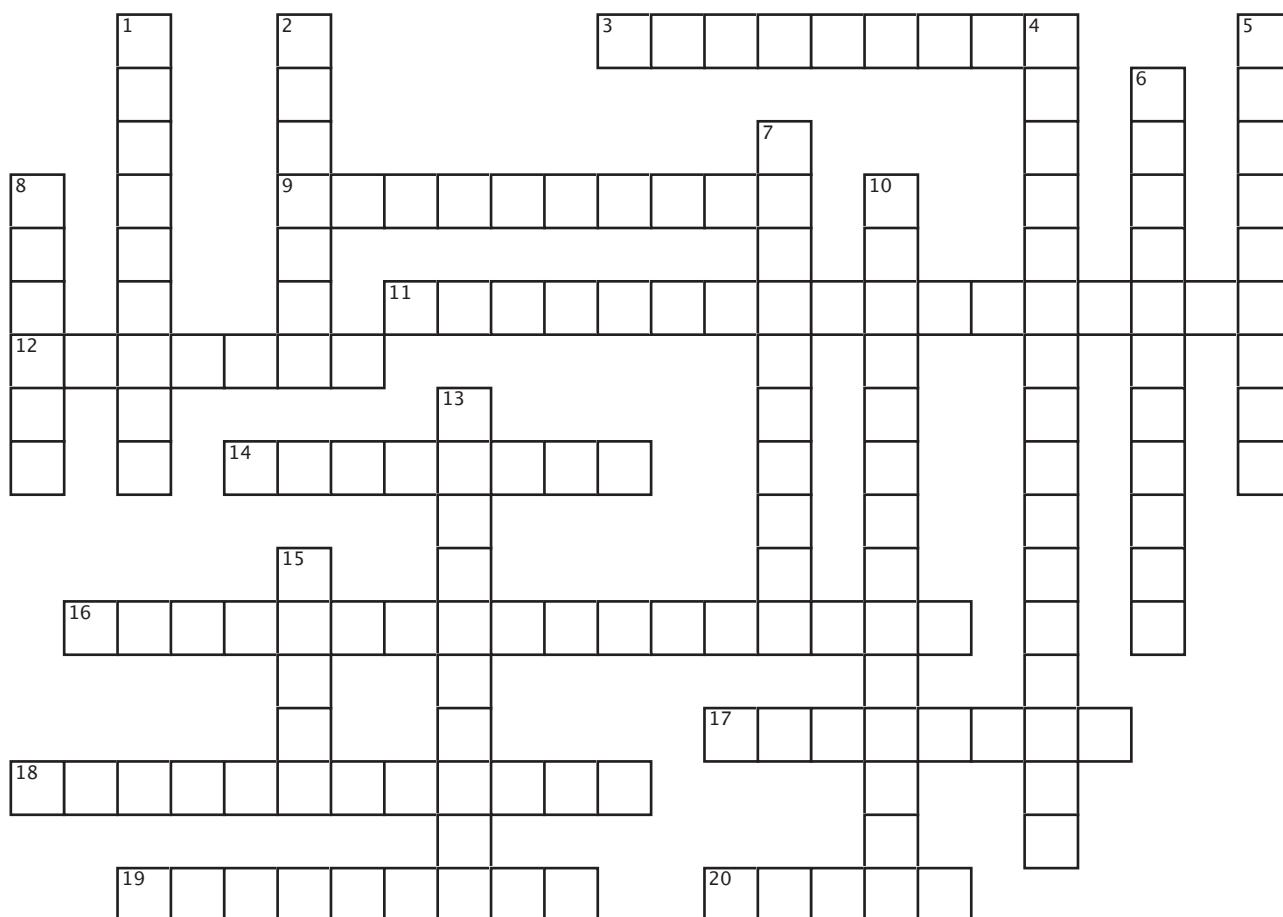
5. Relating to the Universe
6. A large, gas cloud in space
9. Height of energy carrier
10. Visible light is part of this
12. The fusion reactions by which stars convert hydrogen to helium
15. Diagram that shows the relationship between absolute magnitude, luminosity and surface temperature
18. The property of 'occurring at regular intervals'
19. Used in satellite communication
20. Carriers of energy

DOWN

1. Name of person who first measured the speed at which stars and galaxies are approaching us or receding from us
2. Astronomers do this
3. Current theory of the origin of the Universe
4. The change in the wavelength of light, in which the wavelength becomes longer
7. Travels at 300 000 000 km/s
8. Distance between nodes of two waves
11. Degree of brightness of a body in space
13. Name of the galaxy in which Earth resides
14. Form of radiation which is collected from deep space and used in physiotherapy
16. Pulsars emit these rays
17. Furthest penetrating rays

WORD LIST

☆ waves	☆ amplitude	☆ wavelength	☆ gamma	☆ gaze
☆ big bang	☆ light	☆ redshift	☆ microwaves	☆ nebula
☆ electromagnetic spectrum	☆ magnitude	☆ infrared	☆ cosmic	☆ frequency
	☆ Hertzsprung Russell	☆ Xrays	☆ Doppler	
		☆ Milky Way	☆ proton cycle	



ACROSS

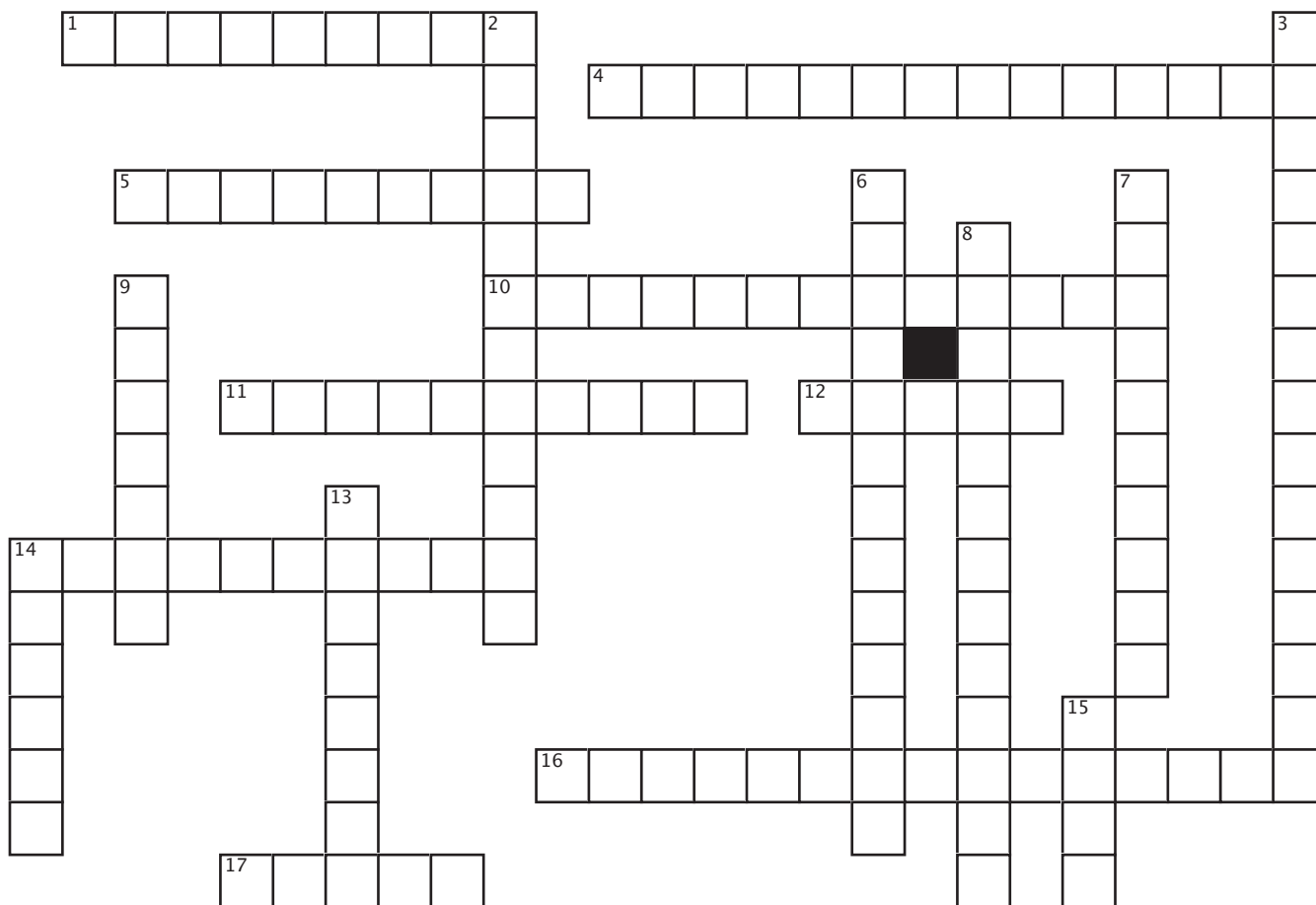
3. Type of boundary where two plates slide past each other
9. Process where a plate overrides another plate and is dragged towards the mantle
11. Magma that is extruded from the mid oceanic ridges causes this phenomenon
12. Name of a force that pulls things down
14. The Pan African, mountain-building event 500 million years ago formed this super continent
16. These geysers are found on the ocean floor along mid oceanic ridges
17. An underwater or other seismic event causes these
18. Type of plate that comprises the fractured surface of the Earth
19. The point on the Earth's surface directly above the focus of an earthquake
20. Earthquake and volcanic zones indicate this boundary

DOWN

1. A slab of crust that is forced down forms ____
2. A rising plume of magma from an oceanic plate that finds its way to the surface
4. Convergence of two continental plates causes this to occur
5. Type of boundary where the plates pull apart
6. Plates that form continents are known as ____
7. Earthquakes that occur within a continent, and away from the plate boundary, are known as this type of earthquake
8. Name of the super continent that formed Gondwana and Laurasia about 350 million years ago
10. This is formed by the rejoining of landmasses due to continental drift
13. The collision of two plates is known as this type of boundary
15. The point at which an earthquake occurs below the surface is known as _____

WORD LIST

☆ lithospheric	☆ transform	☆ plate	☆ Pannotia	☆ seafloor
☆ subduction	☆ continental	☆ hydrothermal	☆ Pangea	☆ spreading
☆ divergent	☆ gravity	☆ vents	☆ mountain	☆ focus
☆ convergent	☆ island arc	☆ tsunamis	☆ building	☆ epicentre
	☆ hot spot	☆ super continent	☆ intra plate	



ACROSS

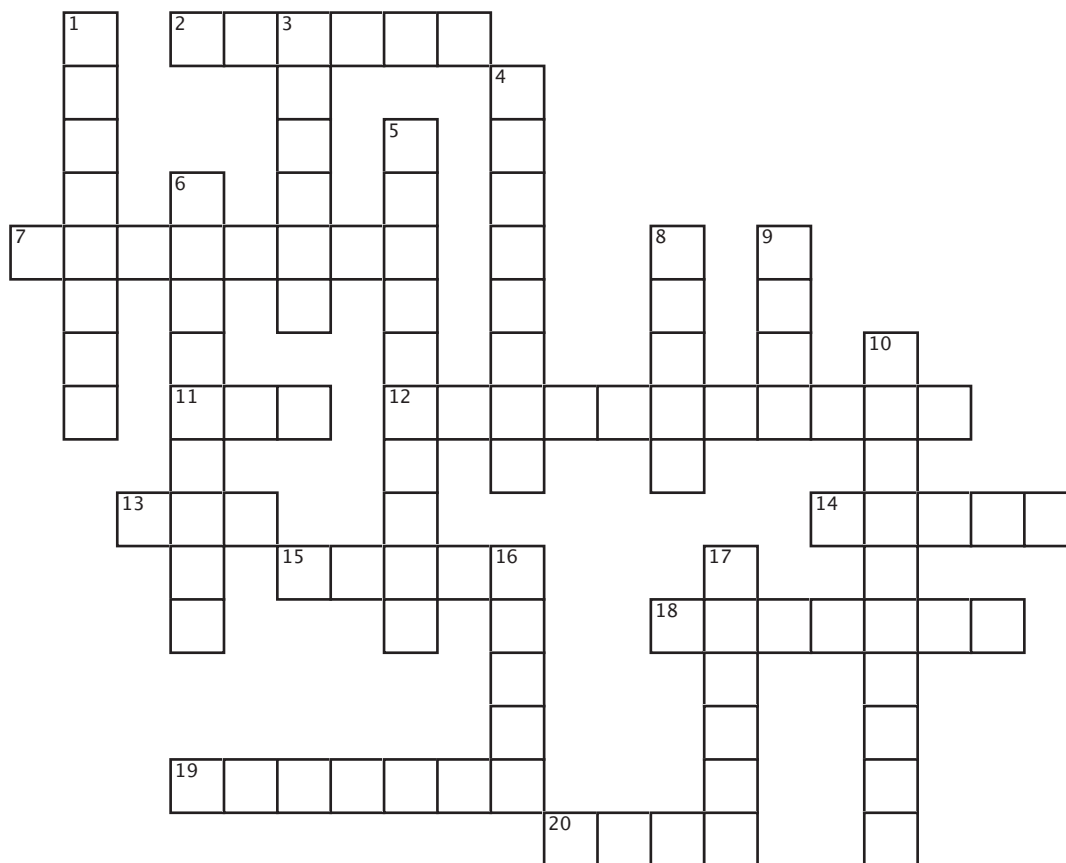
1. Eroded, volcanic plug which belongs to the largest central volcano that straddles the New South Wales-Queensland border
4. Tasmanian Jurassic dolerite mountain, gouged by glaciers
5. Volcanic mountain near Orange, NSW
10. These mountains were uplifted in the Triassic
11. Range in Western Australia made predominantly from iron ore
12. Large sand fan in central Australia that was tilted and uplifted 400 million years ago
14. These mountains are eroded cores of central volcanoes of the Cenozoic age and are found north of Brisbane
16. Central Australian, long, folded, mountain range formed approximately 350 million years ago
17. Describes the largest, dividing, uplifted range along the eastern edge of Australia

DOWN

2. The world's largest reef system of coral reefs, mangrove and estuarine environments
3. Tunnels through which lava flowed from Mt Surprise in North Queensland
6. A group of rounded, beehive-shaped domes of stratified sandstone and conglomerate deposited in Western Australia 375 to 350 million years ago
7. At 2228 metres, this is Australia's highest mountain
8. Remnants of former volcanoes, situated near Coonabarabran, which were active over 13 million years ago
9. Large, limestone caves near Sydney
13. Volcanic island which is part of a submerged volcanic ridge off the coast of NSW
14. One of the most significant and largest meteor impact sites known as a Bluff. It measures approximately 22 kilometres across, and is situated west of Alice Springs
15. A curved cliff face made from granite, measuring 110 metres long and situated 350 kilometres south-east of Perth

WORD LIST

☆ Great	☆ McDonnell Ranges	☆ Bungle Bungle	☆ Mt Warning	☆ Gosses
☆ Kosciuszko	☆ Glasshouse	☆ Blue Mountains	☆ Wave	☆ Lord Howe
☆ Warrumbungles	☆ Jenolan	☆ Canobolas	☆ Undara Lava Tubes	☆ Great Barrier
☆ Uluru		☆ Cradle Mountain		☆ Hammersley



ACROSS

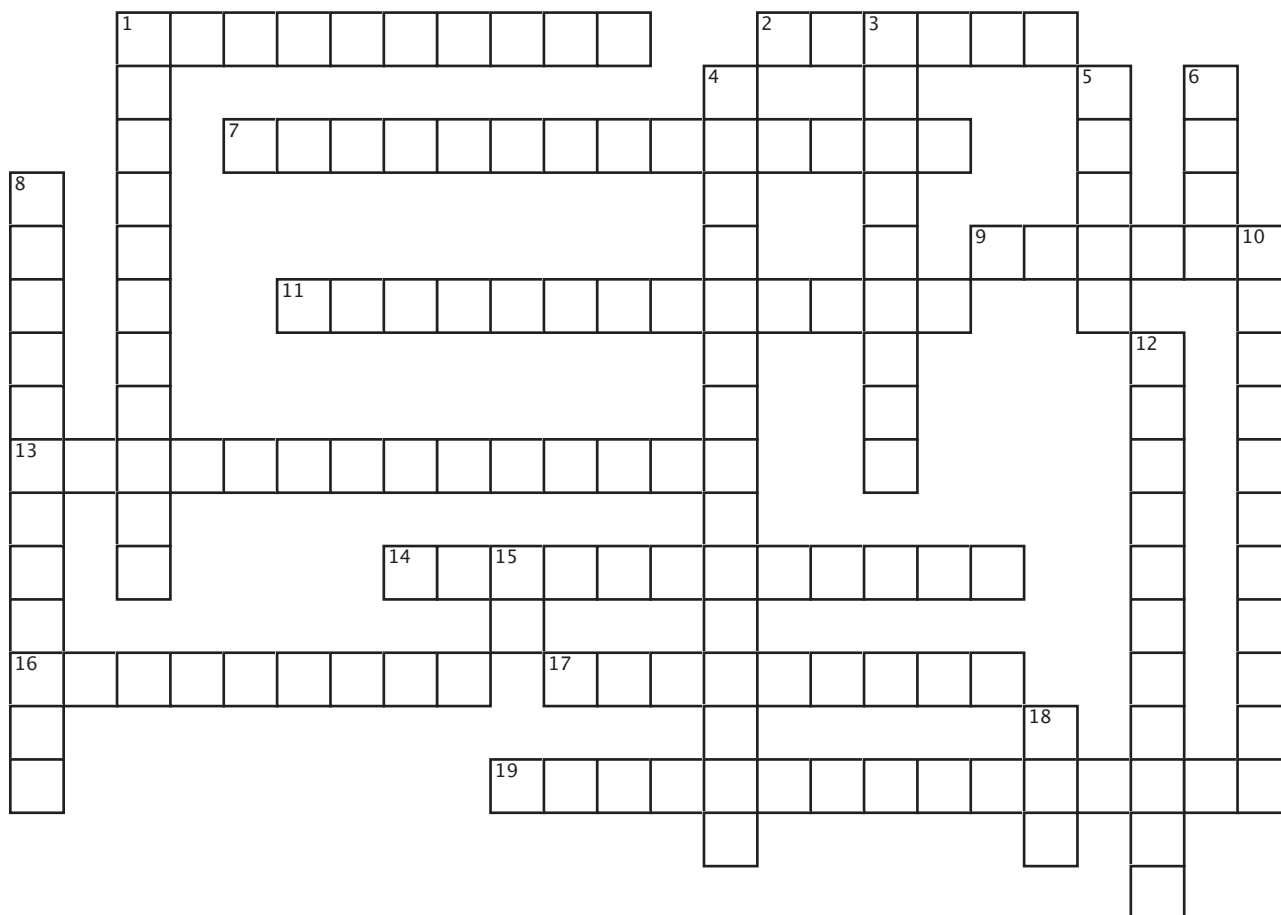
2. Nutrient-rich soils are made from these rocks after weathering
7. Building blocks of rocks
11. A rock body from which a valuable metal can be profitably extracted
12. Layer of the Earth made from upper mantle and crust
13. A finite resource
14. Person who extracts an ore
15. Paper is made from these
18. Mineral mined for nuclear fuel
19. Any natural or synthetic compound made from repeated linked units - silk is an example
20. Energy derived from the ocean

DOWN

1. A very precious metal
3. The colour on a tile after a mineral has been scratched on it
4. Made from heated limestone and clay
5. Derived from living matter of a previous geologic time and used for fuel
6. Person who explores for minerals
8. Made by fusing sand with boric oxide and aluminum oxide
9. An aquatic food resource
10. Not natural
16. Form of sustainable energy that can be converted into heat and electricity
17. A period of human culture between the Stone Age and the Iron Age, where implements were made from an alloy of copper

WORD LIST

☆ minerals	☆ trees	☆ bronze	☆ streak	☆ lithosphere
☆ ore	☆ oil	☆ platinum	☆ miner	☆ fish
☆ artificial	☆ polymer	☆ wave	☆ geologist	☆ glass
☆ uranium	☆ solar	☆ fossil fuel	☆ basalt	☆ concrete



ACROSS

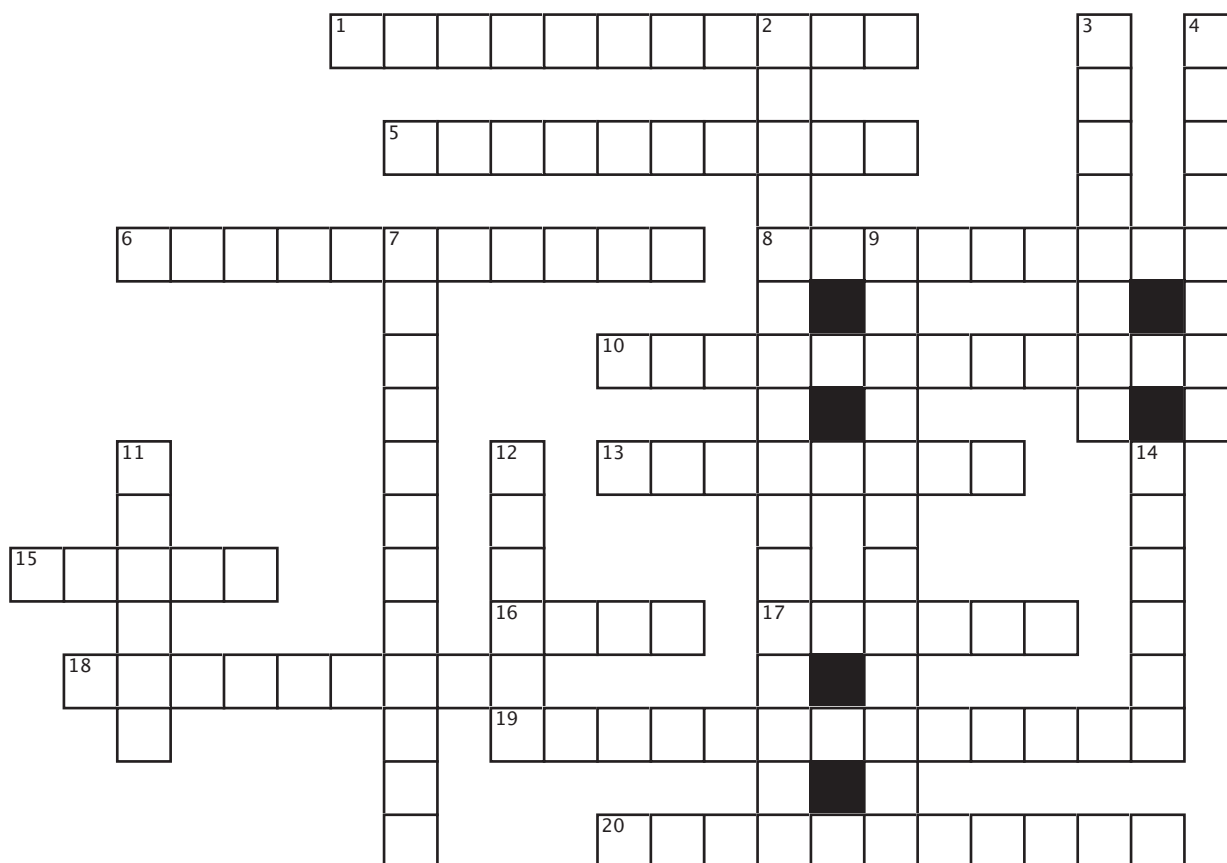
1. Cat and dog faeces are high in this chemical
2. Collective term for junk food containers, plastic bags, etc.
7. Name of bacterial type that indicates faeces in water
9. Person responsible for pollution prevention
11. Name of a smelly gas emitted from fossil fuels
13. Main gas responsible for brown haze in winter
14. Apparatus used to measure the size and concentration of particles in a liquid by analysis of scattered light
16. Grid used to stop rubbish from entering waterways
17. A method of reducing pollution
19. Name given to solids buoyed in water

DOWN

1. Opposite to diffuse source
3. Correct term used to describe muddiness of water
4. High levels of this gas in water indicate a healthy water supply
5. Type of pollution that inflicts damage to the inner ear
6. Unit used to measure dissolved gases, etc.
8. Extremely minute solid emissions
10. Excessive amounts of this causes damage to waterways
12. Two heavy metals that are emitted from exhausts and brake linings
15. Strength of acidity is measured in these units
18. Short for Volatile Organic Compounds

WORD LIST

☆ VOC	☆ carbon monoxide	☆ trash rack	☆ litter	☆ nephelometer
☆ particulates	☆ lead and zinc	☆ fertilisers	☆ phosphorus	☆ noise
☆ sulfur dioxide	☆ suspended solids	☆ faecal coliform	☆ mg/L	☆ recycling
☆ point source		☆ turbidity	☆ dissolved oxygen	☆ myself
		☆ pH		



ACROSS

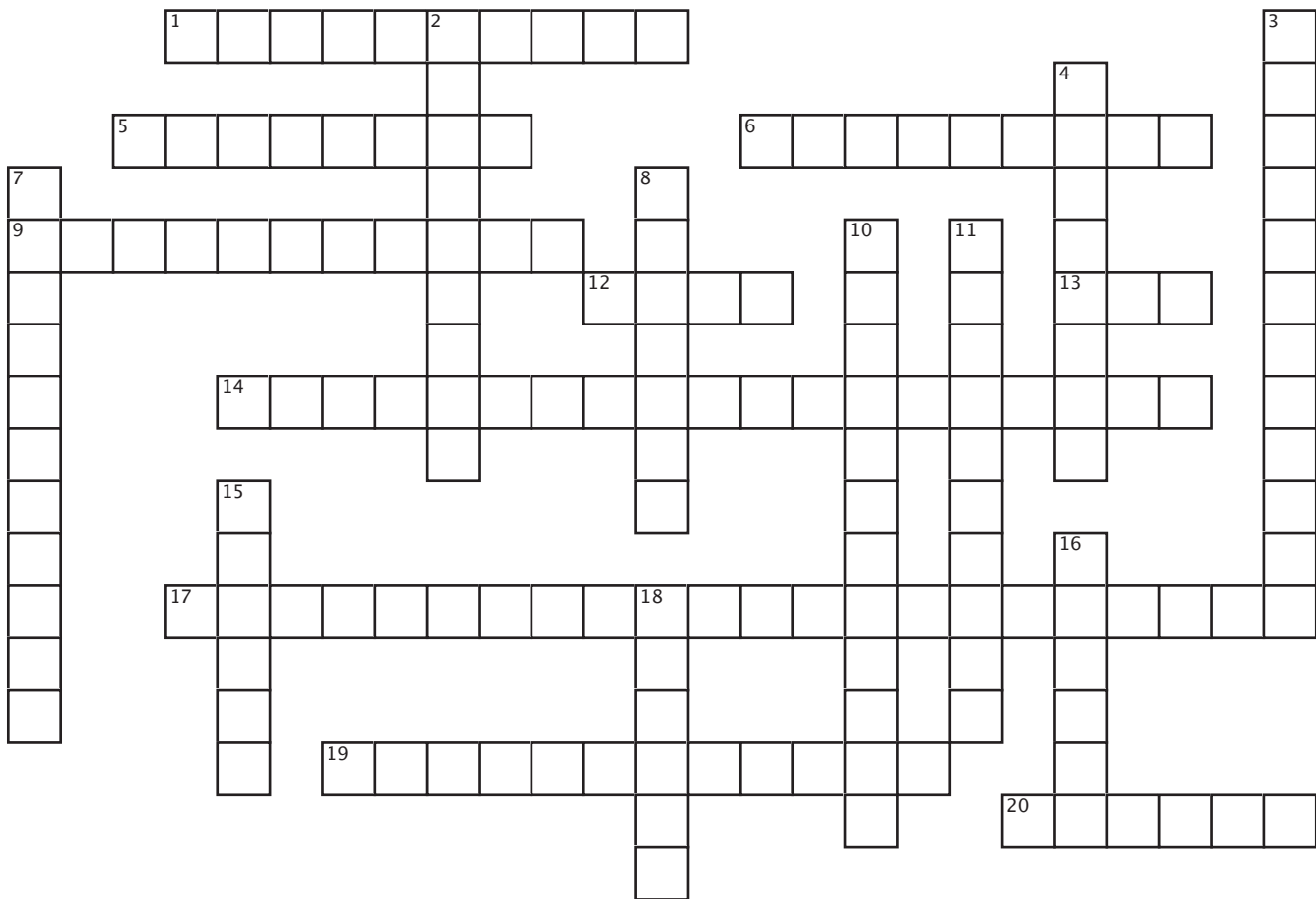
1. The visible surface of the Sun
5. The name of the spectrum in which atoms, under certain conditions, absorb light energy
6. Form of radiation in the blue end of the spectrum emitted by the Sun
8. Other elements within the Sun are debris from this type of explosion
10. Name of the lower atmosphere of the Sun
13. Dark regions associated with magnetic storms, can cause communication problems on Earth
15. Type of rays emitted from the corona
16. The Sun is one of these
17. The Sun's outer atmosphere
18. Name of the heavy hydrogen isotope
19. The spin of the Sun around its axis
20. Name of large loops or shoots from the chromosphere

DOWN

2. Name of light wavelengths given out by the Sun
3. Main element that makes up the Sun's composition
4. Form of electromagnetic radiation that causes sunburn
7. Main form of electromagnetic radiation emitted by the Sun
9. Name of three-stage reaction cycle that uses hydrogen nuclei
11. Name of violent eruptions of gaseous material
12. The fourth state of matter
14. Type of nuclear reaction that occurs in the Sun

WORD LIST

- | | | | | |
|----------------|---------------|-----------------|---------------------|------------------|
| ☆ star | ☆ sunspots | ☆ fusion | ☆ emission spectrum | ☆ axial rotation |
| ☆ photosphere | ☆ prominences | ☆ hydrogen | ☆ absorption | ☆ ultraviolet |
| ☆ chromosphere | ☆ plasma | ☆ proton proton | ☆ deuterium | ☆ visible light |
| ☆ corona | ☆ plagues | ☆ supernova | | ☆ infrared |
| | ☆ X rays | | | |



ACROSS

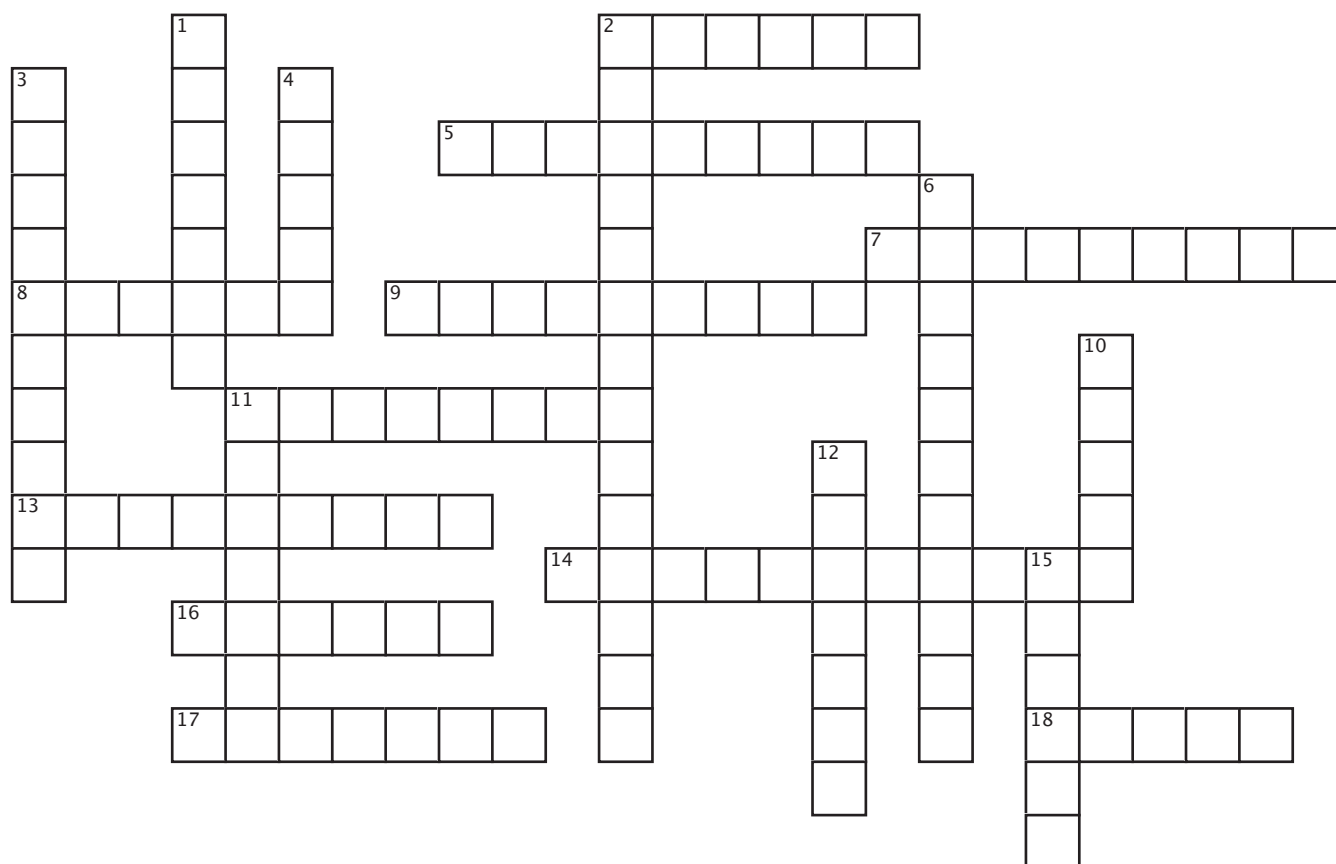
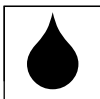
1. Form of agriculture
5. White soil cancer
6. Crop from which a fuel can be derived
9. Cultivation of plants in a solution
12. A soil mixture of sand, clay, silt and organic material
13. Female sheep
14. Crops engineered to be resistant to pests
17. A method of reproduction
19. Form of sustainable agriculture
20. Place where cattle are sold

DOWN

2. Important in soil fertility
3. Name given to planting different crops to improve soil fertility and reduce pests
4. Necessary for nitrogen fixing
7. Nutrient lacking in Australian soils
8. Poultry raised for meat
10. Name of a method where harvestable trees or shrubs are grown among pastures
11. Method of rice growing
15. Type of dairy cow
16. Crop grown for oils
18. Plant type used for nitrogen fixing

WORD LIST

- | | | | | |
|----------------|------------------------|---------------|----------------|---------------------------|
| ◆ jersey | ◆ genetically modified | ◆ legume | ◆ agroforestry | ◆ crop rotation |
| ◆ ewe | ◆ canola | ◆ bacteria | ◆ salinity | ◆ artificial insemination |
| ◆ biodynamic | ◆ sugarcane | ◆ loam | ◆ irrigation | |
| ◆ permaculture | ◆ market | ◆ nutrients | ◆ hydroponics | |
| | | ◆ phosphorous | ◆ broiler | |



ACROSS

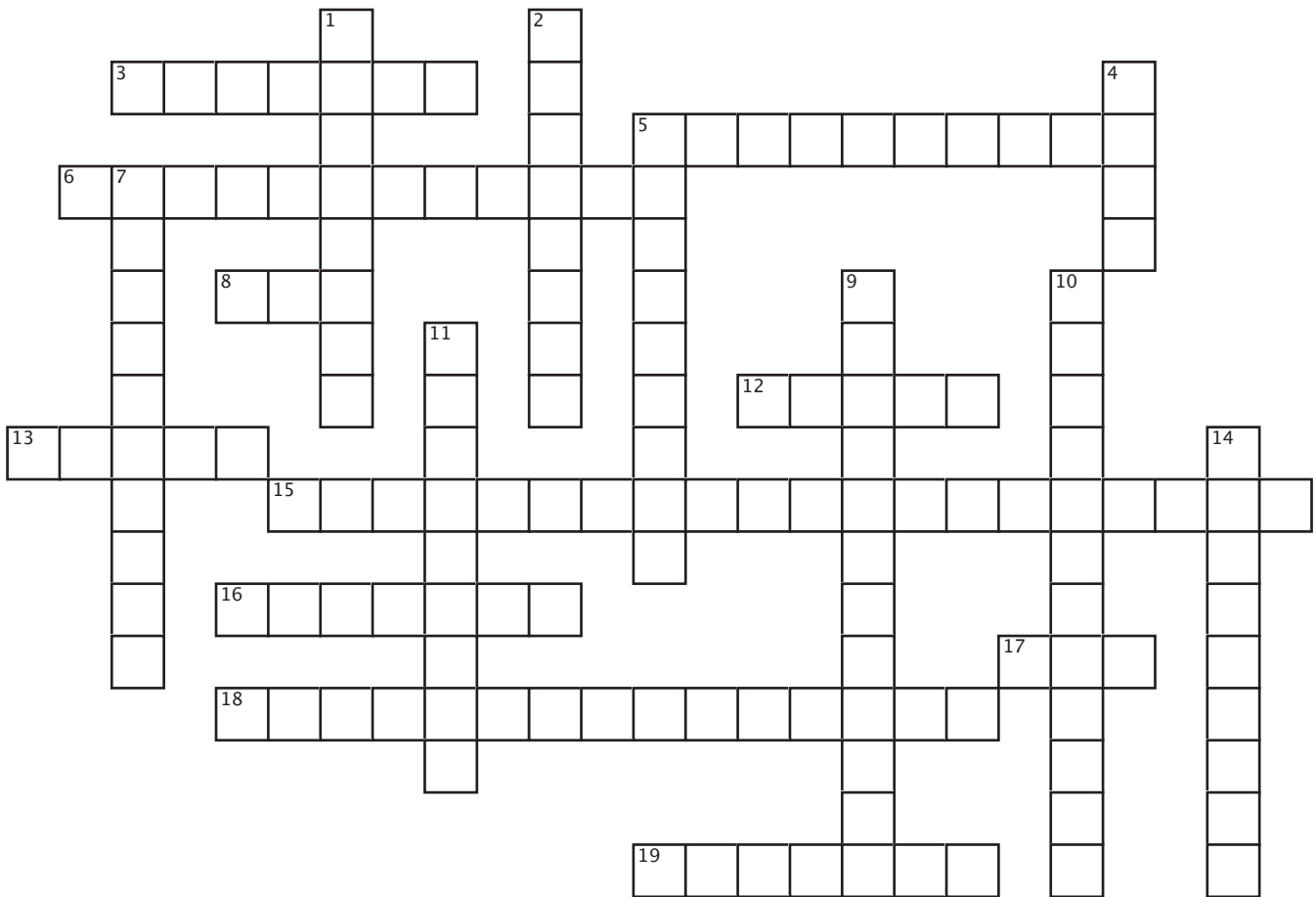
2. A primary division of a kingdom
5. System that regulates bodily functions
7. Movement of a substance from high to low concentration
8. Food-conducting tissue in plants
9. System for processing nutrients
11. Supportive structure in multicellular organisms
13. System used for removing waste products from organs
14. The process involving the production of energy, typically with the intake of oxygen and the release of carbon dioxide
16. A collection of similar cells acting together to perform one or more specific functions
17. The movement of water from areas of high concentration to low concentration through a semi-permeable membrane
18. A group of species that have similar characteristics

DOWN

1. The highest level of classification of organisms
2. The use of sunlight energy to combine water and carbon dioxide to form carbohydrate and oxygen
3. Nearest cousin to humans
4. Water-conducting tissue in plants
6. System to describe blood flow
10. A specialised part of an organism that performs a specific function
11. Organisms that are capable of interbreeding
12. System used for monitoring internal and external environment in multicellular organisms
15. A chemical substance excreted by plants

WORD LIST

- | | | | | |
|------------------|-----------|--------------|---------------|-------------|
| ◆ photosynthesis | ◆ species | ◆ phloem | ◆ skeleton | ◆ nervous |
| ◆ organ | ◆ kingdom | ◆ xylem | ◆ circulatory | ◆ endocrine |
| ◆ tissue | ◆ phylum | ◆ oxygen | ◆ digestive | ◆ diffusion |
| ◆ respiration | ◆ genus | ◆ chimpanzee | ◆ excretory | ◆ osmosis |



ACROSS

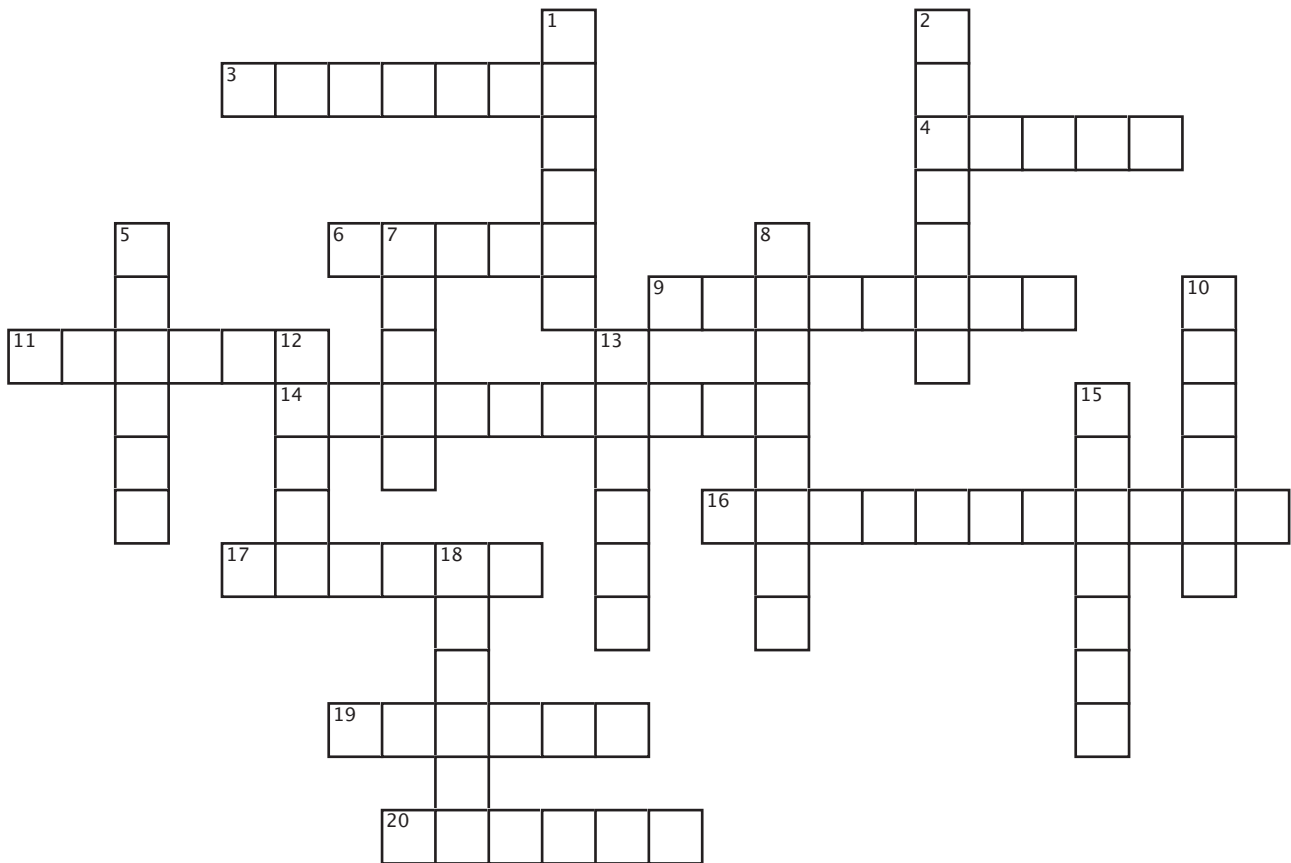
3. Part of the cell that contains the hereditary material and controls the cell's metabolism, growth and reproduction.
5. A threadlike structure found in the nucleus that carries the genes and transmits hereditary information
6. Filaments that extend inwards from the poles of a dividing cell
8. Found as a double helix
12. This apparatus is the protein packaging plant
13. Minute holes in the cell membrane
15. The cell's transport system
16. A cavity in a cell for storing water, food or metabolic waste
17. Abbreviation for ribo nucleic acid
18. The double-layered covering that encloses the nucleus
19. Process of body cell division

DOWN

1. Animal cells don't have this structure
2. The semipermeable covering that encloses the semi-translucent fluid of a cell
4. A hereditary unit in the DNA sequence which occupies a specific location on a chromosome
5. Cell walls contain this tough, fibrous material
7. An organism that does not have a nuclear membrane and whose DNA is not organised into chromosomes.
9. Animal cells do not have these structures
10. An organelle in the cytoplasm that contains enzymes for cell metabolism responsible for the conversion of food to usable energy
11. These manufacture complex proteins
14. A single-celled, or multicellular, organism whose cells contain a membrane-bound nucleus

WORD LIST

◆ cell wall	◆ nucleus	◆ DNA	◆ cellulose	◆ endoplasmic reticulum
◆ membrane	◆ chromosome	◆ gene	◆ pores	◆ golgi
◆ mitochondria	◆ eukaryote	◆ mitosis	◆ nuclear membrane	
◆ vacuole	◆ prokaryote	◆ spindle fibre	◆ RNA	
	◆ ribosomes	◆ chloroplasts		



ACROSS

3. Tube that carries ova from ovary to uterus
4. Paired reproductive organ that produces ova
6. Male gamete
9. Required chemicals in sexual development
11. A developing vertebrate usually from three months after conception, to birth
14. Tube that holds and stores sperm cells
16. Tube from testes to urethra
17. Male part of plant
19. Neck of uterus
20. Male plant equivalent of spermatozoa

DOWN

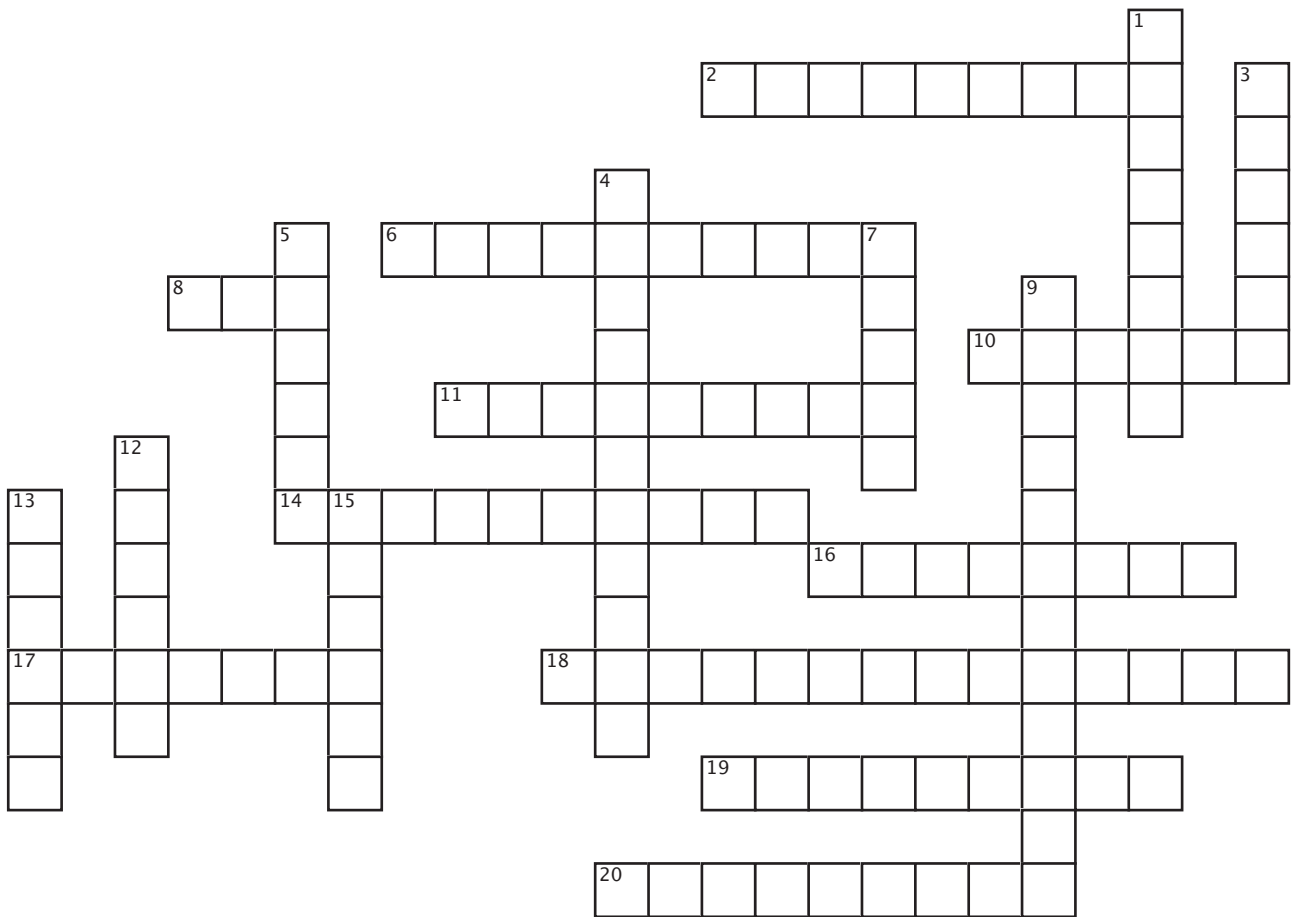
1. Place on the pistil where pollen is deposited
2. Asexual propagation
5. A hollow, muscular organ found in the female pelvic cavity
7. Male sex appendage
8. Name of gland in males that regulates bladder control and secretes a sugary fluid
10. Passage from vulva to cervix
12. Contains spermatozoa and other secretions
13. Fertilised ovum
15. External sac that contains the testes of mammals
18. Fertilised animal cell that grows, differentiates and lays down tissues and organs

WORD LIST

- | | | | | |
|----------------|-----------|--------------|------------|------------|
| ♣ sperm | ♣ semen | ♣ cloning | ♣ foetus | ♣ prostate |
| ♣ vas deferens | ♣ scrotum | ♣ epididymis | ♣ embryo | ♣ vagina |
| ♣ ovary | ♣ pollen | ♣ uterus | ♣ anther | ♣ penis |
| ♣ oviduct | ♣ stigma | ♣ zygote | ♣ hormones | ♣ cervix |



FURTHER REPRODUCTION



ACROSS

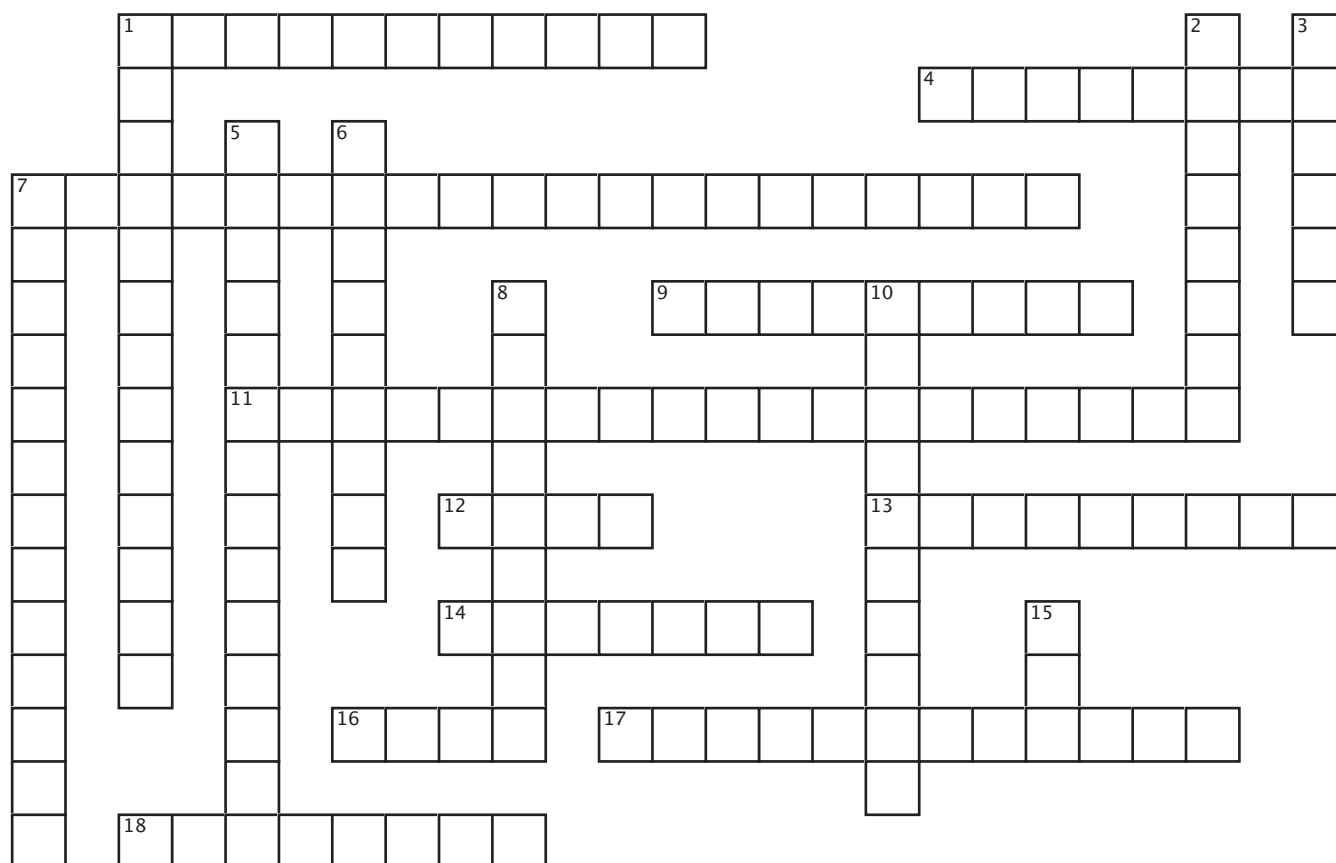
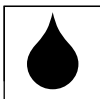
2. Cord that connects the embryo and foetus to the placenta
6. Males continually produce these
8. 200 000 of these are found in each ovary
10. Organs where sperm are produced
11. The permanent cessation of menstruation
14. The expelled placenta is termed ____
16. Arrival of first period
17. Another word for Fallopian tube
18. First stage of labour
19. Release of an ovum from the ovary is called ____
20. Hormone responsible for female, secondary, sexual characteristics

DOWN

1. A vascular organ that develops in female mammals during pregnancy
3. Muscular organ in which the offspring develop
4. The embryo is protected by this
5. Place where sperm are deposited in the female
7. Fluid that contains sperm
9. Shedding of uterine lining and cause of bleeding
12. This expands fully to allow the baby to be delivered
13. Name given to a fertilised ovum
15. Term given to developing egg after two months

WORD LIST

- | | | | | |
|-------------|----------------|----------------|-------------------|---------------|
| ◆ ovulation | ◆ menstruation | ◆ zygote | ◆ umbilical | ◆ afterbirth |
| ◆ ova | ◆ oviduct | ◆ semen | ◆ waters breaking | ◆ cervix |
| ◆ oestrogen | ◆ testes | ◆ foetus | ◆ uterus | ◆ sperm cells |
| ◆ menarche | ◆ vagina | ◆ amniotic sac | ◆ placenta | ◆ menopause |



ACROSS

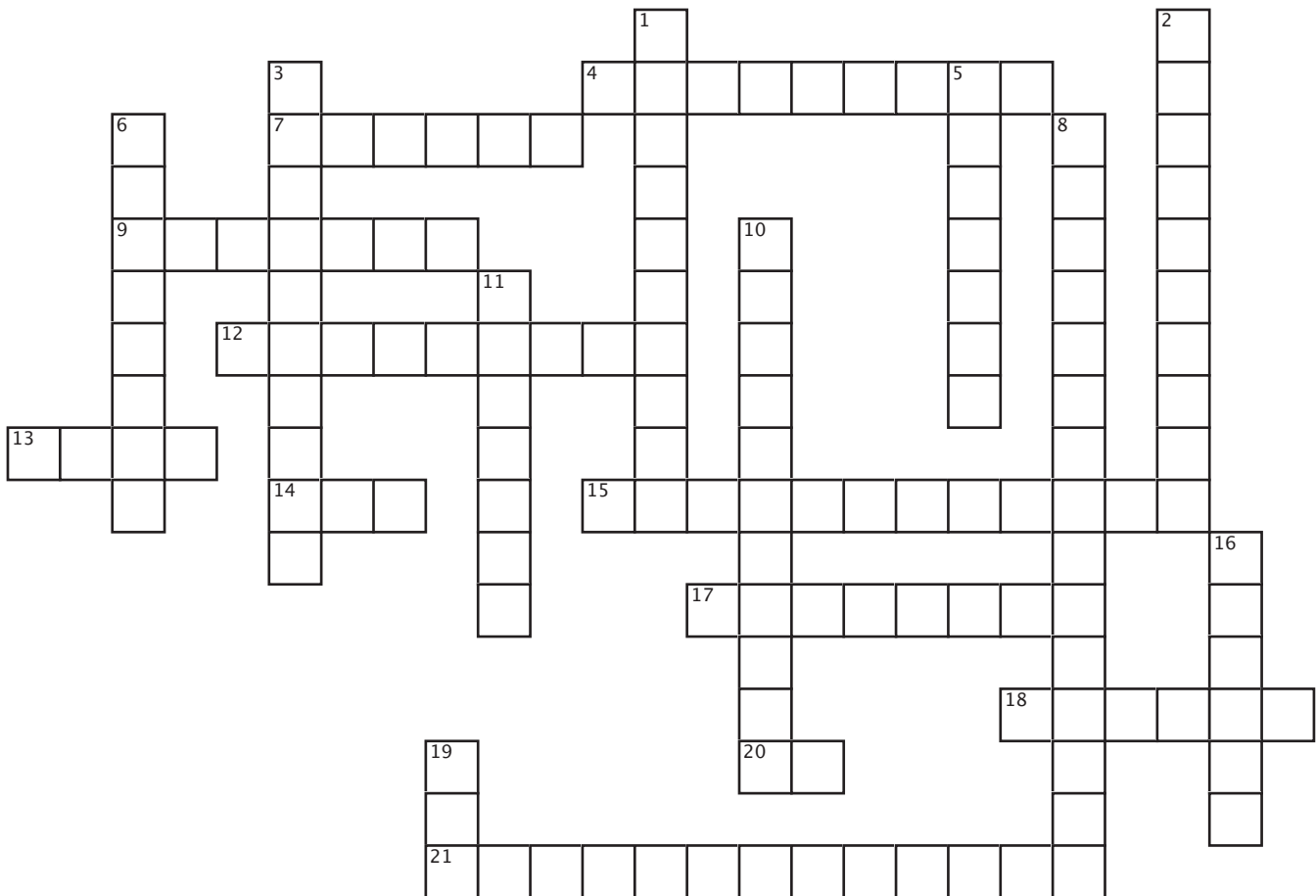
1. Humans have 46 of these
4. Antarctica and Australia were part of this ancient continent
7. The process of natural selection of organisms that are able to adapt to the environment and survive to reproduce
9. Islands of South America, significant to Darwin
11. Term for selective breeding
12. Biological genus classification of humans
13. A successive, genetic change in organisms as a result of natural selection over time, resulting in a new species
14. Darwin studied these birds
16. Hereditary unit
17. Word to describe rock layers
18. Terrible lizard

DOWN

1. Author of 'The Origin of the Species'
2. Geologic period 540 to 500 million years ago
3. Super continent
5. Part of the fossilisation process
6. Meiosis produces this
7. Law that puts the youngest rock layers at the top and the oldest at the bottom
8. Extinct marine arthropod
10. Ancient life era
15. Short for deoxyribonucleic acid

WORD LIST

- | | | | | |
|------------------|---------------------------|---------------|----------------|------------------------|
| ◆ mineralisation | ◆ survival of the fittest | ◆ variation | ◆ trilobite | ◆ finches |
| ◆ Galapagos | ◆ gene | ◆ chromosomes | ◆ Gondwana | ◆ artificial selection |
| ◆ evolution | ◆ DNA | ◆ Cambrian | ◆ Palaeozoic | |
| ◆ Charles Darwin | ◆ superposition | ◆ dinosaur | ◆ Homo | |
| | | ◆ Pangea | ◆ stratigraphy | |



ACROSS

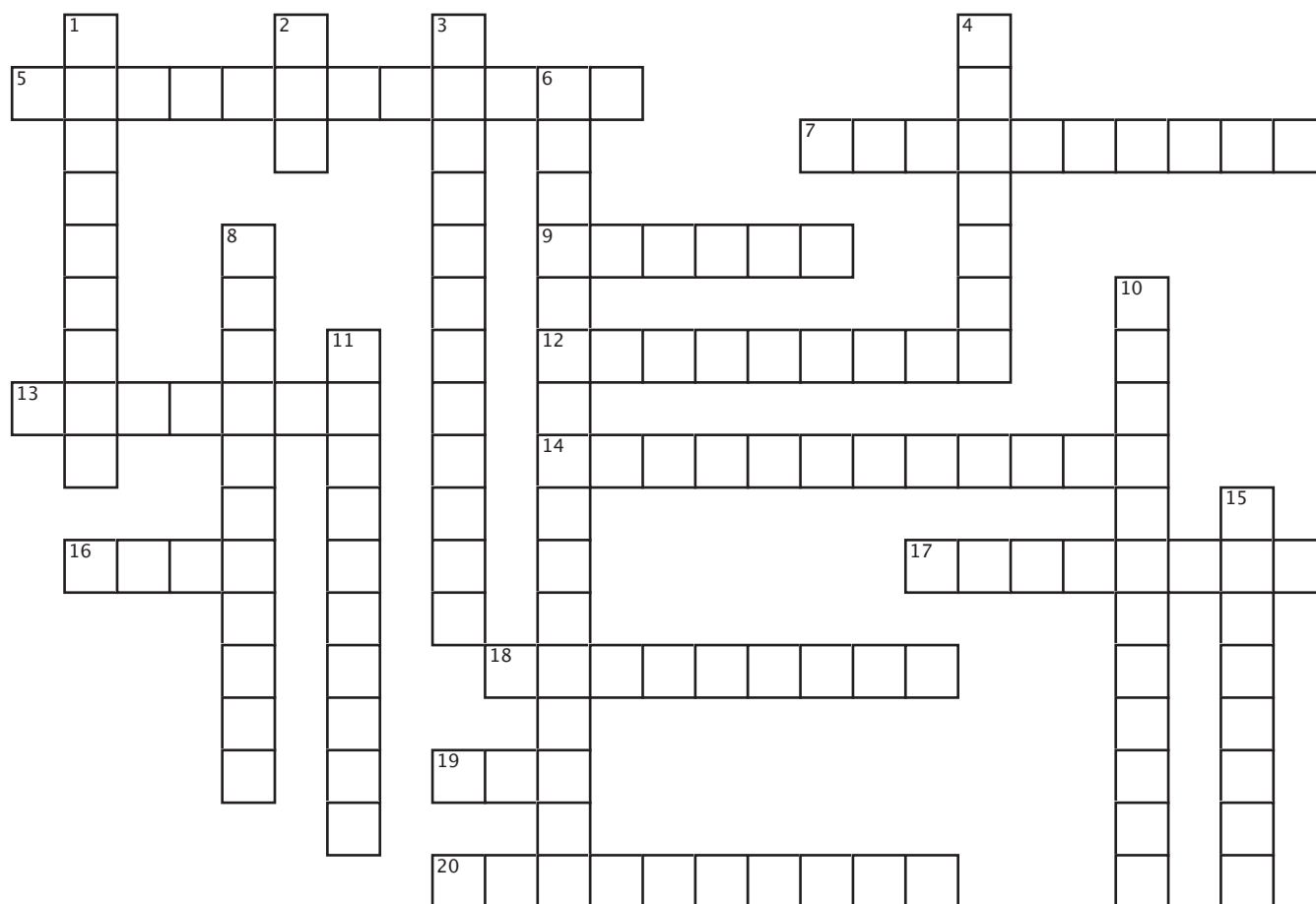
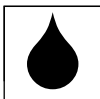
4. The observable physical or biochemical features of an organism
7. The offspring of genetically dissimilar parents
9. Cellular process that allows variation to occur
12. In DNA the compounds adenine-thymine and guanine-cytosine are known as ____
13. Hereditary unit
14. Found in the nucleus of eukaryotic cells
15. Having different genes at one or more specific points on a chromosome
17. The genetic make-up of an organism
18. Father of genetics
20. Chromosome pair responsible for maleness
21. Deletion or addition of chromosomes produce ____

DOWN

1. Component in a nucleus that contains genetic information
2. The same genes on the same chromosomal points
3. These are separated during cell division to make new chromosomes
5. Person who created these squares for determining offspring genotypes
6. Gene that masks a recessive gene
8. Birth of a new individual from an unfertilised egg; common in arthropods
10. Structure of DNA
11. Person who passes on a trait but is usually not affected by it
16. An alternate form of a gene
19. Similar to, but not, DNA

WORD LIST

- | | | | | |
|-------------|----------------|--------------|-------------------|-----------|
| ◆ Mendel | ◆ DNA | ◆ meiosis | ◆ double helix | ◆ carrier |
| ◆ phenotype | ◆ heterozygous | ◆ base pairs | ◆ abnormalities | ◆ allele |
| ◆ genotype | ◆ homozygous | ◆ RNA | ◆ parthenogenesis | |
| ◆ gene | ◆ hybrid | ◆ chromosome | ◆ chromatids | |
| | ◆ punnett | ◆ XY | ◆ dominant | |



ACROSS

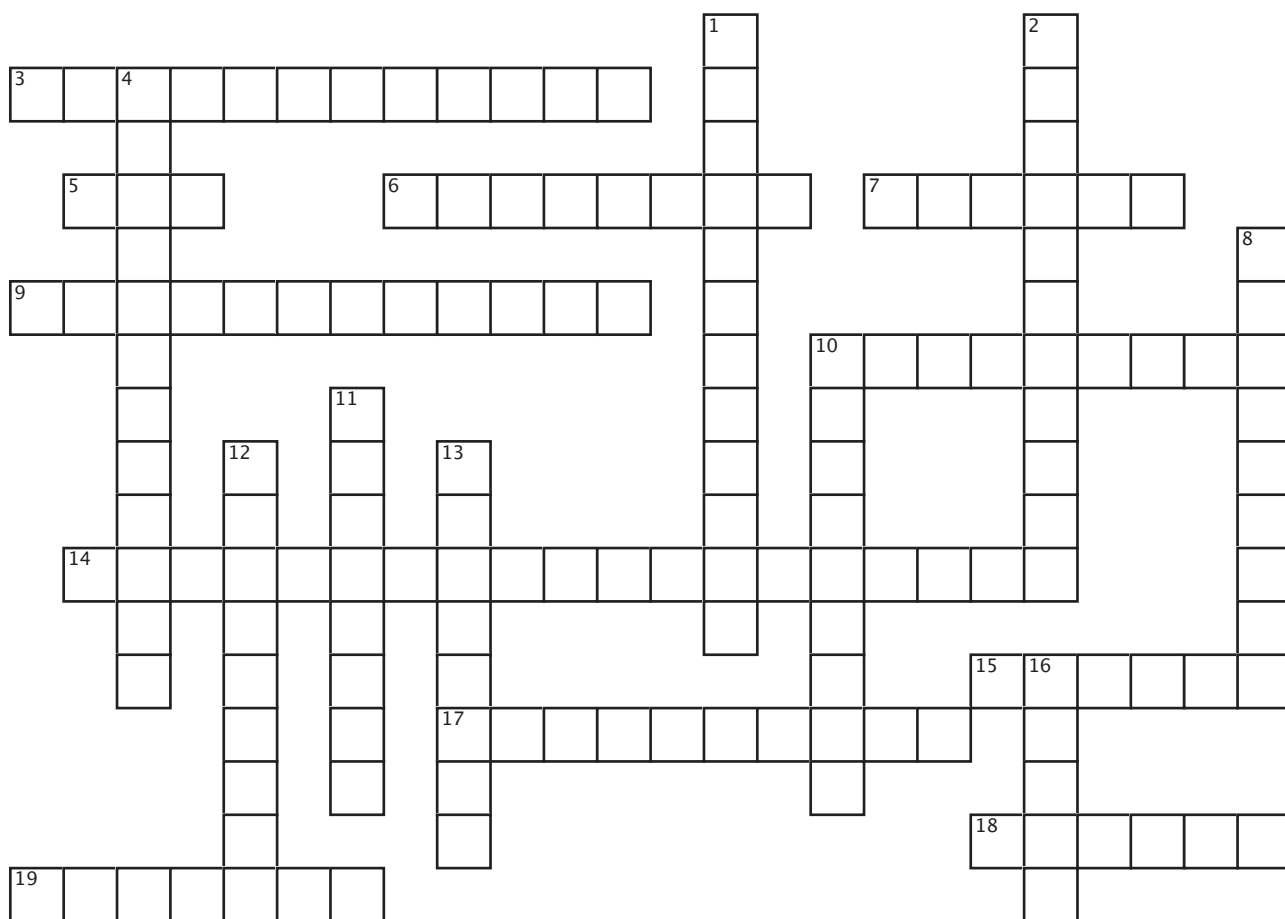
5. Hormone responsible for the development and maintenance of male secondary sex characteristics
7. Extension of the hind brain
9. An involuntary response to a stimulus
12. Hormone that increases the rate of cell metabolism and regulates growth
13. Word for nerve cell
14. Hormone that controls water loss in a body
16. Long, swollen portion of a nerve cell
17. Chemicals that target specific cells
18. Known as the master gland
19. A chemical component needed to make hormones
20. Fight or flight hormone

DOWN

1. Hormone necessary for ovulation to occur
2. Short for follicle-stimulating hormone
3. Hormone needed to prepare the uterus for pregnancy and to promote development of the mammary glands
4. Gap between dendrite and body of nerve cell
6. A chemical substance that transmits nerve impulses across a synapse
8. Automatic nerve function
10. Affecting the mind or mood or other mental processes
11. Part of the brain responsible for the regulation and coordination of complex, voluntary, muscular movement, and the maintenance of posture and balance
15. Part of the brain that controls and integrates motor, sensory, and higher mental functions

WORD LIST

- | | | | | |
|---------------|----------------|---------------|--------------------|----------------|
| • hormones | • pituitary | • neurone | • oestrogen | • testosterone |
| • cerebellum | • adrenaline | • axon | • neurotransmitter | • fat |
| • cerebrum | • antidiuretic | • synapse | • thyroxine | • progesterone |
| • spinal cord | • reflex | • involuntary | • FSH | • psychoactive |



ACROSS

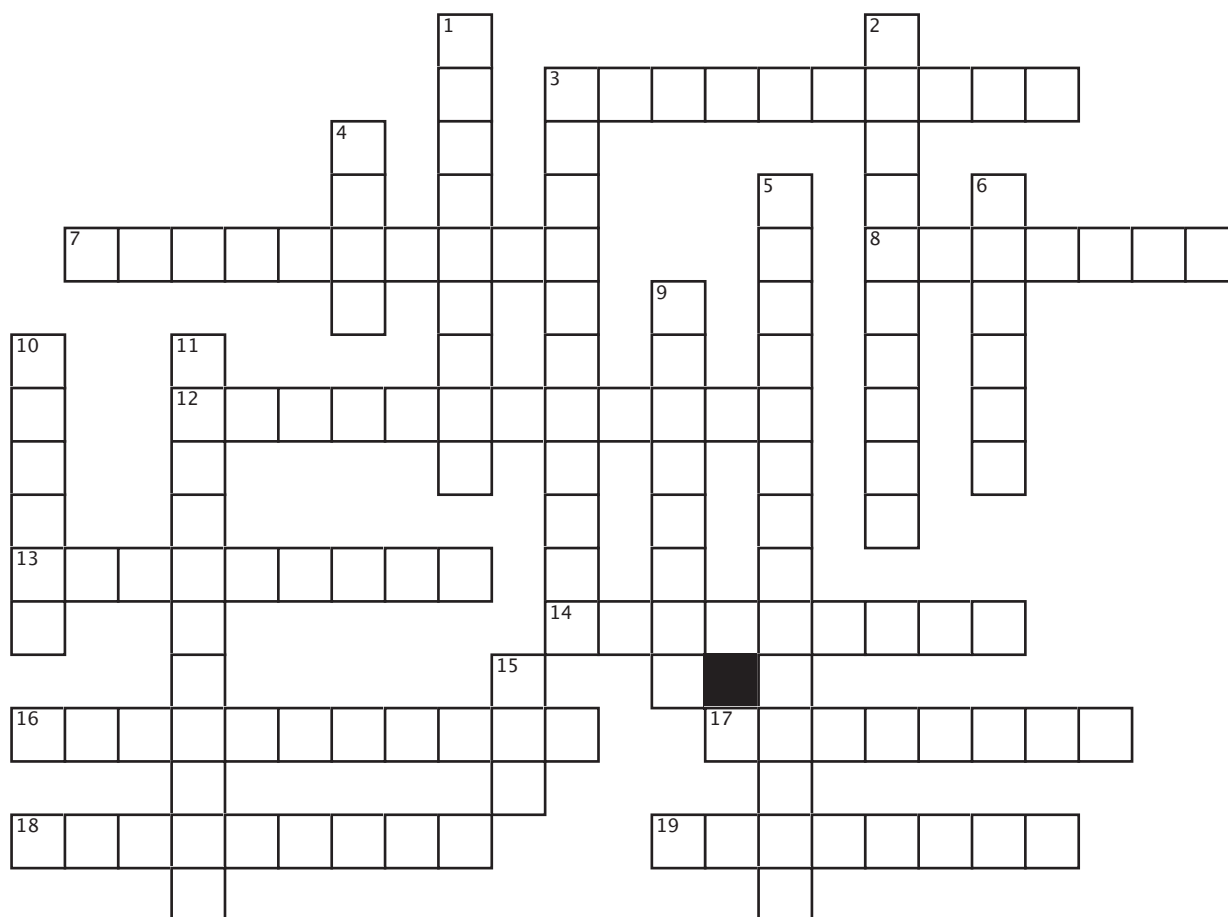
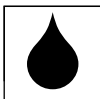
3. Hormone that acts on the kidney to preserve fluid and electrolytic balance
5. Abbreviation for melanocyte-stimulating hormone, which is responsible for skin tone
6. Hormone involved in immune suppression and stress response
7. Brain gland that produces melatonin
9. Hormone necessary for the maintenance of pregnancy
10. Best known as the master gland, found at the base of the brain
14. Hormone that stimulates ovarian follicles and sperm production
15. Element essential in thyroid hormones
17. Known as the fight or flight hormone
18. Hormone that stimulates growth
19. Gland that produces the hormone adrenaline

DOWN

1. Gland in brain that produces antidiuretic hormone
2. Hormone produced by the pituitary gland that stimulates production of oestrogen and testosterone
4. Hormone responsible for secondary sexual characteristics and sperm production
8. Hormone needed to develop brain, reproductive tract and to regulate metabolism
10. Hormone necessary for lactation to occur
11. Organ that produces insulin
12. Hormone that regulates the biological clock
13. System needed by body to regulate hormones
16. Organ that produces oestrogen

WORD LIST

- | | | | | |
|---------------|----------------|----------------|--------------|------------|
| ◆ luteinising | ◆ progesterone | ◆ melatonin | ◆ pituitary | ◆ pineal |
| ◆ follicle | ◆ testosterone | ◆ antidiuretic | ◆ feedback | ◆ cortisol |
| ◆ stimulating | ◆ ovary | ◆ thyroxine | ◆ iodine | |
| ◆ pancreas | ◆ growth | ◆ adrenal | ◆ adrenaline | |
| | ◆ prolactin | ◆ hypothalamus | ◆ msh | |

**ACROSS**

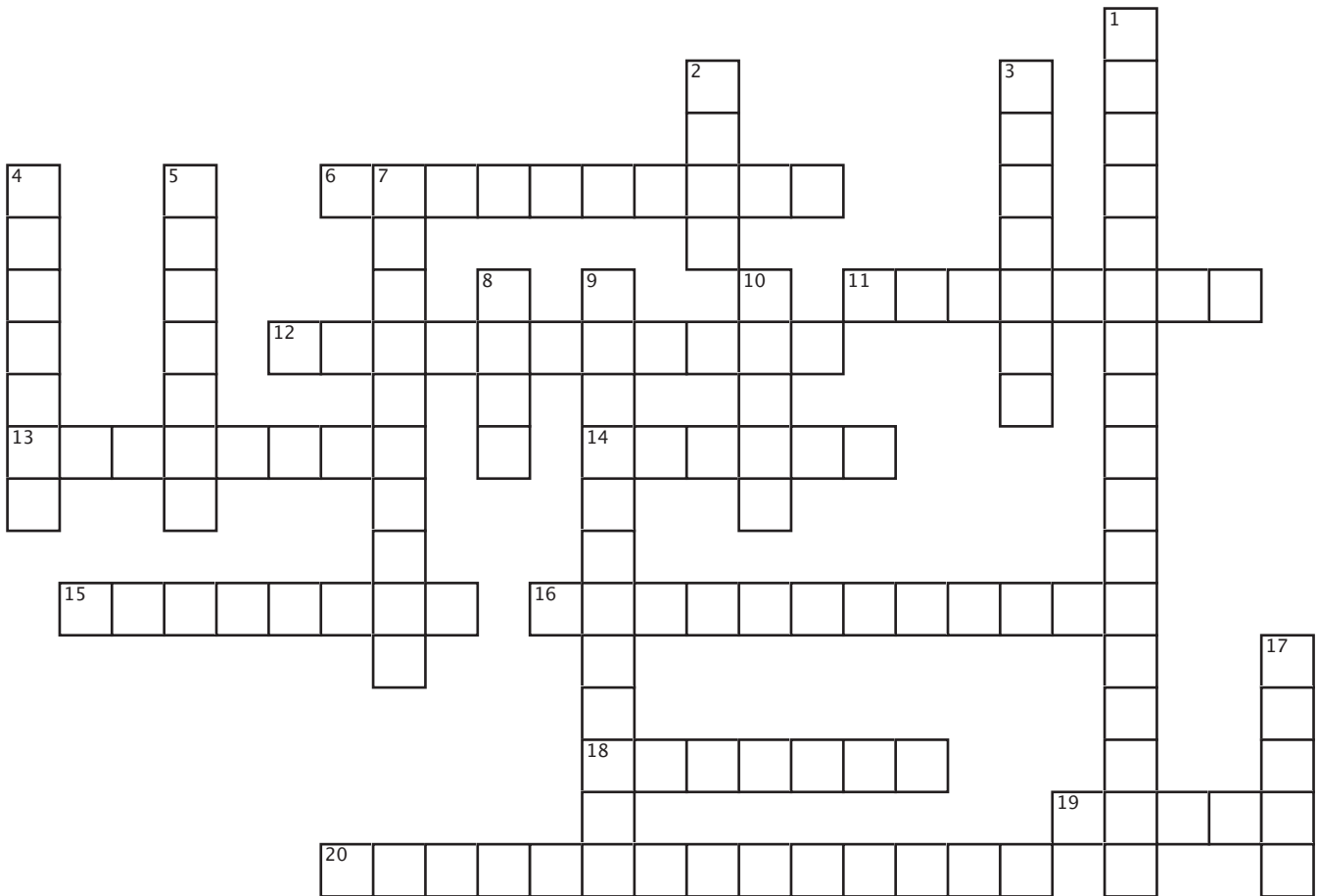
3. Lowering skin temperature and maintaining normal internal heat is carried out by these organisms
7. Fur and feathers are examples of this type of heat-control mechanism
8. The insulating layer of fat in large marine animals
12. Increased activity levels in humans generate more ____
13. Large distance movement of moths, butterflies, some birds and animals in order to find food and avoid temperature extremes
14. Term for night time activity
16. Term used for hibernating when conditions are hot
17. Birds and reptiles excrete this to conserve water efficiently
18. Means 'cold blooded'
19. Marsupial that is more efficient in energy usage when bounding compared to any four-legged animal

DOWN

1. A behaviour that reduces water loss in animals
2. High food intake increases the level of ____
3. Impervious covering of arthropods and molluscs, used to conserve water
4. By changing their posture, lizards are able to control which abiotic factor?
5. The system of having arteries and veins close together is known as this type of heat exchange
6. Many warm-blooded animals do this to control heat loss in cold conditions
9. System used by animals to maintain a constant internal temperature
10. Heat loss and conservation is dependent on the ratio of surface area to ____
11. The slowing of metabolism and lowering of body temperature in animals is called ____
15. Kangaroos and wallabies reduce water loss by sweating only when they ____

WORD LIST

- | | | | | |
|-------------|---------------|--------------|--------------|-------------------|
| ◆ kangaroo | ◆ exoskeleton | ◆ nocturnal | ◆ ectotherm | ◆ counter current |
| ◆ hop | ◆ hibernation | ◆ huddle | ◆ insulators | ◆ metabolism |
| ◆ burrowing | ◆ migration | ◆ heat | ◆ blubber | ◆ feedback |
| ◆ uric acid | ◆ aestivation | ◆ endotherms | ◆ volume | ◆ internal heat |



ACROSS

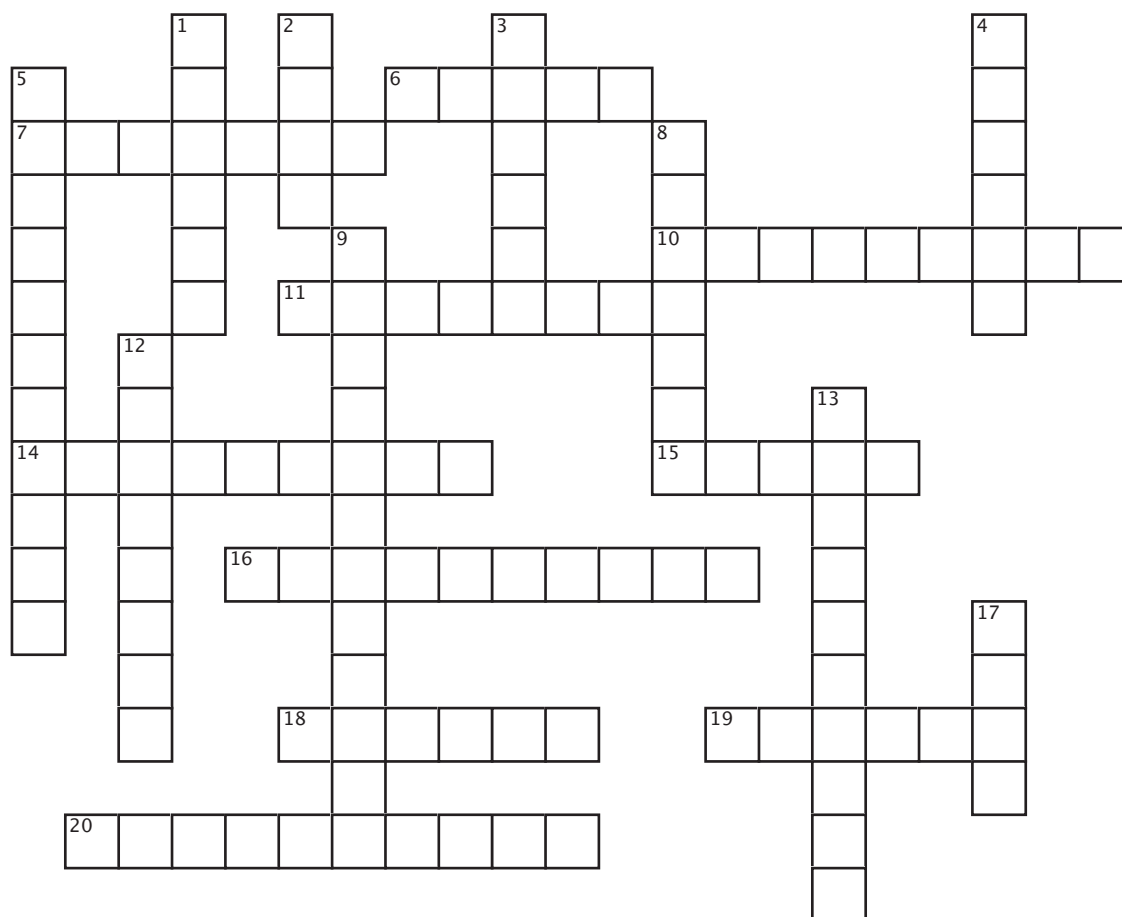
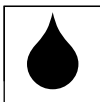
6. Characterised by high humidity, closed canopy, warm to hot temperatures and high biodiversity
11. The most complex and productive system that includes saltwater marshes, mangroves, sea grasses and breeding grounds for fish and birds
12. Major herbivore found in arid ecosystems
13. This zone relates to the banks of a natural course of water
14. A delicately balanced, small and isolated ecosystem
15. A system subjected to seasonal changes in water depth, flow, turbidity and width
16. An ecosystem with few trophic levels, heath, bogs and harsh weather
18. A system where small pockets of plants and animals survive
19. Tree-dwelling marsupial, found in dry sclerophyll forest
20. Characterised by many trees, nutrient-poor soils and a high biodiversity

DOWN

1. Maintains biodiversity and sustainability of riverine and wetland areas
2. Exists in shallow seas with high biodiversity that recycles most of the nutrients
3. System that relates to oceans and seas
4. A transitional ecosystem from sea to land which is influenced by tidal flows and rivers
5. Pertaining to zones of great depths in oceans or lakes, where light does not penetrate
7. This system is sparsely populated, has poor, ancient soils and six vegetation types
8. A system formed underground by water, and characterised by limestone structures
9. This system has no diversity, high fertiliser input, leached soils and often toxic outputs
10. A system that is extremely cold with no rainfall, and specialised organisms
17. This system has no biodiversity, no natural recycling and extinction of many species

WORD LIST

- | | | | | |
|----------------|-------------------|----------------|-----------------|-----------------|
| ◆ urban | ◆ mangrove | ◆ remnant | ◆ environmental | ◆ alpine tundra |
| ◆ agricultural | ◆ dry sclerophyll | ◆ red kangaroo | ◆ flow | ◆ cave |
| ◆ rainforest | ◆ arid inland | ◆ riparian | ◆ pelagic | |
| ◆ riverine | ◆ island | ◆ koala | ◆ abyssal | |
| | ◆ reef | ◆ estuary | ◆ polar | |



ACROSS

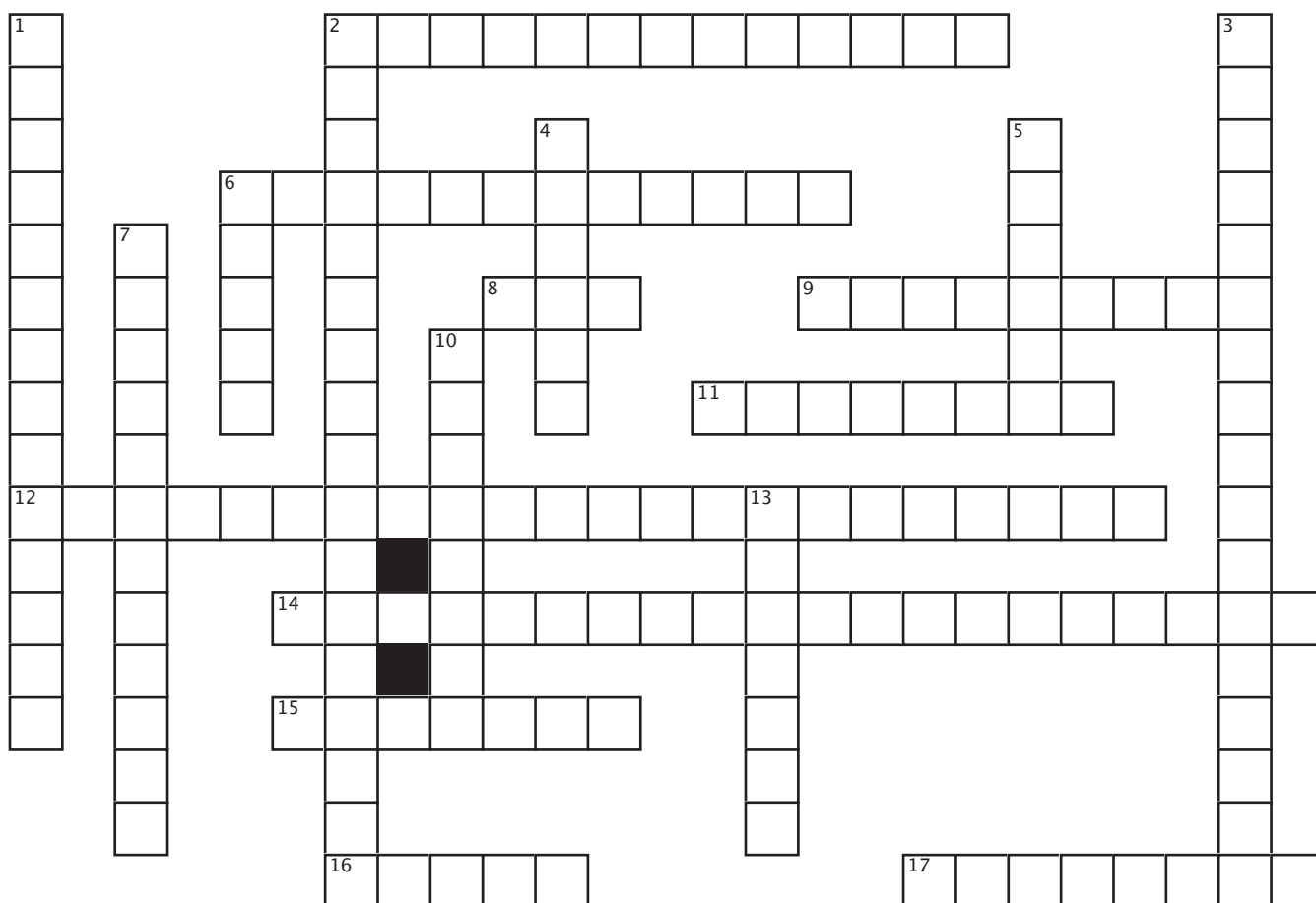
6. A source of electrical energy derived from the sun
7. Word for non-living
10. A strategy used by waste managers
11. Essential element in protein manufacture
14. The contamination of soil, water or the atmosphere by the discharge of harmful substances
15. Abiotic substances that make up the atmosphere
16. Atmospheric pollution enhances this effect
18. Name of gas extracted from waste
19. Plants are an essential part of this cycle
20. Arrows in a food chain, or food web, from one organism to another indicate ____

DOWN

1. Living
2. Type of rain that is produced by excessive amounts of nitrogen, carbon and sulfur oxides in the air
3. Name of a warming, circulation pattern that causes drought in Australia
4. Type of fuel emission from industry and motor vehicles
5. Abundant greenhouse gas
8. Method of hunting and agriculture as used by Aboriginals
9. Logging, housing and agriculture can reduce this, if not managed properly
12. Deep-rooting plants can alleviate this problem
13. Heat energy derived from the Earth
17. Name of energy harnessed from differences in atmospheric pressure

WORD LIST

- | | | | | |
|-------------|--------------|---------------|----------------|-------------|
| ◆ abiotic | ◆ geothermal | ◆ solar | ◆ biotic | ◆ acid |
| ◆ nitrogen | ◆ salinity | ◆ biogas | ◆ water vapour | ◆ wind |
| ◆ gases | ◆ fossil | ◆ energy flow | ◆ El Nino | ◆ burning |
| ◆ pollution | ◆ carbon | ◆ greenhouse | ◆ biodiversity | ◆ recycling |



ACROSS

2. Name of process that uses plants to produce specific compounds such as nylon, plastics, lubricants, etc.
6. Removal of one set of genes, which is replaced by another set of genes
8. Most important polymer for life
9. Used to extract and recover metal from ores and control acid drainage
11. Type of organism used in yoghurt
12. A method of reproduction used in some animals and some humans
14. Process where sperm and ova are fertilised in a Petri dish before insemination
15. The transplantation of a nucleus from a body cell into an ovum which develops into an embryo
16. Type of organism used in wine production
17. Unspecialised cell that makes a specific, specialised cell

DOWN

1. The use of biological agents, such as bacteria or plants, to remove or neutralise pollutants
2. The making of micro-electronic, microchip and microcircuit about one-thousandth of a micrometre in size, using biological techniques
3. Method of controlling a pest species by another species
4. These organisms can be modified to remove toxic chemicals from the ground
5. Name of a virus used to eradicate rabbits
6. Found in DNA
7. Name of paints that reduce marine fouling, and resist mould or bacteria
10. A sudden change in a DNA sequence
13. A noticeable segment of DNA

WORD LIST

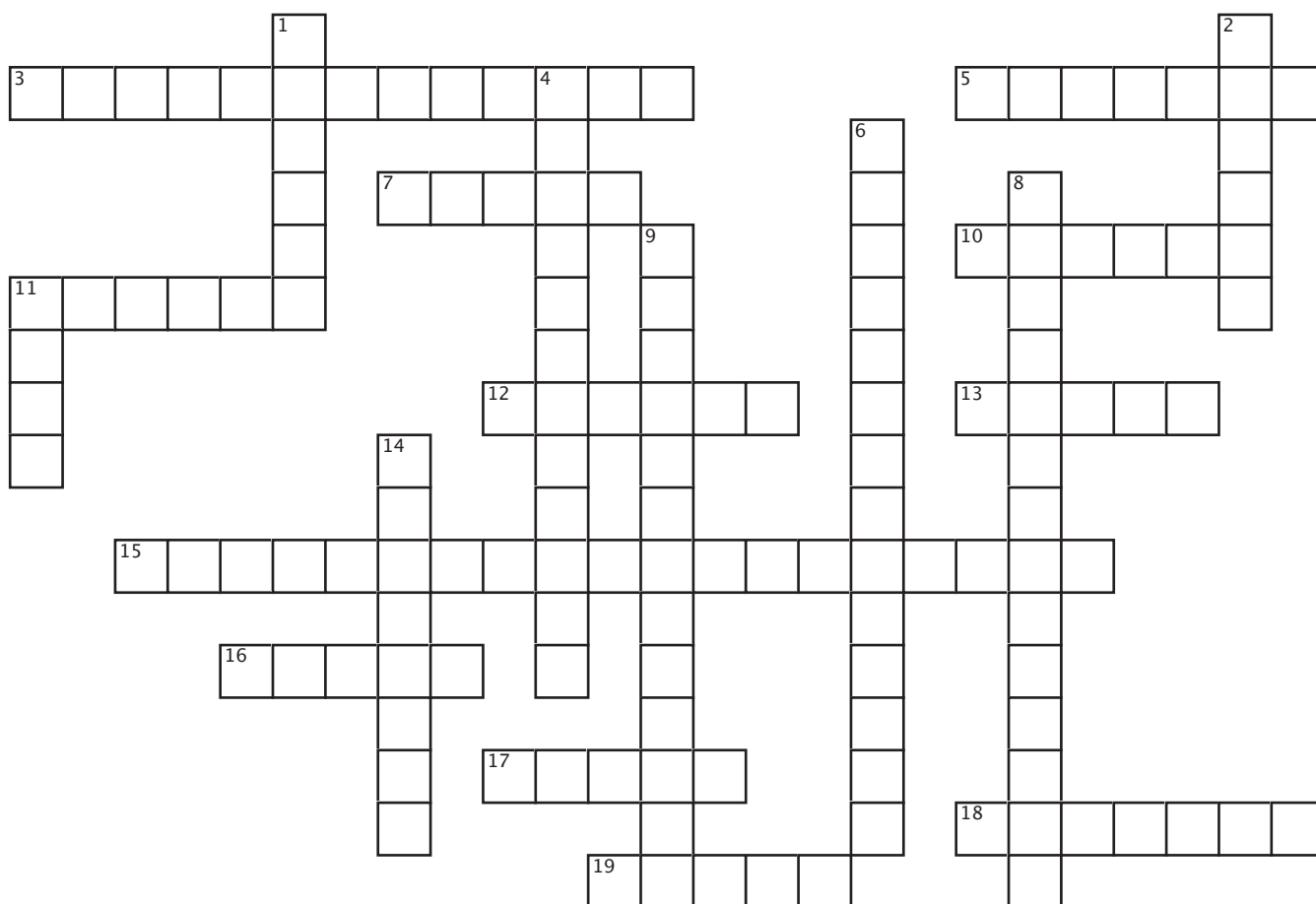
♦ bacteria
 ♦ yeast
 ♦ cloning
 ♦ Calici

♦ plants
 ♦ artificial
 insemination
 ♦ biological
 control

♦ gene splicing
 ♦ biomining
 ♦ in vitro
 fertilisation
 ♦ biomaterials

♦ stem cell
 ♦ bionanote
 chnology
 ♦ bioprocessing
 ♦ bioremediation

♦ DNA
 ♦ mutation
 ♦ markers
 ♦ genes



ACROSS

3. A device without moving parts
5. The support point on which a lever pivots
7. A vector that produces acceleration of an object
10. The moment of a force that produces torsion and rotation about an axis
11. A counterbalance in a machine
12. A wheelbarrow is this order of lever
13. A bar that is pivoted and can transmit a force and weight
15. The ratio of the output force to the applied input force of a machine
16. Unit of work
17. The order is represented by the following sequence – fulcrum effort load
18. The unit of force
19. A double-sided inclined plane

DOWN

1. The force of attraction to the Earth or another body
2. A simple machine with a grooved, rimmed wheel, used to move a load
4. A simple machine with an acute angle to the horizontal, used to raise or lower an object
6. A kitchen whisk is an example of this
8. Increases the force applied
9. One kilogram times acceleration due to gravity
11. Force times distance equals ____
14. Hauling or lifting device, found on ships, 4WDs, wells, etc.

WORD LIST

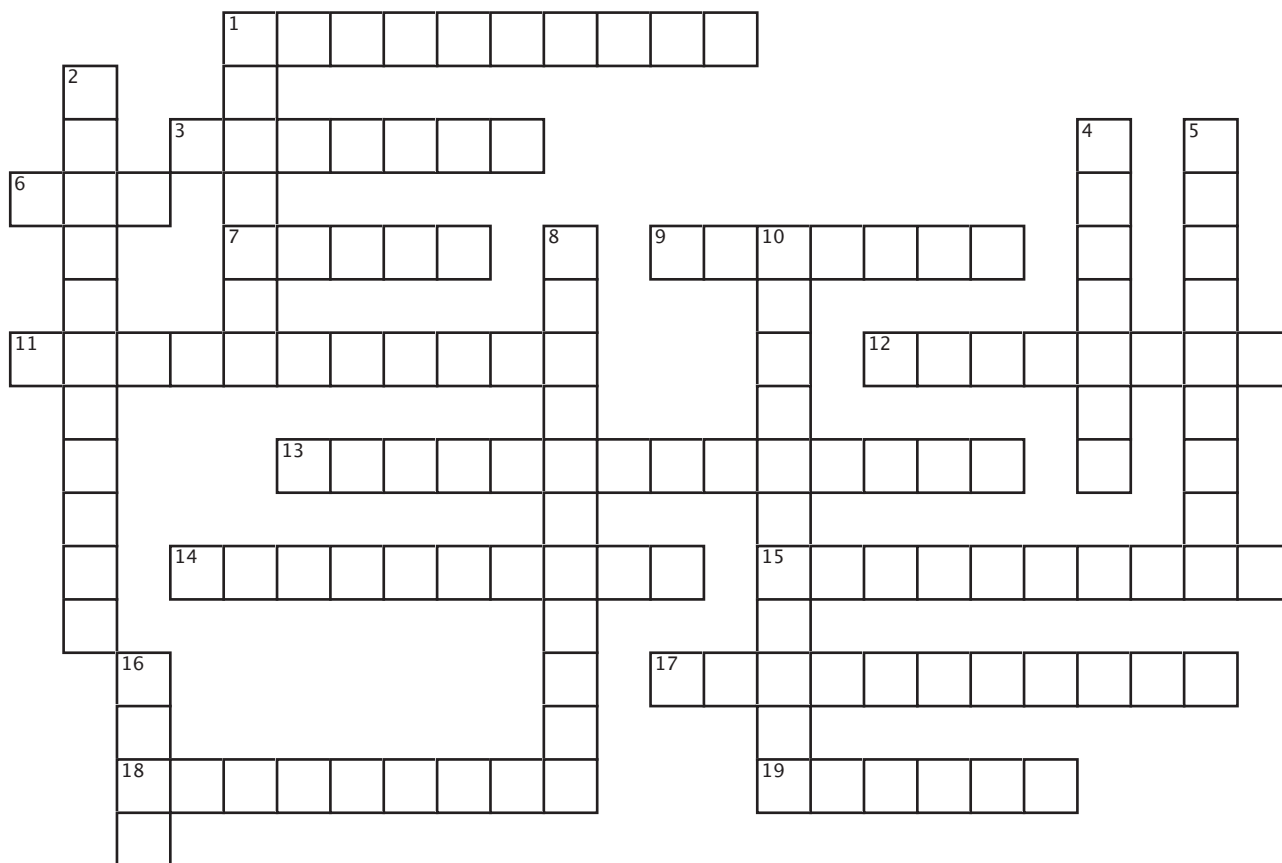
simple machine
 lever
 incline plane
 fulcrum

second
 third
 work
 force
 weight

pulley
 torque
 mechanical
 advantage
 weight

force magnifier
 speed magnifier
 wedge
 windlass
 kilogram force

Newtons
 joule



ACROSS

1. Place where crime occurred
3. Substance used for taking impressions of footprints, tyre marks, etc. found in soil
6. Found as a double helix structure in most organisms
7. Often found at crime scenes
9. Person who, through an inquest, investigates an unnatural death
11. The impression of the ridges of a fingertip, left on a surface
12. Bluish discolouration of skin and membranes, indicating lack of oxygen
13. Process of separating ingredients from a liquid or fluid
14. The motion of projectiles, or the processes that occur within a firearm
15. Study of bite marks
17. Person who undertakes an autopsy
18. Name of computer process of obtaining an image of a perpetrator
19. A source of DNA

DOWN

1. Internal bore diameter of a fire arm
2. The sequence of nucleotides in a living organism's organic polymer
4. Common name of an organism used to determine age of decomposition
5. Term used to determine a suspect's behavioural characteristics
8. Type of radiation used to detect banknote or passport forgeries
10. Muscle stiffening after death
16. Dead part of body that can be used to determine poisoning

WORD LIST

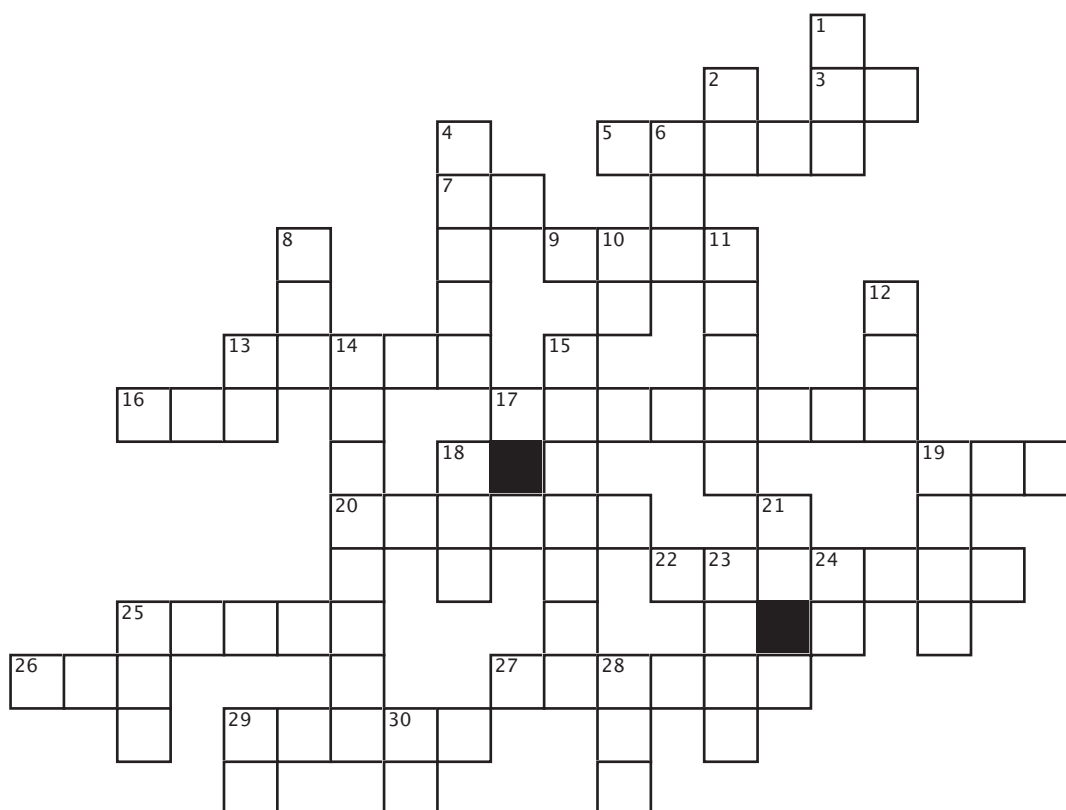
crime scene
finger print
chromatography
calibre

pathologist
genetic code
DNA
ballistics

saliva
hair
identikit
rigor mortis

coroner
ultraviolet
profiling
plaster

maggots
odontology
cyanosis
blood



ACROSS

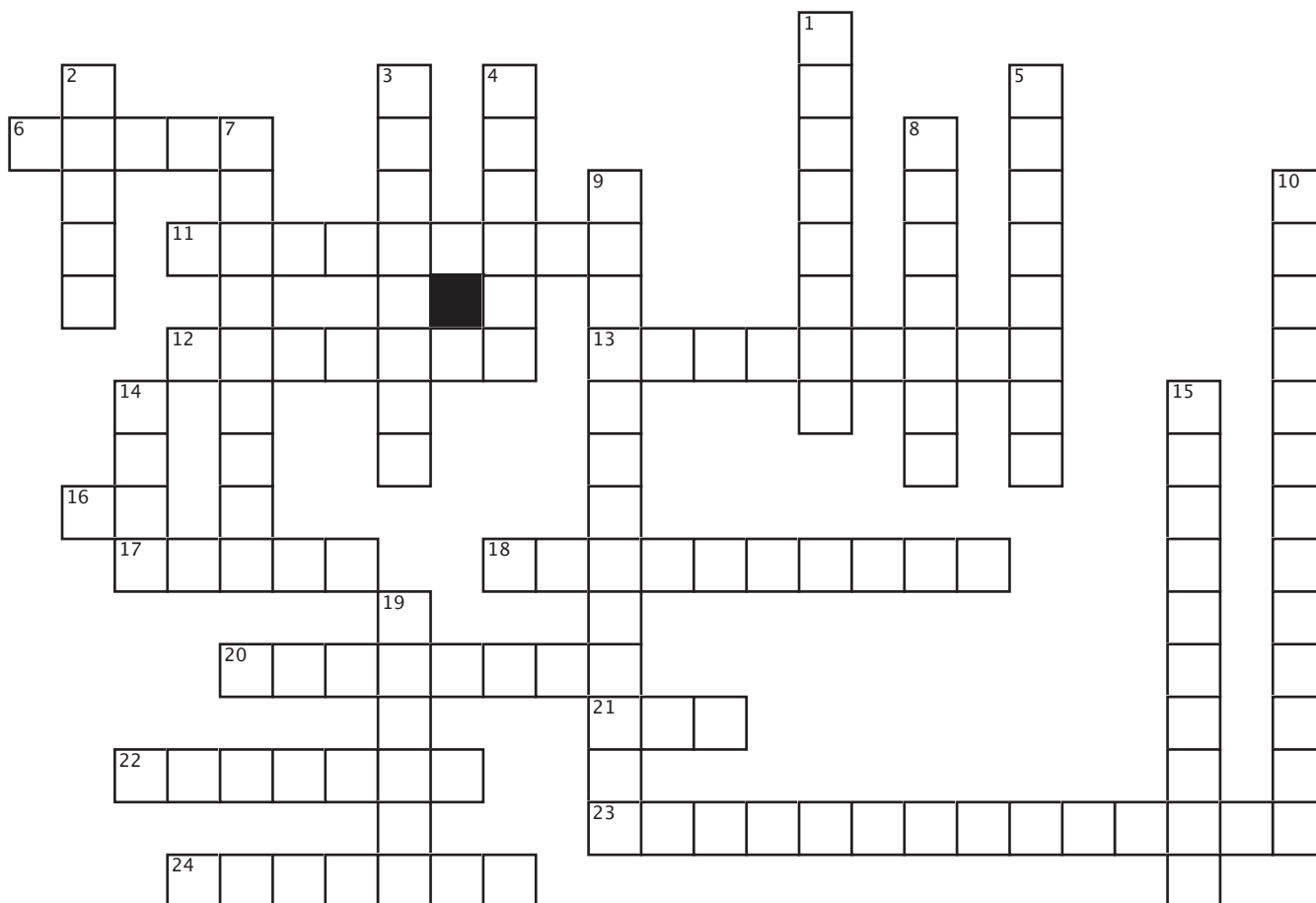
3. Helium
5. Potassium permanganate
7. Iodine
9. Sodium hydroxide
13. Phosphoric acid
16. Bromine
17. Barium nitrate
19. Ammonia
20. Sodium sulfite
22. Ethanoic acid
25. Calcium carbonate
26. Potassium oxide
27. Ethanol
29. Aluminium oxide

DOWN

1. Methane
2. Zinc
4. Silicon tetrachloride
6. Magnesium oxide
8. Sulfur trioxide
10. Gold
11. Sulfuric acid
12. Chlorine gas
13. Hydrogen gas
14. Lead nitrate
15. Calcium hydroxide
18. Water
19. Nitrogen pentoxide
21. Ozone
23. Nitric acid
24. Copper
25. Carbon dioxide
28. Hydrochloric acid
29. Silver
30. Oxygen gas

WORD LIST

H ₂ O	Pb(NO ₃) ₂	H ₂	Cl ₂	Au	SiCl ₄	Ba(NO ₃) ₂
H ₂ SO ₄	CaCO ₃	O ₂	N ₂ O ₅	Ag	NH ₃	He
CH ₃ COOH	MgO	O ₃	C ₂ H ₅ OH	Al ₂ O ₃	HCl	K ₂ O
NaOH	Na ₂ SO ₃	CO ₂	I ₂	Cu	Ca(OH) ₂	Zn
	KMnO ₄	SO ₃	CH ₄	H ₃ PO ₄	HNO ₃	Br ₂



ACROSS

6. Bond formed between an alkali metal and a non-metal element
11. A substance that indicates the change in concentration of another substance, usually by colour
12. Particle with approximately the same mass as a proton, but no electrical charge
13. A polyatomic ion made from carbon and oxygen
16. Symbol for hydroxide
17. Short for reduction and oxidation
18. Name of person who pioneered the orbital theory and discovered positive charges in the nucleus
20. Bond formed by sharing electrons
21. Particle with a positive or negative charge
22. A polyatomic ion made from sulfur and oxygen
23. The reaction between an acid and a base to form a salt and water
24. The combining capacity of a particle determined by the number of electrons that it will lose, add or share when it reacts

DOWN

1. Name of the person who devised the 'Plum Pudding Model'
2. Collective name of gases in the eighth column of the Periodic Table
3. Negatively electrically charged particle
4. Positively electrically charged particle in nucleus
5. Chlorine belongs to this group of elements
7. In chemical reactions, matter is always ____
8. Name of polyatomic ion NO_3^-
9. The process of separating a substance from a solution as a solid
10. When a compound is broken into simpler components
14. The person who proposed that electrons orbited the nucleus
15. Electrical decomposition of water into hydrogen and oxygen gases
19. Name of person who devised the 'Billiard Ball' model

WORD LIST

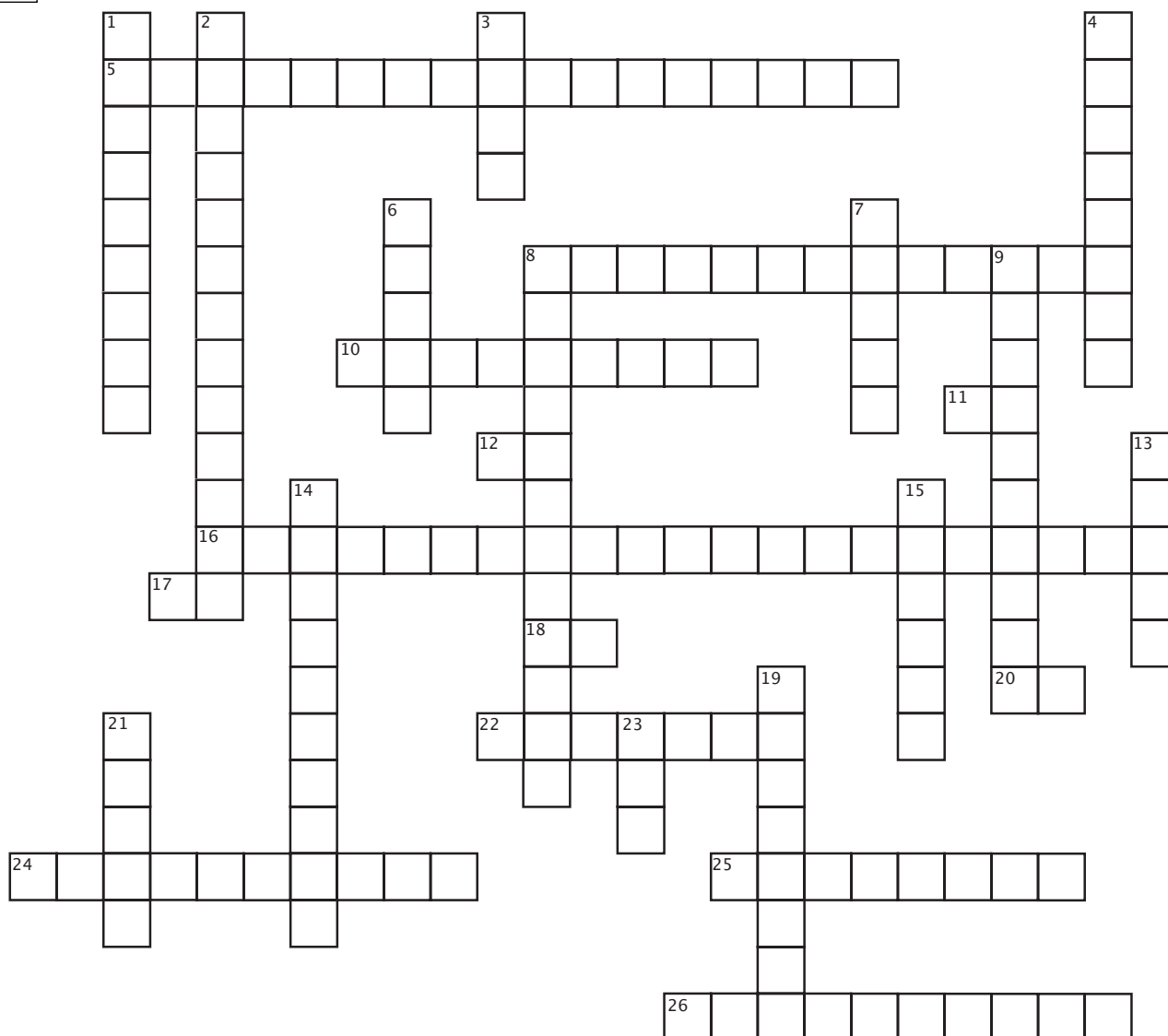
⌚ proton
 ⌚ electron
 ⌚ neutron
 ⌚ Thompson

⌚ Dalton
 ⌚ Rutherford
 ⌚ Bohr
 ⌚ Noble
 ⌚ ion

⌚ halogens
 ⌚ nitrate
 ⌚ redox
 ⌚ decomposition
 ⌚ hydrolysis

⌚ conserved
 ⌚ valency
 ⌚ precipitation
 ⌚ neutralisation
 ⌚ indicator

⌚ OH
 ⌚ sulfate
 ⌚ carbonate
 ⌚ ionic
 ⌚ covalent



ACROSS

5. The tendency of an atom or radical to attract electrons
8. Helium nucleus
10. The gradual destruction of a metal by oxidation or chemical action
11. Symbol for lead
12. Symbol for the third element in the Periodic Table
16. Written as 2.8.1
17. Symbol for a liquid non-metal
18. Symbol for magnesium
20. Symbol for sodium
22. Rows in the Periodic Table
24. A property of metals
25. Chemical bond formed by sharing electrons
26. Protons + neutrons

DOWN

1. Element in emerald
2. Device used for measuring radioactivity
3. Name of high-speed electron
4. Plastics, DNA and silk are examples of these
6. A mixture of two or more metals
7. Number of electrons in the outer shell provides this
8. Elements are ranked in this order
9. Chemical word for burning
13. Chemical bond formed by electrostatic attraction
14. In metal atoms, the number of electrons in the outer shell governs this
15. Columns in the Periodic Table
19. Atoms that have the same atomic number but different mass numbers
21. Name for O₃
23. Charged particle

WORD LIST

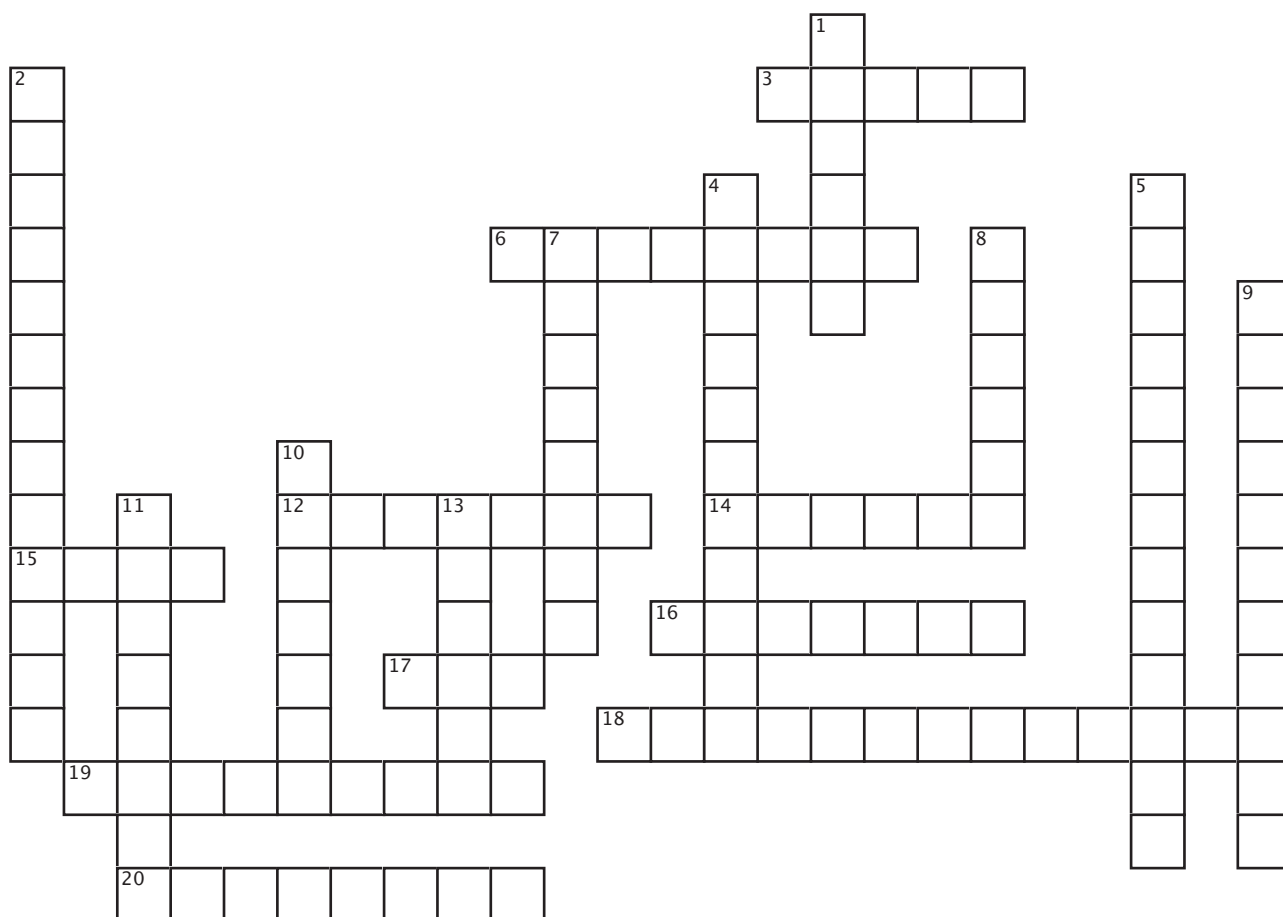
⌚ mass number
⌚ isotope
⌚ electron
⌚ configuration
⌚ atomic number

⌚ electronegativity
⌚ Geiger counter
⌚ periods
⌚ groups
⌚ group
⌚ alpha particle

⌚ beta
⌚ corrosion
⌚ combustion
⌚ polymers
⌚ ionic
⌚ covalent

⌚ alloy
⌚ conduction
⌚ reactivity
⌚ Li
⌚ Na
⌚ ion

⌚ Pb
⌚ Br
⌚ beryllium
⌚ Mg
⌚ ozone



ACROSS

3. A synthetic, silk-like, polymer fabric made from cellulose
6. A colourless, toxic, flammable liquid that may be used as antifreeze or as a solvent
12. Chemical made by fermentation of sugars
14. Name of any hydrocarbon molecule containing only carbon and hydrogen atoms connected by single bonds
15. Protein polymer extruded by larvae and spiders
16. A molecular unit that joins with similar units to form a larger molecule with the same units
17. Abbreviation for 'Polyethylene Terephthalate'
18. Major pollutant from hydrocarbon emissions
19. Many types of clothes and plastics are made from this kind of polymer
20. Major source of hydrocarbons

DOWN

1. Group 4 element essential to organic chemistry
2. Type of plastic that can be reheated and cooled without undergoing chemical change
4. Greenhouse gas given off when ethanol combusts with oxygen
5. Buttons are made from this type of plastic
7. Correct systematic chemical name for acetic acid
8. Name of hydrocarbon molecules with a triple bond between carbon atoms
9. A translucent strong-smelling hydrocarbon made from coal tar or petroleum and used as an insect fumigant
10. Gas derived from decomposing organic material
11. Name for carboxylic acids
13. Ethene is the first in this series

WORD LIST

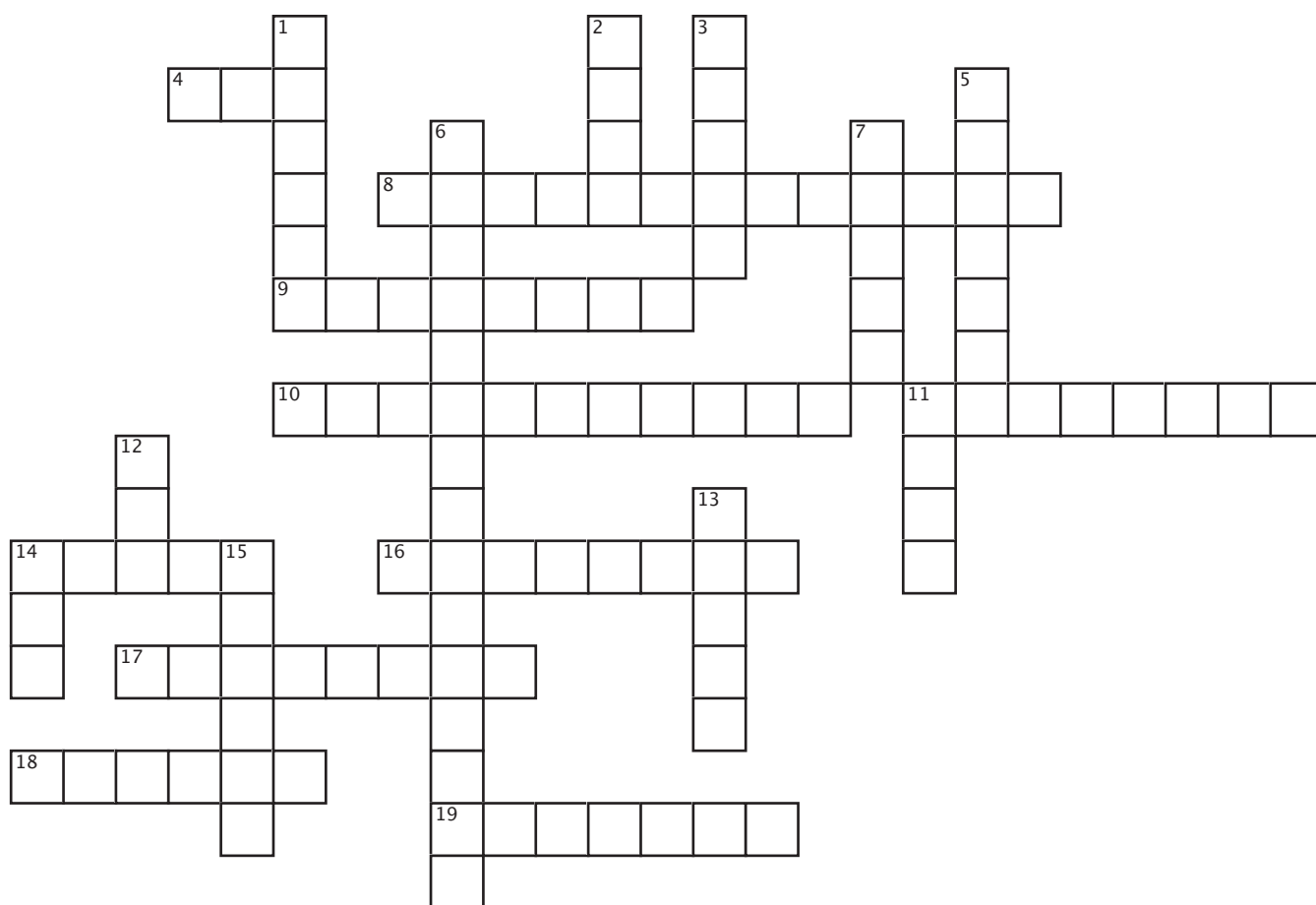
monomer
alkane
alkene
alkyne

ethanoic
polyester
crude oil
PET

alkanoic
thermoplastic
thermosetting
silk

ethanol
methanol
rayon
water vapour

carbon dioxide
naphthalene
carbon
methane



ACROSS

4. Maximum number of electrons found in the first shell
8. A positively charged particle with two protons and two neutrons
9. Particle not found in nucleus
10. A proton is this
11. A type of chemical bond in which a hydrogen atom of one molecule is attracted to an electronegative atom
14. A chemical bond between two ions with opposite charges
16. The type of bond formed when electrons are shared
17. The type of bond between mobile electrons and metallic ions
18. Particle that provides the positive charge when electrons are lost
19. Shell used commonly in bonding

DOWN

1. The type of bond formed when four electrons are shared between two atoms
2. Name of a high speed electron
3. Type of covalent bond where there is an uneven charge distribution
5. Said to be the combining power of an element
6. Opposite of electronegative
7. Number of electrons in the outer shell of an inert atom
11. This is released when bonds are broken
12. Formed when complete electron transfer occurs
13. Name of a particle that gains electrons
14. A charged particle
15. Name of a particle that loses its electrons

WORD LIST

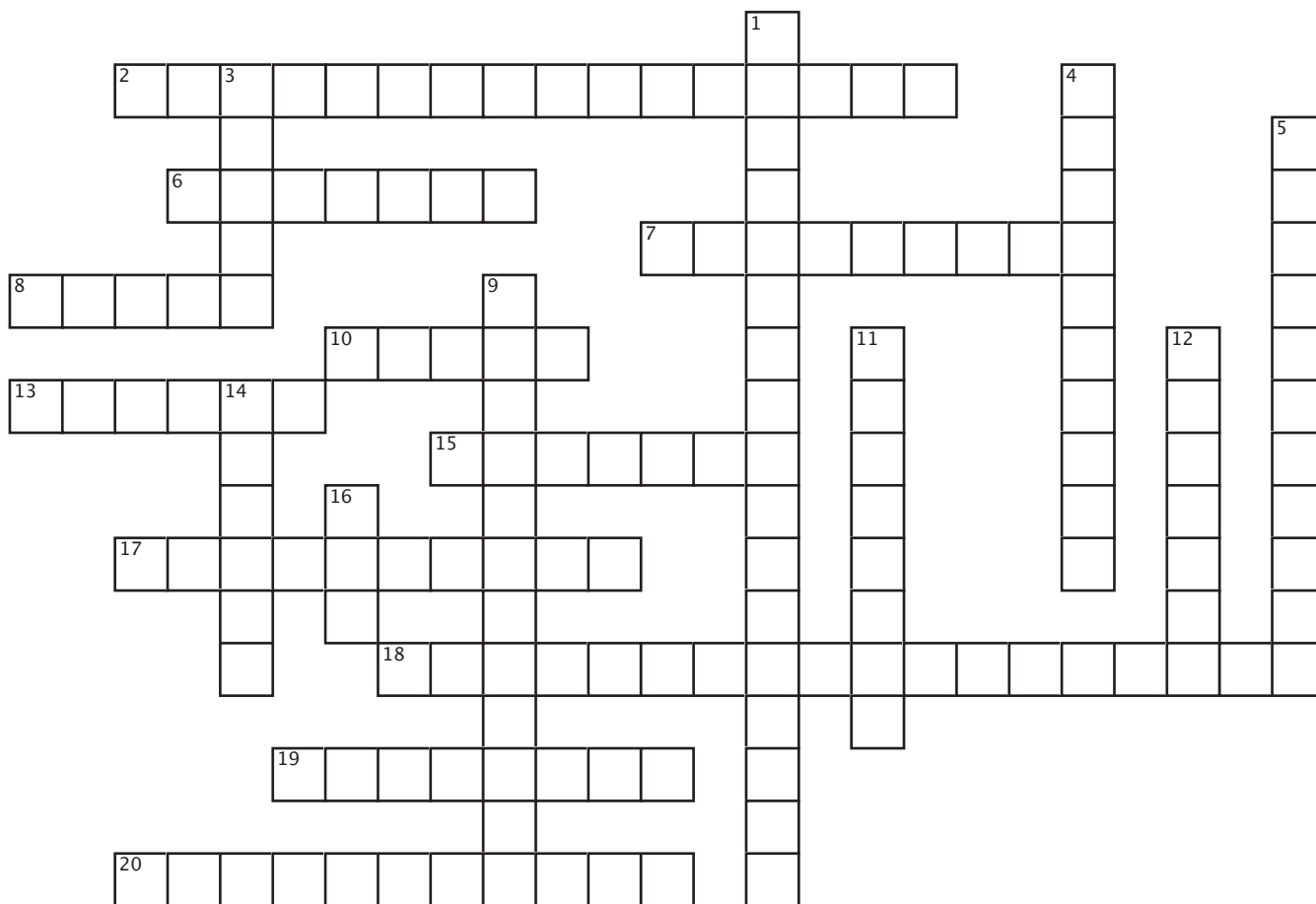
covalent
polar
metallic
valence

hydrogen
ionic
electropositive
double
heat

ion
two
valency
hydrogen ion
beta

eight
alpha particle
ion
cation
anion

electron
proton



ACROSS

2. Name of the professor who invented the Periodic Table
6. Name of the horizontal rows in the Periodic Table
7. A series of radioactive elements ranging from atomic numbers 89 to 103
8. Number of allowable electrons in the outer shell
10. The name given to the gaseous elements in the eighth column
13. Name of vertical columns
15. Name of outer shell electrons
17. What type of elements are boron, silicon, germanium, arsenic, antimony, tellurium, polonium and astatine?
18. Name of metallic elements that have multiple valences
19. Particle approximately 1/2000 of a proton's mass
20. The modern Periodic Table is based on this researcher's work

DOWN

1. The tendency for an atom to attract electrons
3. Group of elements not found in the original Periodic Table
4. Total number of nuclear particles in an atom
5. Name given to rare earth elements
9. Name of Group II elements
11. Means 'salt maker'
12. Particle not found in the hydrogen atom, but found in deuterium, tritium and all other elements
14. Name of the particle that provides the atomic number
16. Name of the first three electron shells

WORD LIST

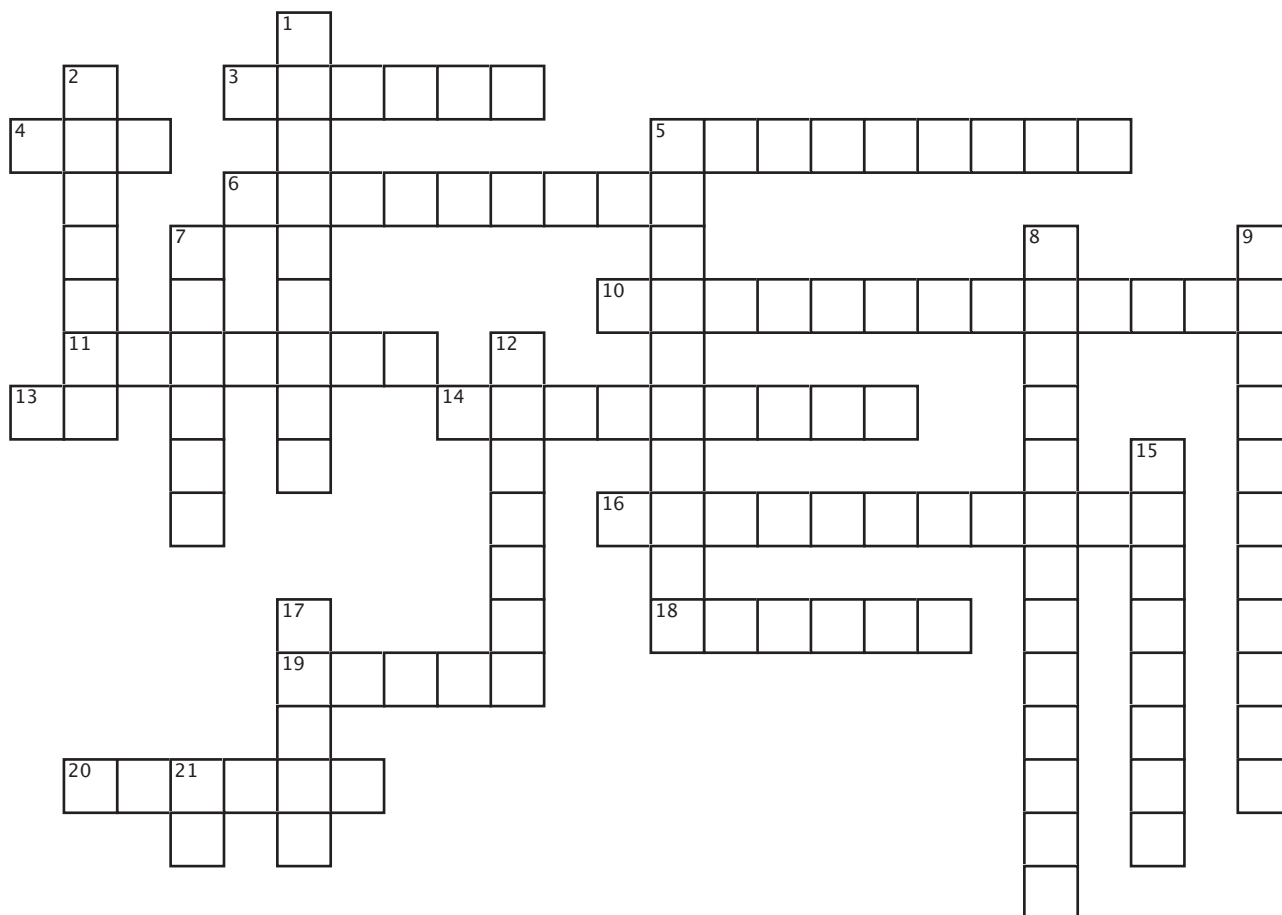
periods
groups
halogens
noble

mass number
proton
Dmitri
Mendeleev
inert

Henry Mosley
electronegativity
alkali earths
KLM
valence

electron
neutron
actinides
lanthanides

transition
elements
eight
metalloids



ACROSS

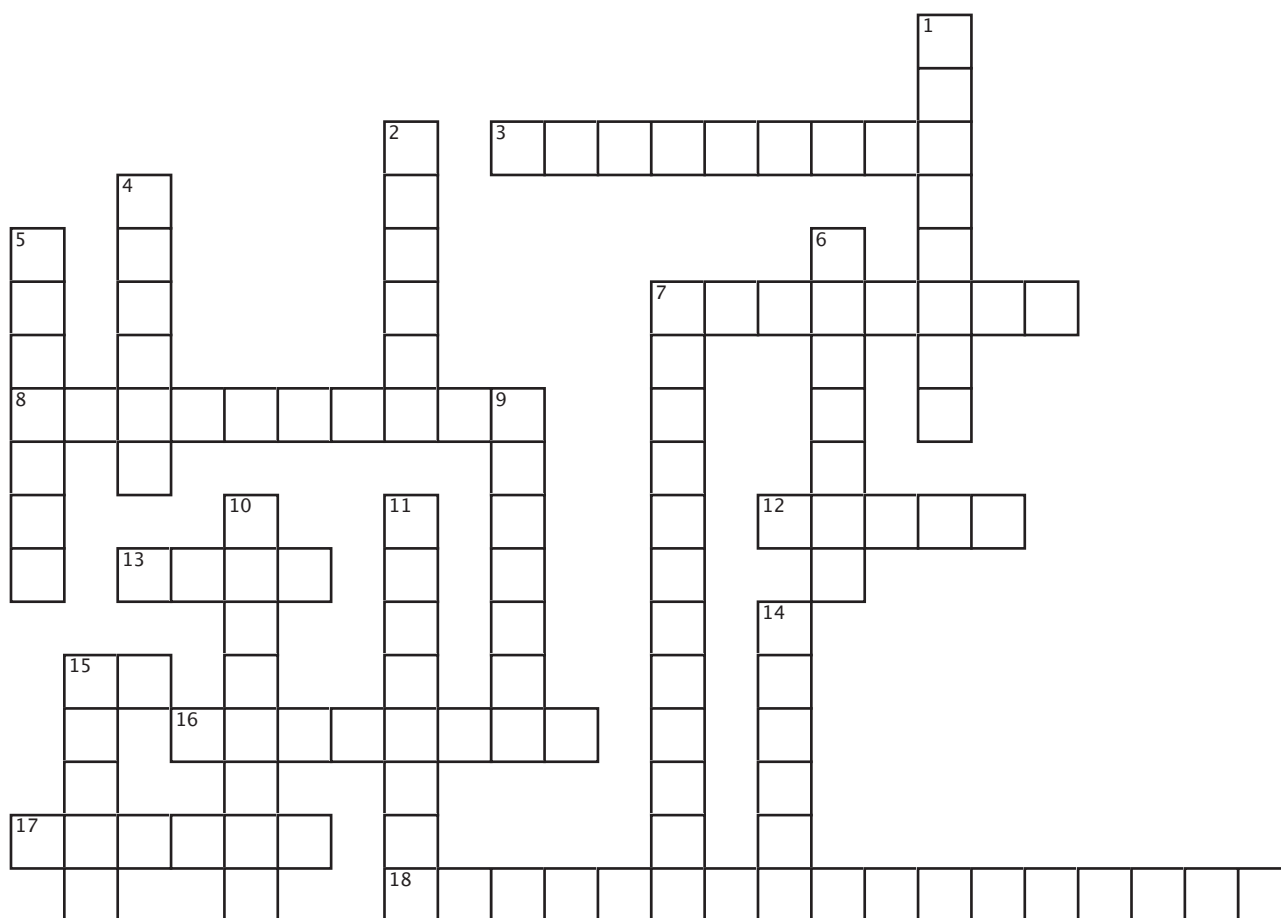
3. This element has an electronic configuration of 2.8.1
4. The suffix of most metals
5. Combusts with oxygen, and yields a very bright light
6. Element number 25
10. Metal sulfate used in medical diagnosis
11. Used in batteries
13. The symbol for this element comes from the word 'ferrous'
14. The fourth element in the Periodic Table
16. A property of metals
18. Best conductor of electricity
19. A solid mixture of metals
20. When flame tested, gives a green flame

DOWN

1. Group 1 element used in fertiliser
2. Metal which is not brittle and can be easily formed or bent
5. Another name for semi-metals
7. An ion, or group of ions, having a positive charge
8. The oxides of calcium, strontium, radium and barium are termed ____
9. Not restricted to a particular location
12. Heavy liquid metal
15. Some metals produce this gas when reacted with acid
17. The metal hydroxides are termed ____
21. The symbol for this element is derived from the word 'plumbum'

WORD LIST

⌚ manganese	⌚ alloy	⌚ mercury	⌚ high density	⌚ Pb
⌚ silver	⌚ magnesium	⌚ lithium	⌚ ductile	⌚ ium
⌚ barium sulfate	⌚ copper	⌚ potassium	⌚ hydrogen	⌚ metalloids
⌚ sodium	⌚ beryllium	⌚ delocalised	⌚ alkaline earth	
	⌚ Fe	⌚ cation	⌚ bases	



ACROSS

3. Chemical term for locality-fixed electrons
7. Most abundant gas in atmosphere
8. Comes as red, yellow or white, and is used in matches
12. Term used for Group 8 elements
13. Billboard lights are usually filled with this gas
15. Symbol for argon
16. Most electronegative element
17. Amorphous, purplish solid
18. The tendency of an atom to attract electrons to form an ionic bond

DOWN

1. All inorganic acids start with this non-metal
2. Yellow, powdery element with several allotropes
4. Graphite is one of the softest forms of this element, while diamonds are the hardest
5. Where Superman was born
6. Liquid halogen
7. A property of non-metals
9. Found bound with oxygen on beaches
10. Non-metal to non-metal bond
11. Has the electronic configuration of 2.8.7
14. Best gas for non-plants
15. Name for negative ions

WORD LIST

oxygen
silicon
bromine
inert

electronegativity
fluorine
carbon
Krypton

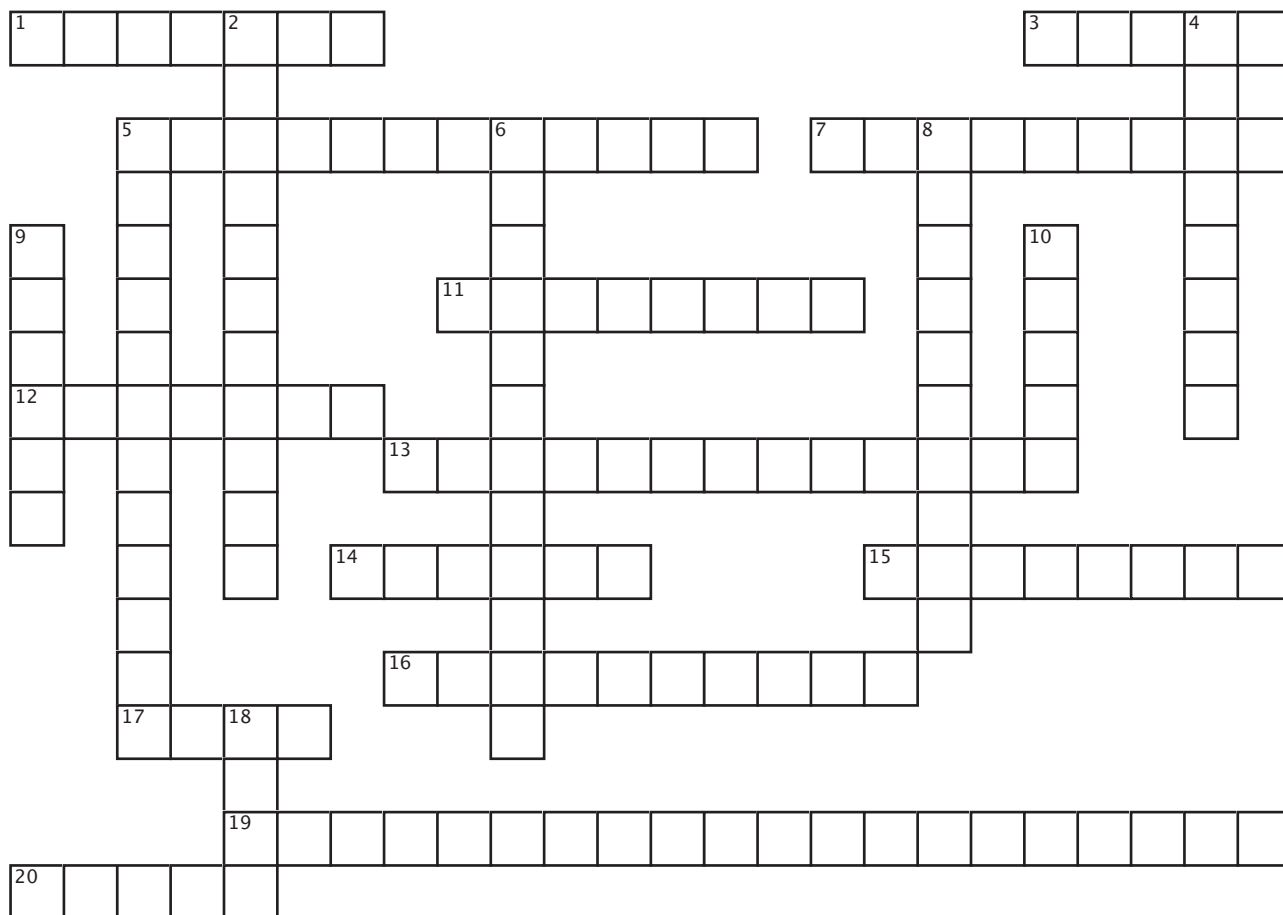
sulfur
anion
phosphorus
Ar

chlorine
neon
nitrogen
covalent

localised
iodine
hydrogen
non conductor



MOTION



ACROSS

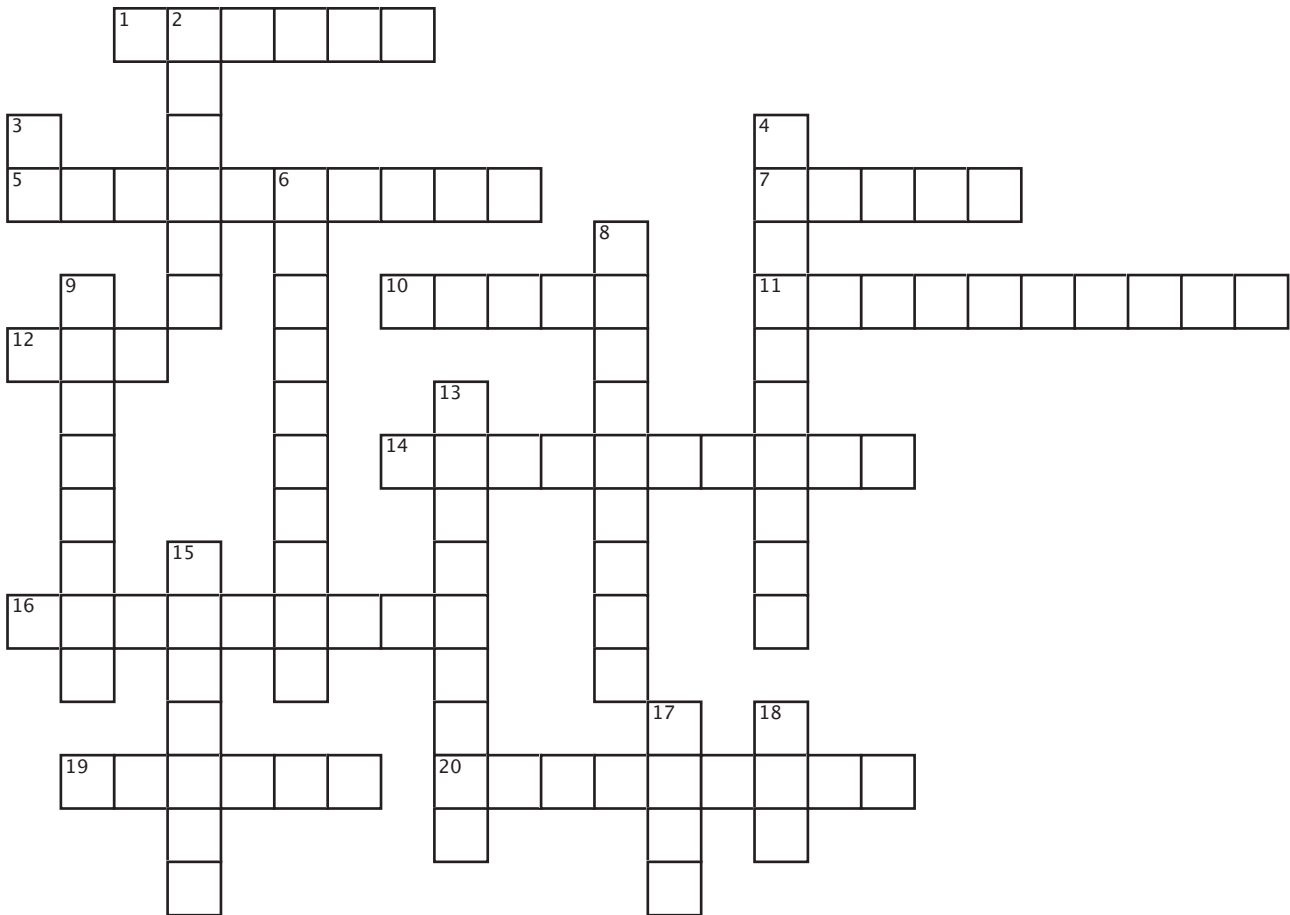
1. The tendency of a body to resist acceleration
3. Plot used to graphically show relationships or trends
5. Opposite of acceleration
7. Mathematical representations of physical phenomena are called ____
11. Newton's third law states: 'For every action there is always an equal and opposite ____'
12. Newton wrote the law about this
13. Used to measure weight
14. 100 g equals one of these
15. Similar to speed, but with direction
16. A moving object subjected to constant gravitational force
17. Speed equals distance per unit ____
19. Regular back and forth vibrations
20. Horizontal line of a two-dimensional, Cartesian coordinate system

DOWN

2. Device for recording acceleration
4. An object suspended from a fixed support so that it swings freely under the influence of gravity
5. Capital S is the symbol for ____
6. 9.8 m/s/s: Gravitational ____
8. The type of force in Newton's first law
9. The force with which a body is attracted to Earth
10. Mass X acceleration equals this
18. The amount of matter in a substance

WORD LIST

time	gravity	x axis	reaction	deceleration
velocity	acceleration	force	spring balance	simple harmonic motion
mass	weight	displacement	pendulum	
ticker timer	Newton	equations	inertia	
	graph	unbalanced	projectile	



ACROSS

1. A substance through which something else is transmitted, or carried on
5. The electromagnetic waves used in cooking and mobile phones
7. Visible form of the electromagnetic spectrum
10. A form of energy that can be changed into heat or electrical energy
11. The transfer of hot to cold in a circular manner
12. Short for Ultra High Frequency
14. The transfer of heat along a specific pathway
16. The transfer of heat through a medium in all directions
19. Stored energy
20. The size of a wave

DOWN

2. Something that does work or causes change
3. Short for Frequency Modulation
4. The flow of electric charge through a conductor
6. The distance between two wave crests or troughs
8. The number of times a wave passes in a specified interval
9. Energy that comes from substances
13. The energy that exists in something as a result of its position or condition
15. Motion energy
17. The conversion of moving air energy into electricity
18. All energy goes back to this bright object

WORD LIST

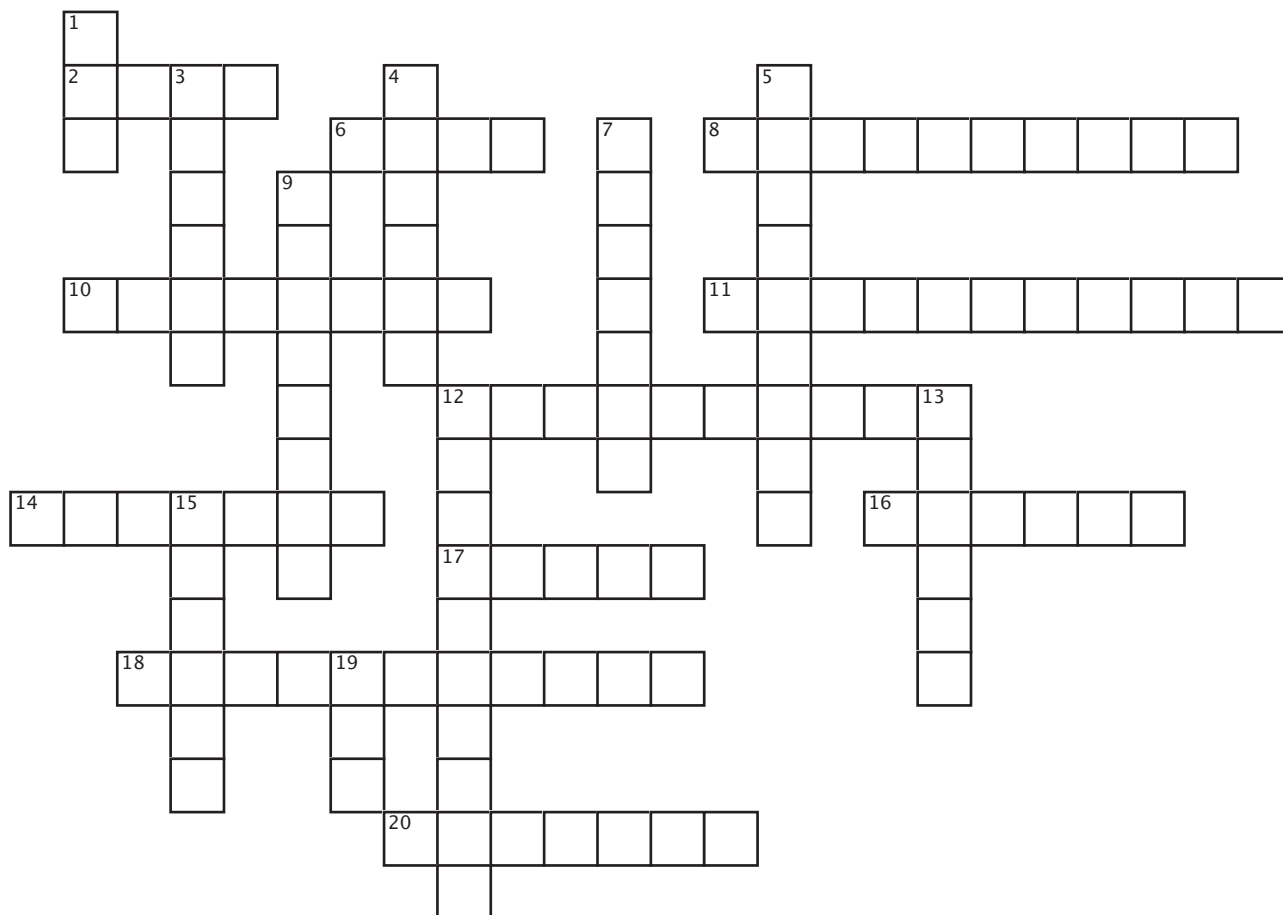
energy
 kinetic
 potential
 electrical

light
 solar
 wind
 medium

Sun
 chemical
 latent
 convection

radiation
 conduction
 microwaves
 amplitude

wavelength
 frequency
 FM
 UHF



ACROSS

2. Energy phase between electrical and light energy
6. Used to focus light rays
8. The opposition of a substance to a current passing through it
10. A circuit in which the current divides into two or more paths before recombining
11. A device for changing the number of volts and amperes
12. Term used for the bending of light
14. An analogy for the flow of electricity
16. Used for turning electrical energy flow on and off
17. A form of radiation produced by nuclear fission or fusion
18. A change in the direction and intensity of waves after passing through an opening about the same size as the wave's wavelength
20. Term used to express the force in electricity

DOWN

1. Name of the person who found electrical resistance as a relationship between voltage and current
3. Unit for measuring current
4. A non-branching electrical circuit
5. A device that converts mechanical energy into electrical energy
7. Item used for storing electricity
9. Thin wire in a light globe
12. The return of light or sound waves from a surface
13. Name of the person who found that white light, when passed through a prism, made a colour spectrum
15. The part of the eye that converts light energy to electrical energy
19. Abbreviation for the colour language of computers

WORD LIST

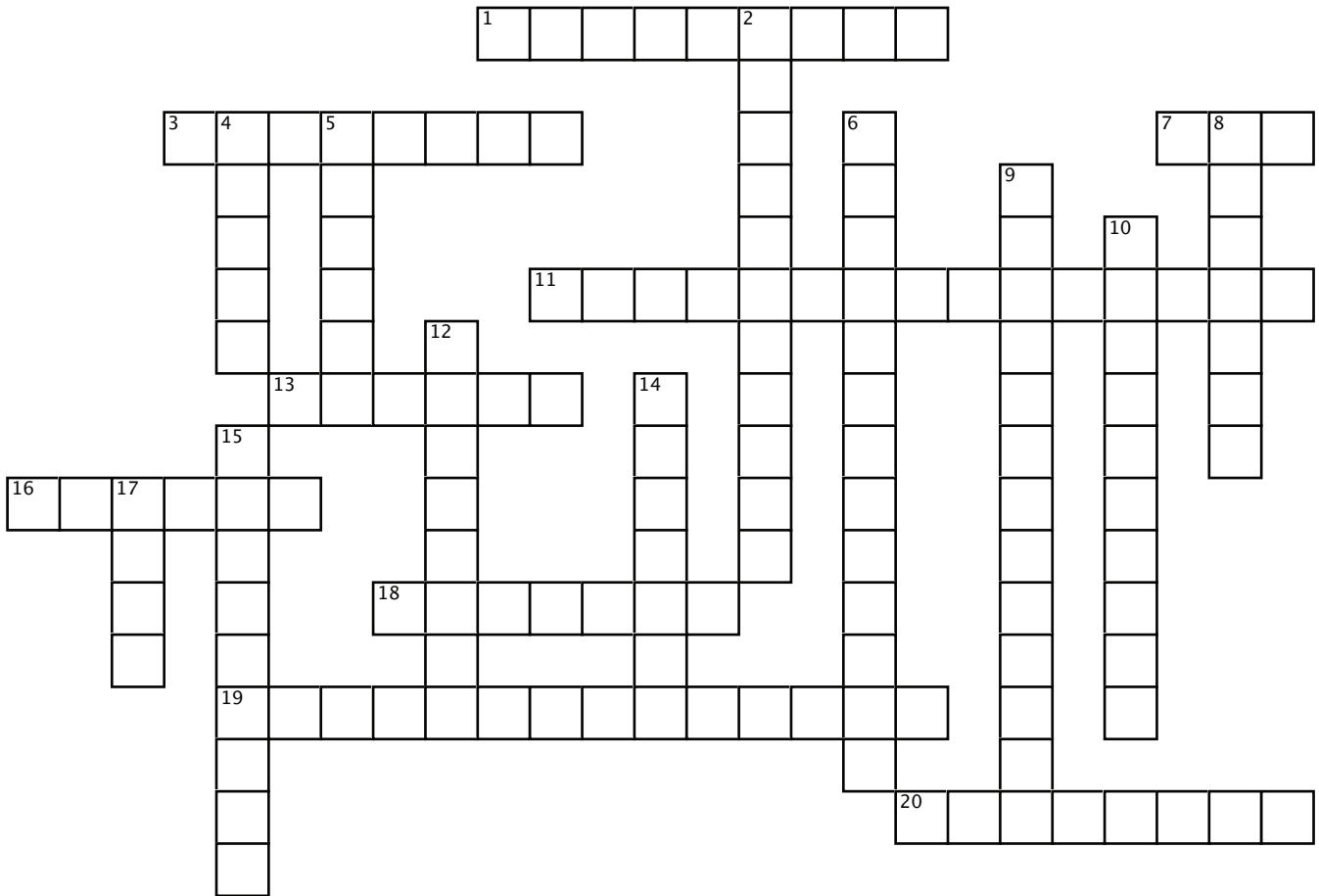
⌚ voltage
⌚ current
⌚ resistance
⌚ ohm

⌚ ampere
⌚ refraction
⌚ reflection
⌚ parallel
⌚ series

⌚ transformer
⌚ switch
⌚ battery
⌚ filament
⌚ generator

⌚ light
⌚ heat
⌚ lens
⌚ retina
⌚ diffraction

⌚ Newton
⌚ RGB



ACROSS

1. Not a particle, but a form of electromagnetic radiation
3. The gradual radioisotope decay rate of an element is known as its ____
7. Celestial object in which thermonuclear fusion reactions occur
11. A very expensive and long-lasting process of shutting down and maintaining a nuclear reactor after it has served its purpose
13. Mass times the velocity of light squared equals ____
16. A radioisotope used in medicine
18. Term used for splitting atoms
19. Controlled chain reactions take place in a ____
20. One of Australia's richest uranium mine sites in the Northern Territory

DOWN

2. Nuclear dating method to determine the age of things
4. These particles are charged helium nuclei
5. Process of joining lighter atoms to make a heavier one
6. The sequential, or random, splitting of atoms
8. Name of the ore mined for nuclear energy
9. Instrument used to detect radiation
10. The process of enhancing the amount of uranium 235 content
12. A moderator made of carbon
14. An atom with a different number of neutrons in its nucleus to other varieties of the same element
15. Highly toxic, radioactive isotope used in nuclear weapons
17. These particles are high-speed electrons

WORD LIST

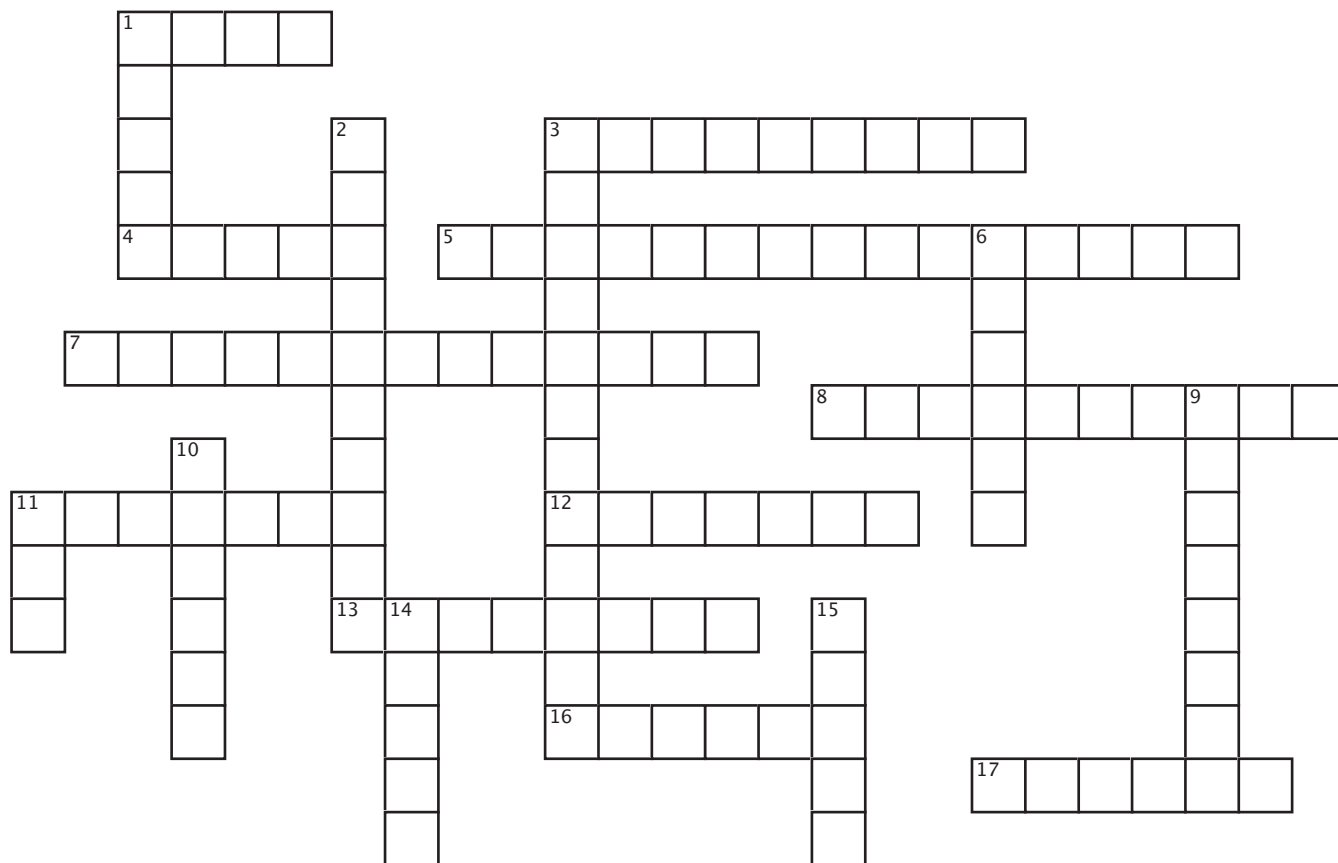
⌚ alpha
 ⌚ beta
 ⌚ gamma rays
 ⌚ geiger counter

⌚ isotope
 ⌚ fission
 ⌚ chain reaction
 ⌚ nuclear reactor
 ⌚ graphite

⌚ fusion
 ⌚ energy
 ⌚ uranium
 ⌚ half life
 ⌚ radiometric

⌚ plutonium
 ⌚ cobalt
 ⌚ Sun
 ⌚ decommissioning

⌚ Narbalek
 ⌚ enrichment



ACROSS

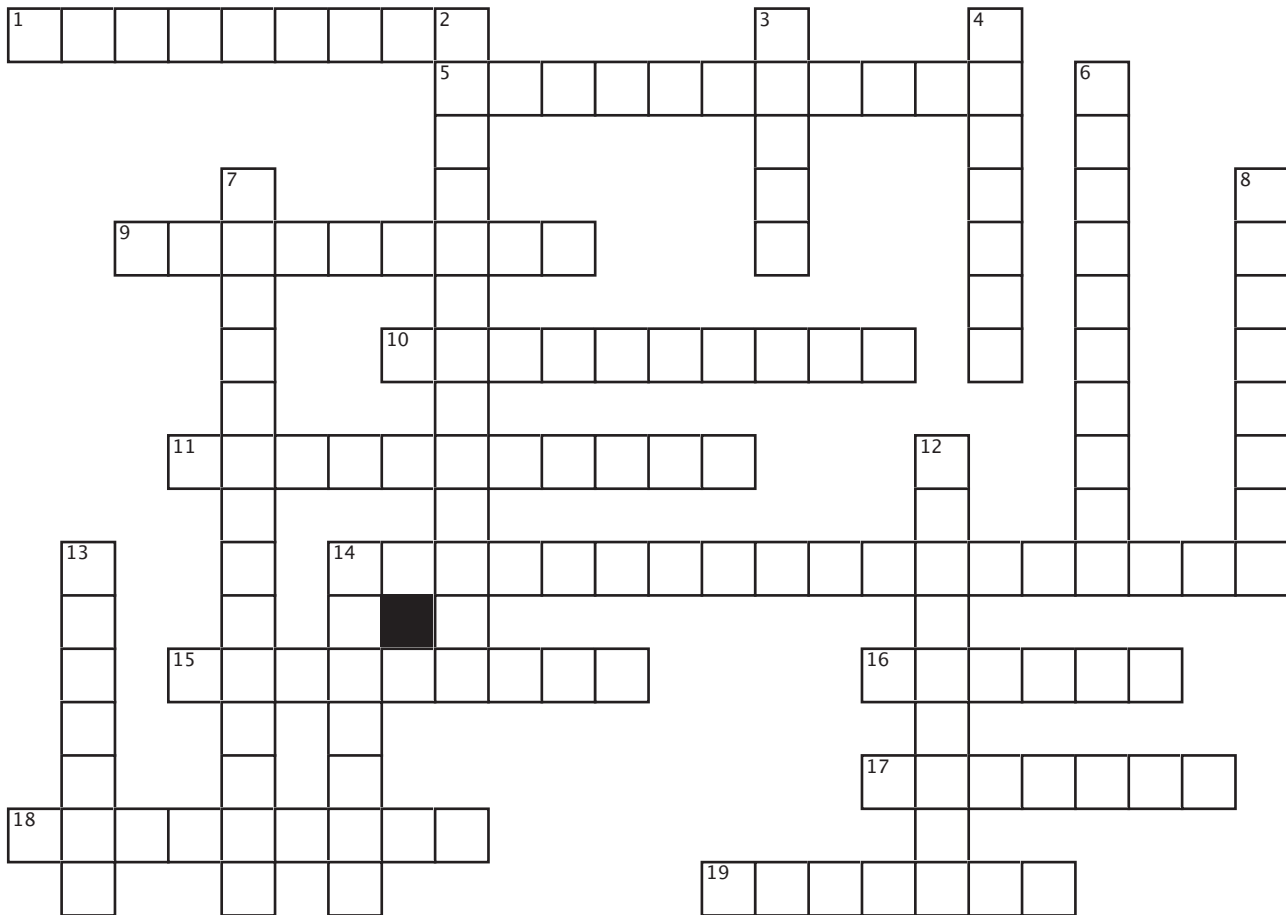
1. Light produced when blue and green are mixed
3. Not primary
4. The number of visible colours in a rainbow
5. Light is a form of this type of radiation
7. Name given when a primary and secondary colour produce white light
8. An image made on light sensitive material
11. Negative colour film images are always in ____
12. A secondary colour made from blue and red light
13. White light is split into this
16. Light sensitive membrane in the eye
17. Name of device that absorbs certain parts of the spectrum

DOWN

1. Structures in the eye that allow humans to see colour
2. This is needed to focus light, to produce a pure spectrum
3. Instrument used to analyse light
6. Person who first split light into a spectrum
9. Term given to the mixing of primary colours
10. Red and green light make which secondary colour
11. Abbreviation for the colour language of computers
14. Transparent object that can split light
15. Objects that absorb all colours are ____

WORD LIST

⌚ RGB	⌚ prism	⌚ complementary	⌚ Newton	⌚ reverse
⌚ spectrum	⌚ black	⌚ cyan	⌚ seven	⌚ cones
⌚ spectrometer	⌚ magenta	⌚ filter	⌚ electromagnetic	⌚ convex lens
⌚ secondary	⌚ additive	⌚ yellow	⌚ photograph	⌚ retina



ACROSS

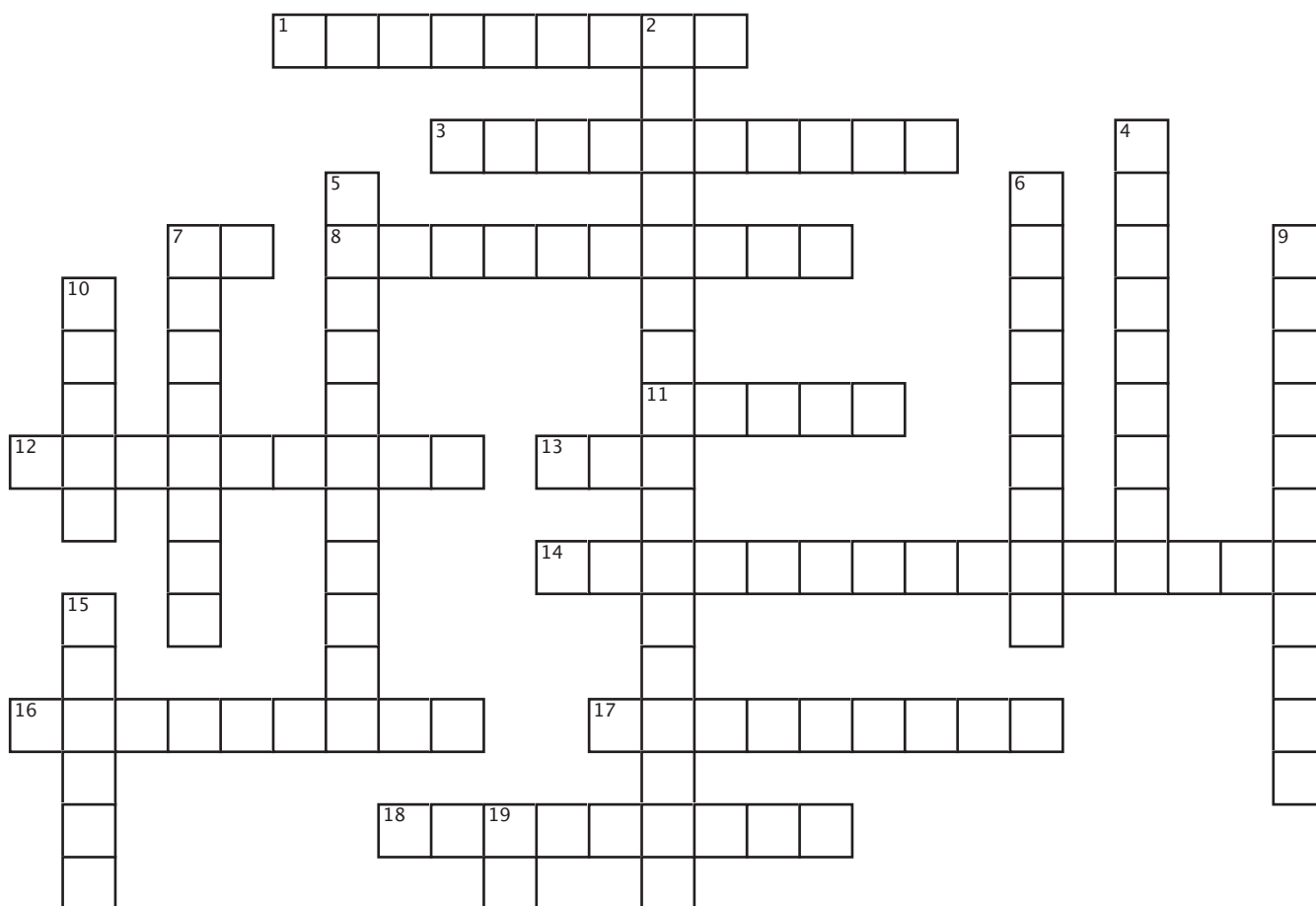
1. Cylindrical magnets are used to ensure equal frequency and ____
5. Device that vibrates to produce sound
9. Device that is the reverse of an electric motor
10. Device that converts sound to electrical energy
11. Device that generates and amplifies a carrier wave, and radiates a signal from an antenna
14. The meaning of AC
15. Name of the component to which a coil is fixed in a loudspeaker
16. The type of fuel from which energy in power stations is usually derived
17. Person who discovered electromagnetic induction
18. Type of magnet in a loudspeaker
19. These items cause an electrical current to be produced

DOWN

2. Device that changes electrical energy to kinetic energy
3. The unit of cycles per second or frequency
4. These carbon components allow an electric current to flow into a coil
6. A cylindrical arrangement of insulated, metal bars connected to the coils of a DC electric motor or generator
7. Flow of electricity in a direct current device
8. Direction in which a coil in a loudspeaker moves
12. Mobile phones use these waves to transmit a signal
13. Device used to measure current
14. Units used to measure electric current

WORD LIST

⌚ electric motor	⌚ brushes	⌚ alternating current	⌚ amplitude	⌚ transmitter
⌚ generator	⌚ commutator	⌚ ammeter	⌚ Hertz	⌚ microwave
⌚ microphone	⌚ magnets	⌚ amperes	⌚ Faraday	
⌚ loudspeaker	⌚ unidirectional	⌚ diaphragm	⌚ permanent	
	⌚ fossil		⌚ in and out	



ACROSS

1. Type of wave used by police, fire brigade, ham radios and military communication
3. Longitudinal waves can be ____
7. Abbreviation for amplified modulation
8. Waves of less than 50 MHz are reflected by this atmospheric layer
11. The measurement unit of the number of cycles per second of a wave
12. Wave used in radar, mobile phones and cooking
13. Abbreviation for extra low frequency
14. All waves are part of this spectrum
16. The measurement of the number of times that a repeated event occurs per unit of time
17. Waves that are bounced back are ____
18. A wave that reaches another medium at an angle can be ____

DOWN

2. Wave used for high quality sound
4. The height of a wave is termed ____
5. The longer the waves the greater the ____
6. Wave frequency used in television and GPS navigation
7. When waves enter another medium they are ____
9. Space between two compression points in a longitudinal wave
10. Type of wave used by aircraft and weather radar
15. The time taken for a wavelength to pass a given point
19. A carrier wave that has its frequency modified; often used for radio

WORD LIST

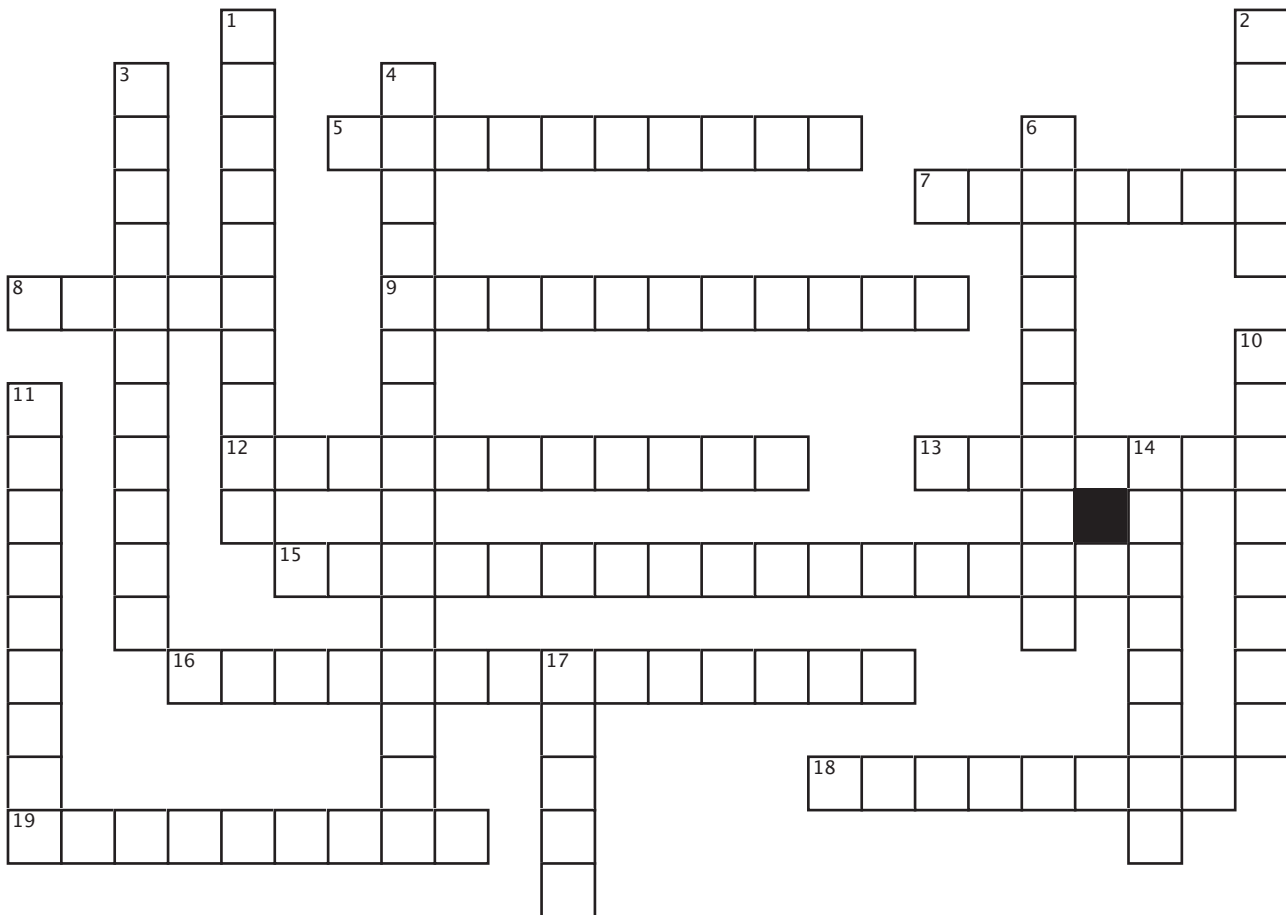
⌚ electromagnetic
⌚ amplitude
⌚ frequency
⌚ ELF

⌚ very high frequency
⌚ microwave
⌚ Hertz
⌚ period

⌚ refracted
⌚ absorbed
⌚ reflected
⌚ compressed
⌚ rarefaction

⌚ AM
⌚ ultra high
⌚ short wave
⌚ FM
⌚ radio

⌚ ionosphere
⌚ diffraction



ACROSS

5. Name for a variable that may influence the results of what is being tested
7. The measurable effect of observable, or felt, improvement in health, not attributed to treatment
8. The presentation of the results of the data as a diagram
9. Needed to reduce variability, increase mathematical significance and confidence, and draw strong conclusions from the data
12. The variable that you change on purpose
13. A standard by which any change can be measured
15. The main purpose of design is to rule out ____
16. This type of design involves doing a controlled experiment twice with the same subjects, to reduce the effect of any uncontrolled differences
18. Using fine measuring instrument scales
19. The name of the action that is done to an experiment

DOWN

1. A possible explanation for an observation or scientific problem that can be tested by further investigation
2. An unwanted side effect in an experiment which may produce the effect rather than what is done to the experiment
3. An experiment that consistently produces the same results
4. The deliberate imposition of a treatment on a group of subjects
6. The property of completely chance events, that are not predictable
10. Making sure that what you have is the correct test for that hypothesis
11. The variable which is measured as a result of the experiment
14. Required background information
17. Type of control group test where neither the evaluator nor the subject knows which items are controls

WORD LIST

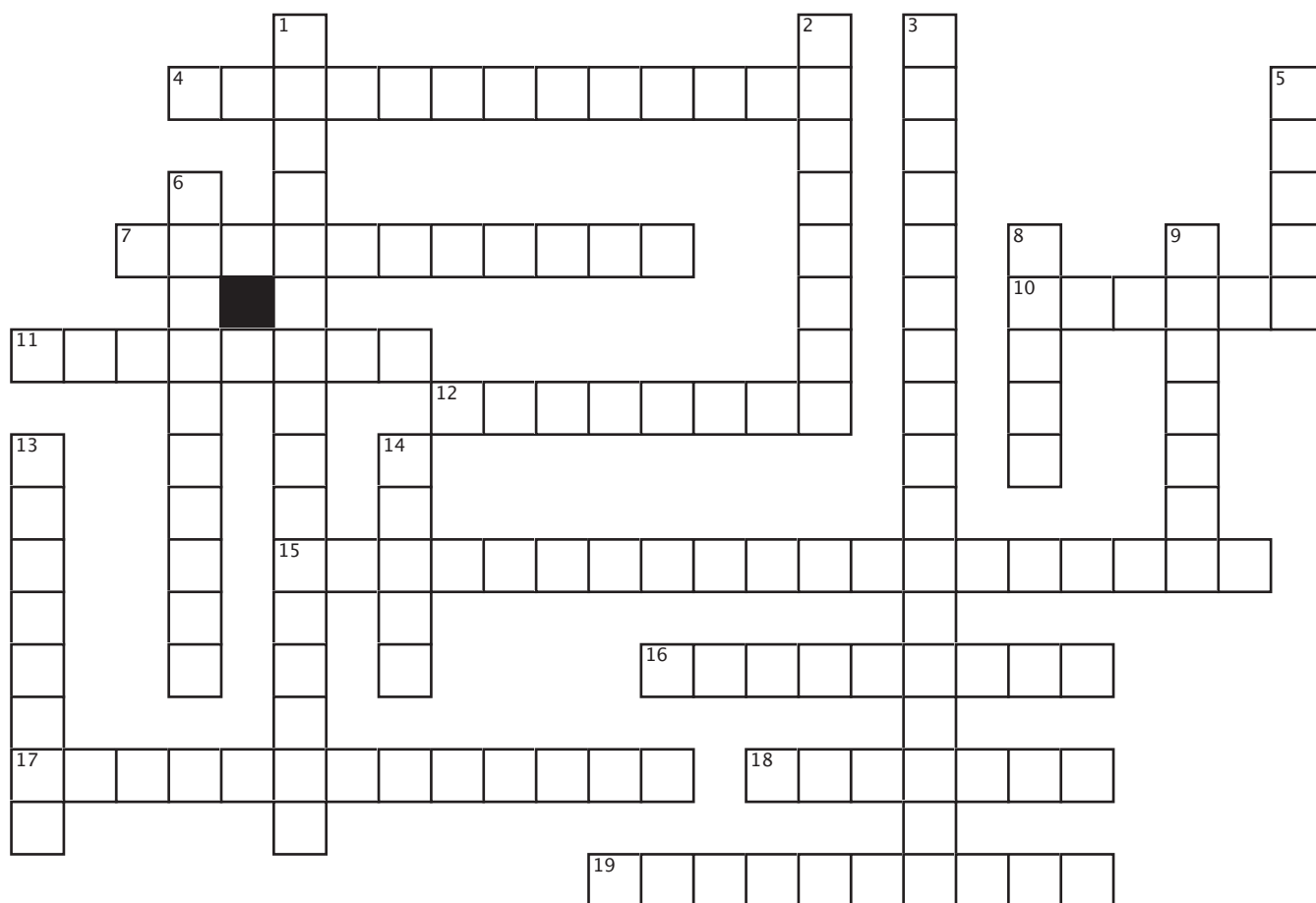
experimentation
 alternative
 causes
 control

extraneous
 randomness
 error
 replication

blind
 placebo
 reliability
 validity

accuracy
 hypothesis
 independent
 dependent

treatment
 graph
 counterbalance
 research



ACROSS

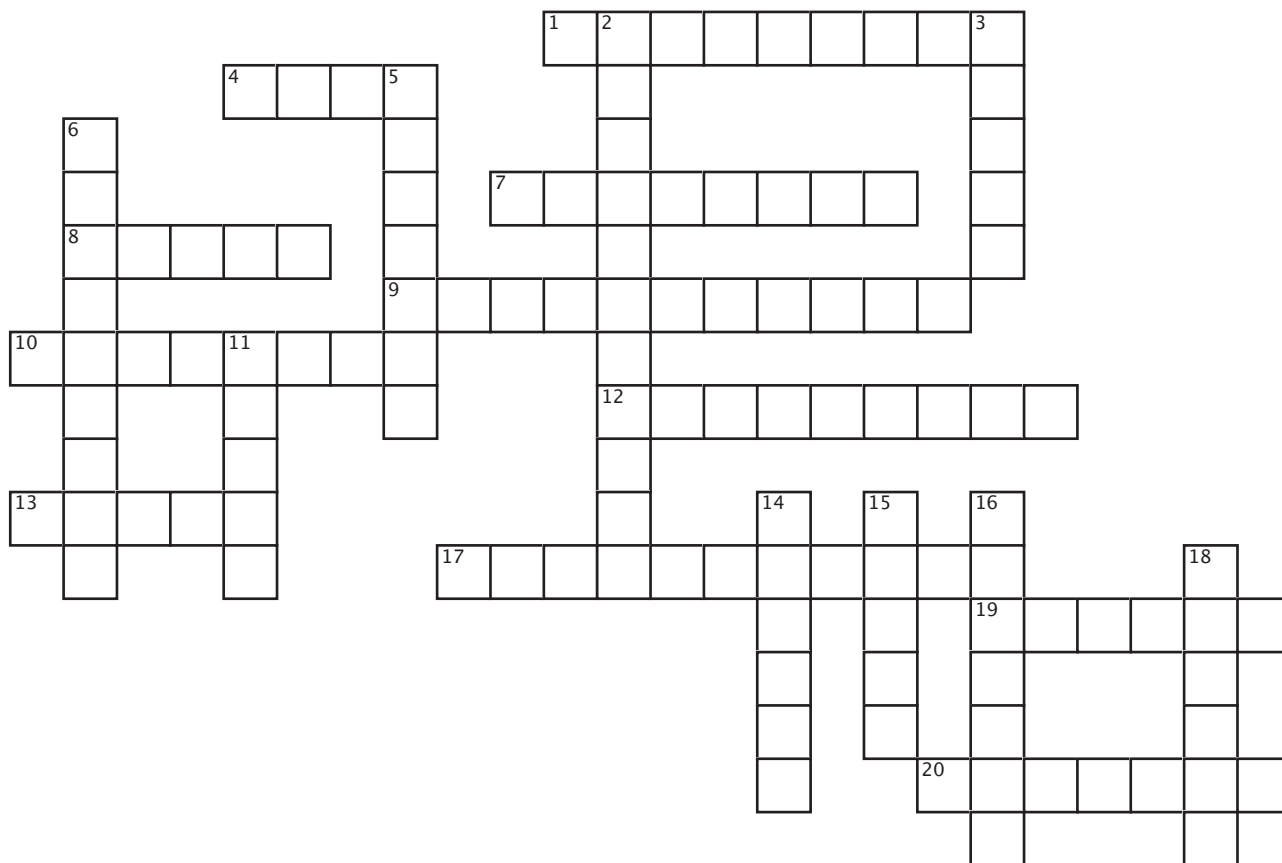
4. The record of observations and measurements taken accurately over a number of trials
7. Consistently produces the same results
10. To write down observations or data
11. To achieve this, many repeats in an experiment are needed
12. To judge carefully
15. A change in something where the values are independent of all other changes; usually constants such as time
16. Used for expressing relationships
17. To express information in such a way that it is readily and clearly understood
18. The progressive record of an experiment over time
19. A trial explanation for an observation, phenomenon or scientific problem that can be tested by further investigation

DOWN

1. The processes needed to skilfully conceptualise, apply, analyse, synthesise and evaluate information to reach a conclusion
2. A collection of data arranged for ease and speed of search and retrieval
3. The observed change in an experiment, as measured against a constant in that experiment
5. Well grounded in evidence
6. To ensure safety
8. Image of a relationship between two or more mathematical results
9. To regulate a scientific experiment by conducting a parallel experiment, against which other elements in the experiment are measured
13. Scientists weigh this for and against a hypothesis
14. A trial piece of work that serves as a plan from which a final product is to be made

WORD LIST

thumbs up equations	thumbs up graph	thumbs up critical thinking	thumbs up control	thumbs up first hand data
thumbs up accuracy	thumbs up dependent variable	thumbs up database	thumbs up reduce risk	thumbs up model
thumbs up reliability	thumbs up independent variable	thumbs up evaluate	thumbs up evidence	
thumbs up hypothesis		thumbs up valid	thumbs up record	
		thumbs up communication	thumbs up log book	



ACROSS

1. The presumed effect of the independent variable
4. Not all graphs start at this point
7. Any measured characteristic or attribute that differs from others
8. The sequence of numbers or units with equal intervals between them
9. The variable that is actively controlled or changed
10. Lines that start at the bottom left and go to the top right of the graph indicate this kind of relationship
12. Graph that shows trends and patterns
13. Name given to the quantity and units being plotted on an axis
17. To make a prediction within a range of measurements
19. Graph that shows data as fractions or percentages
20. A graph shows numerical data as a ____

DOWN

2. To make a prediction beyond the range of measurements
3. Before graphing, data is usually recorded in one of these
5. A single observation far away from the rest of the data
6. Name of a column graph that shows frequency occurrence at its height
11. Statement given either at the top or bottom of a graph, summarising its function
14. Type of graph that compares measurements
15. Line on which the independent variable is placed
16. A line that best represents the data on a scatter plot or graph, which can be mathematically expressed
18. A bell-shaped distribution curve that concentrates in the centre and decreases on either side

WORD LIST

thumbs up column
thumbs up line graph
thumbs up sector
thumbs up interpolate

thumbs up extrapolate
thumbs up variable
thumbs up independent
thumbs up dependent

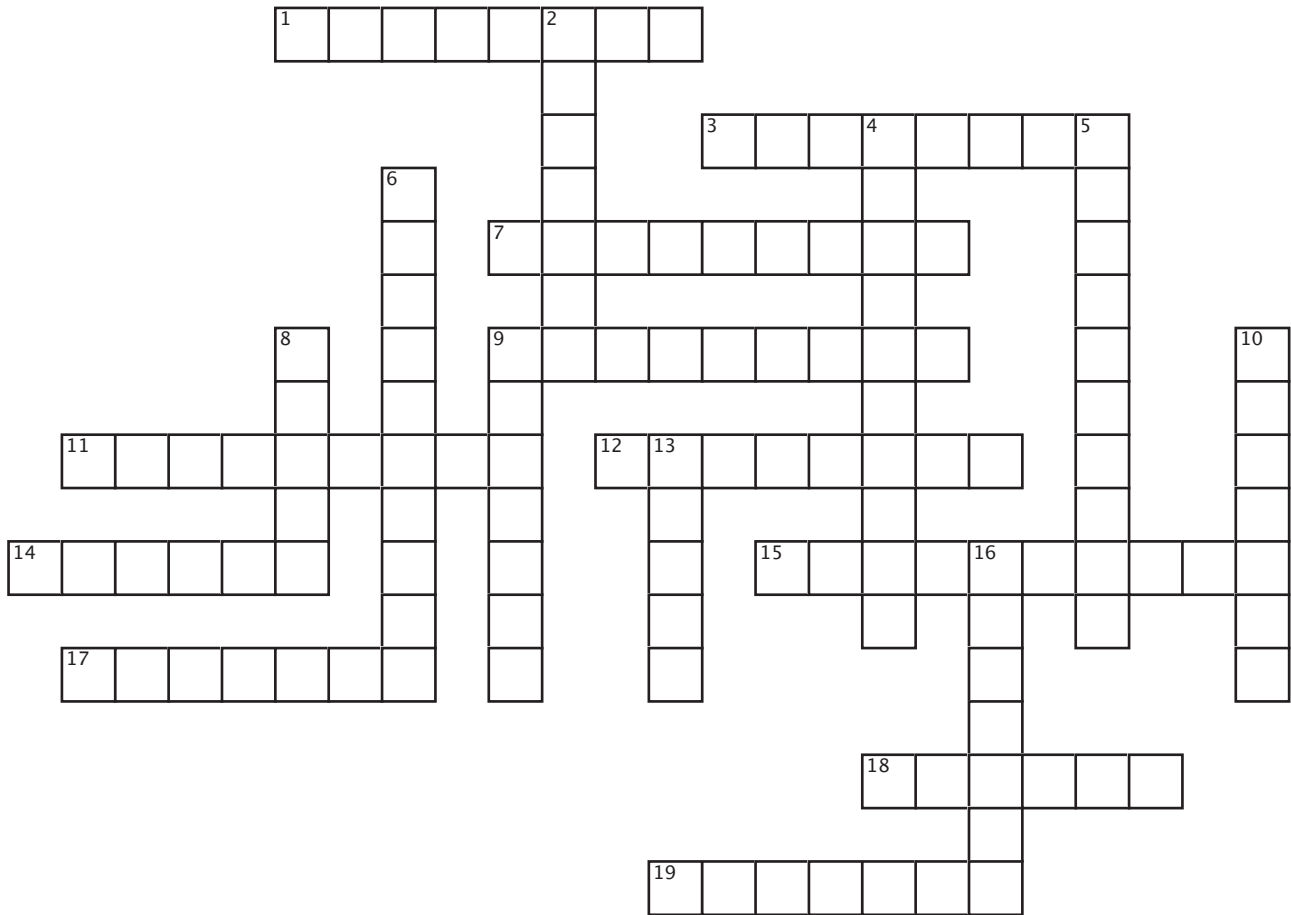
thumbs up x axis
thumbs up label
thumbs up scale
thumbs up title

thumbs up table
thumbs up best fit
thumbs up normal
thumbs up zero

thumbs up diagram
thumbs up histogram
thumbs up positive
thumbs up outlier



USEFUL TERMS



ACROSS

1. The heat content of a substance
3. Downward bending fold
7. To move apart in different directions
9. Inherited characteristics that are determined by the X chromosome
11. The number, size and shape of chromosomes in an individual cell
12. Disease-carrying organism
14. An alternative form of a gene
15. Disease caused by direct or indirect contact between organisms
17. The total mass of organic material in an ecosystem
18. A positive ion
19. The feeding level in an ecosystem

DOWN

2. Ordered arrangement of particles in three dimensions
4. The adaptive evolution of similar structures in different organisms
5. A chemical reaction that releases heat
6. Individual with identical alleles for a given gene
8. Exact copy of an organism by asexual reproduction
9. Individuals of the same type that can interbreed to produce fertile offspring
10. A process of cell division that results in the new cells having half the number of chromosomes
13. A negative ion
16. Non-living part of the environment

WORD LIST

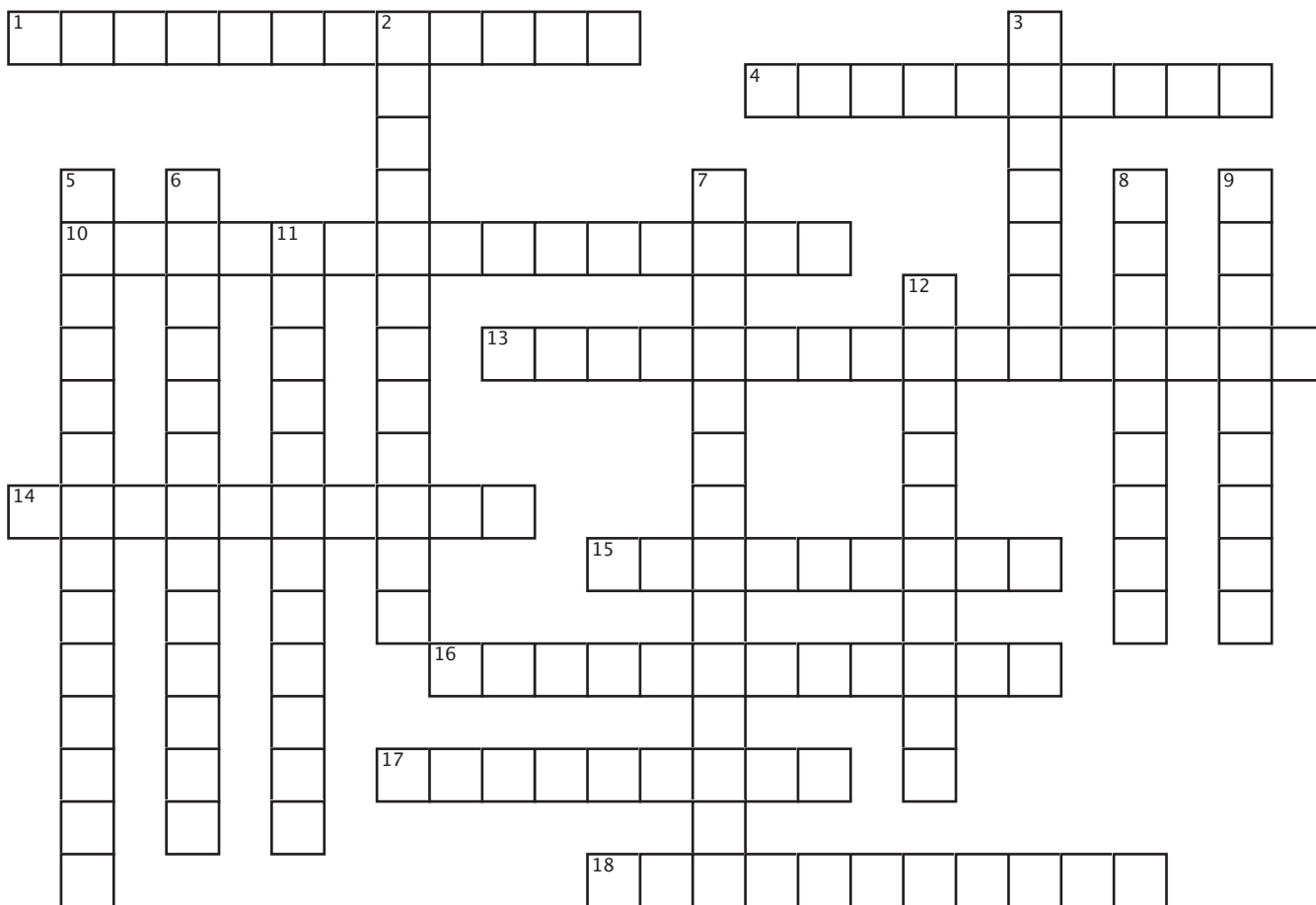
abiotic
 anion
 cation
 biomass

convergent
 divergent
 contagious
 pathogen

clone
 homozygous
 allele
 enthalpy

exothermic
 karyotype
 meiosis
 lattice

trophic
 syncline
 species
 sex linked



ACROSS

1. Person who studies the physics of the Earth
4. Person who studies chemical substances and vital processes in living organisms
10. Person who scientifically examines fossils
13. Scientist involved in protecting the natural environment from destruction or pollution
14. A specialist in viruses and viral diseases
15. Person who studies the relationships between organisms and their environments
16. Person who studies the functions of living organisms
17. Scientist who examines the structure, physiology, development and classification of animals
18. A doctor who interprets and diagnoses the changes caused by disease in tissues and body fluids, as well as deceased persons

DOWN

2. Scientist involved in studying the immune system
3. Scientist who diagnoses, prevents and treats diseases of the teeth
5. Doctor who studies the causes, distribution and control of disease in populations
6. Person who studies climates and their phenomena
7. A specialist biologist who studies micro-organisms and their effects on other living organisms
8. Person who studies the origin, history and structure of the Earth
9. Person who studies the interaction between matter and energy
11. Person who studies insects
12. Person involved in the study of heritable traits

WORD LIST

biochemist
 climatologist
 dentist
 ecologist

entomologist
 environmentalist
 epidemiologist
 geneticist

geologist
 geophysicist
 immunologist
 palaeontologist

virologist
 pathologist
 microbiologist
 physiologist

physicist
 zoologist

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Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Challenging Science Crosswords – Middle Secondary

This series of *Challenging Science Crosswords* has a unique, student-friendly approach to increasing science literacy. They have been thoroughly trialled in classrooms, with positive feedback from students who find the crosswords great for revision and fun to do. Teachers have found the crosswords to be excellent springboards for promoting discussion of science concepts and issues, and for introducing or revising core curriculum. The crosswords have been designed so that teachers can target every student's ability and confidence level with an easy photocopiable feature that makes the activities simpler or more challenging.

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ABN 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

